Doctorat de l'Université de Toulouse

Interaction magnétiques isotropes et anisotropes et leur manipulation par le champ électrique: du complexe mononucléaire au matériau Herbertsmithite

Thèse présentée et soutenue, le 12 décembre 2025 par

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Acknowledgements

First of all, I would like to thank my two thesis supervisors, Nathalie and Nicolas. Nicolas, who witnessed my academic beginnings with my first atomistics classes during my first year. And Nathalie, who steered me back onto the right path of theoretical chemistry during my master's degree after I had changed my course to physics. For all these discussions on scientific or more diverse topics, I thank you. I would also like to thank all the members of the team and the laboratory who welcomed me during these three years, plus my master's internship. I must give special mention to Nadia Ben Amor, who showed me the ropes of MOLCAS and who never flinched, no matter what type of error I asked her to fix. I thank Coen de Graaf as well, who, in addition to helping me with my thesis project, welcomed me to the "Departement de Quimica Fisica i Inorganica" during an internship in Tarragona. I also wish to give my thanks the University of Toulouse, where I completed my entire academic career over nine long years. Thanks are extended to the members of the jury and reviewers for agreeing to evaluate this work. Finally, my warmest thanks go to my parents and friends who supported and encouraged me throughout this journey.

List of abreviations

AMFI Atomic Mean-Field integrals

AIMP *Ab initio* Model Potential

ANO-RCC Atomic Natural Orbitals Relativistically Core correlated

AO Atomic Orbitals

BO Born-Oppenheimer

CASSCF Complete Active Space Self-Consistent Field

CASPT Complete Active Space Perturbation Theory

CI Configuration Interaction

DDCI Difference-Dedicated -Configuration-Interaction

DFT Density Functional Theory

DMI Dzyaloshinskii-Moriya Interaction

ECP Eeffective Core Potential

EPR Electron Paramagnetic Resonance

HDvV Heisenberg-Dirac-Van Vleck

HF Hartree-Fock

HS High Spin

JT Jahn-Teller

LMCT Ligand-to-Metal Charge Transfert

LS Low Spin

MLCT Metal-to-Ligand Charge Transfert

MRCI	Multi-Reference Configuration Interaction
NEVPT	N-electron Valence Perturbation Theory
MO	Molecular Orbitals
QSL	Quantum Spin Liquid
SCF	Self-Consistent Field
SDCI	Single and Double excitation Configuration Interaction
SMM	Single Molecule Magnet
SO	Spin-Orbit
SOC	Spin-Orbit Coupling
SO-SI	Spin-Orbit-State-Interaction
WFT	Wave-Function Theory
ZFS	Zero-Field Splitting

Thesis summary

This thesis is part of the ongoing effort to understand and control the magnetic properties of molecular materials, whose growing interest arises from their potential applications in information storage, spintronics, and quantum computing. Using ab initio approaches based on wavefunction theory and density functional theory, it explores isotropic and anisotropic magnetic interactions in systems ranging from mononuclear complexes to strongly correlated materials such as Herbertsmithite. The general objective is to determine how the nature of the orbitals, the spin–orbit coupling, and external perturbations, particularly electric fields, affect the effective parameters describing magnetic behavior. The study relies on the analysis of the electronic wavefunction, obtained through advanced numerical methods, in order to establish a direct connection between the electronic structure and measurable spin quantities. The first part presents the theoretical and methodological foundations required for the rationalization of model Hamiltonians and their validation from the exact electronic Hamiltonian. Particular attention is devoted to the treatment of dynamic correlation and to the use of effective Hamiltonian theory to relate *ab initio* results to the spin parameters employed in phenomenological models. Mononuclear complexes provide a privileged framework for the analysis of spin–orbit coupling and magnetic anisotropy. Their finite size allows a complete description of the system and direct comparison with experimental data. This work investigates how the ligand field, the symmetry, the electronic configuration, and spin-orbit coupling determine the magnitude and nature of anisotropy. Two main aspects are developed : the influence of first-order spin–orbit coupling on the parameters of the Zero-Field Splitting model, and the possibility of modulating these parameters through an external electric field, opening the way toward magnetoelectric control of

spin states. The study is then extended to polynuclear systems and model complexes, where exchange interactions become dominant. A detailed analysis is proposed to identify the mechanisms underlying these isotropic and anisotropic exchange interactions and to evaluate their evolution under the influence of first-order spin–orbit coupling and applied electric field. Finally, the application of this methodology to a real system, Herbertsmithite, $ZnCu_3(OH)_6Cl_2$, makes it possible to address the physics of quantum spin liquids. In this Kagomé material, geometric frustration prevents the establishment of magnetic order, leading to a quantum disordered state. Density functional theory calculations combined with multireference embedded fragment approaches were used to determine the isotropic and anisotropic interactions identified as responsible for this behavior, illustrating the ability of *ab initio* methods to describe correlated quantum systems from their local electronic structure. This work contributes to the development of a rigorous methodology for understanding magnetic anisotropies through a detailed electronic description. By bridging quantum chemistry and the physics of correlated magnetism, it provides a unified framework for modeling spin interactions and open the way for the rational design of molecular systems and quantum materials with magnetic properties controlled by external perturbations.

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Résumé de Thèse

L'anisotropie magnétique est à l'origine du comportement des aimants moléculaires ("single molecule magnet", SMM) qui a été observé pour la première fois en 1993 par le groupe de Roberta Sessoli dans le complexe $Mn_8^{III}Mn_4^{IV}O_{12}(CH_3COO)_{16}(H_2O)_4$, en court Mn_{12} [1]. Depuis les premières études puis celles menées sur des composés similaires [2], de nombreux autres SMM ont été synthétisés, caractérisés et étudiés théoriquement permettant une bien meilleure compréhension du phénomène. Le domaine fait cependant encore l'objet de nombreux travaux tant dans le domaine fondamental de la mécanique quantique (quantum tunnelling, quantum interferences, etc.) que dans des domaines plus appliqués (spintronique [3], quantum computing [4], high data storage, etc.). L'ambition de réaliser un qubit et de le contrôler afin de stocker et manipuler l'information quantique à l'échelle de la molécule, paraissait démesurée il y a 30 ans. Aujourd'hui, cela semble accessible et de récents travaux ont même permis de proposer des portes logiques [5] ou des transistors [6] à l'échelle de la molécule unique. L'anisotropie magnétique conduit en outre à des propriétés singulières et très intéressantes dans d'autres types systèmes à fermions fortement corrélés, comme par exemple les matériaux ferroélectriques (*i.e.* couplant propriétés magnétique et électrique) ou se comportant comme des liquides de spin. Le travail présenté dans cette thèse vise à améliorer la compréhension des mécanismes microscopiques à l'œuvre dans des molécules ou matériaux dans lesquels l'anisotropie magnétique joue un rôle crucial à l'aide de méthodes de la chimie théorique, WFT et DFT. Plus précisément, il consiste à déterminer les interactions isotropes et anisotropes dans des systèmes magnétiques de natures différentes : molécules modèles ou réelles, mono- ou polynucléaires et un matériau fortement corrélé. Tous ces systèmes sont dits corrélés et leur étude au moyen de

techniques expérimentales ou par la théorie est en général complexe. Pour cette raison, les expérimentateurs chimistes et physiciens ainsi que les théoriciens de la physique de solide ont recours à des Hamiltoniens plus simples que le hamiltonien électronique exact (que seuls les chimistes théoriciens manipulent) appelés Hamiltoniens modèles. Ces Hamiltoniens ne considèrent que les sites magnétiques, les réduisent à leurs seuls électrons célibataires, et souvent au seul degré de liberté de spin, ce qui permet de réduire considérablement la base dans laquelle ils s'expriment. Les interactions de ces modèles sont des interactions effectives et peuvent correspondre à des mécanismes complexes. La réduction de la base qui souvent suffit pour les matériaux moléculaires où les molécules sont de taille raisonnable (quelques centres magnétiques) ne suffit pas lorsqu'il s'agit d'un matériau fortement corrélé, pour lequel la base est infinie, et l'utilisation de méthodes de traitement des effets collectifs (comme les tensors network [7, 8] ou machine learning [9]) au moyen de ces modèles sont alors nécessaires. Ces dernières sont essentiellement les outils des physiciens théoriciens et nous n'aurons pas la prétention de décrire les propriétés collectives des matériaux fortement corrélées. En revanche, la détermination des modèles, leur validation et la quantification de leurs interactions effectives sont des enjeux de la chimie théorique dans le domaine du magnétisme et sont le cœur des travaux réalisés durant ce travail de doctorat tant sur des molécules que sur un matériau fortement corrélé. En effet, ces modèles sont souvent anticipés sur la base de l'intuition physico-chimique et peuvent être très complexes comme celui qui suit :

$$\hat{H}_{AB} = J(\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B) + \hat{\mathbf{S}}_A \cdot \overline{\overline{D}}_{AB} \cdot \hat{\mathbf{S}}_B + \mathbf{d}_{AB} \cdot (\hat{\mathbf{S}}_A \times \hat{\mathbf{S}}_B) \tag{0.0.1}$$

où J_{AB} est la constante de couplage isotrope, $\overline{\overline{D}}_{AB}$ le tenseur symmétrique d'échange anisotrope et $\mathbf{d}_{AB}$ le pseudo-vecteur de Dzyaloshinskii-Moriya. Les interactions apparaissant dans l'équation 0.0.1, sont nombreuses et ne sont pas des observables. Les spécialistes de caractérisation des systèmes magnétiques réalisent des fits de leurs valeurs permettant de reproduire au mieux les propriétés mesurables (susceptibilité magnétique, courbe d'aimantation, spectre RPE, etc.). Cependant, plusieurs jeux de valeurs des paramètres peuvent parfois reproduire avec la même qualité les quantités mesurées et rien ne permet d'assurer la validité du modèle utilisé. C'est ici que le chimiste théoricien entre en jeu. Parmi ses atouts, la fonction d'onde du hamiltonien électronique exact, qu'il est le seul à pouvoir déterminer, joue un rôle crucial dans la validation des modèles utilisés et dans la proposition de nouveaux modèles si nécessaire. Les fonctions d'onde calculées sont également essentielles pour la rationalisation des propriétés étudiées. Ces rationalisations qui utilisent aussi la théorie du champ cristallin et/ou des modèles orbitalaires, permettent d'établir un dialogue qui s'est souvent avéré très utile et productif avec des expérimentateurs spécialistes de synthèse. Elles ont servi à maintes reprises à guider les synthèses dans la conception de nouveaux composés aux propriétés améliorées.

Cette recherche de nouveaux composés ferroélectriques est particulièrement cruciale dans le domaine du quantum computing qui nécessite de manipuler des spin qubits au moyen d'une perturbation extérieure. En effet, si l'utilisation du champ magnétique est appropriée à l'échelle macroscopique, elle se révèle problématique à l'échelle nanoscopique car le champ magnétique est difficilement focalisable sur une seule molécule (son application entrainerait la manipulation de plusieurs spin qubits simultanément) et il convient de se tourner vers l'utilisation du champ électrique, qui lui est aisément focalisable. Les systèmes magnétiques étant souvent dépourvus de polarisation électrique, leur manipulation par l'application d'un champ électrique reste malgré tout un défi. Les travaux des spécialistes des systèmes multiferroïques ont montré que les matériaux hybrides (par exemple constitués de couches successives de matériaux magnétiques et ferroélectriques) présentent une modification du moment magnétique sous l'effet d'un champ magnétique et, symétriquement, une modification du moment électrique sous l'effet d'un champ électrique mais qu'ils présentent rarement une réponse magnétique au champ électrique. Pour que ce soit le cas, il est en effet nécessaire que les mêmes électrons soient ceux impliqués dans les réponses aux deux champs et donc que le système présente un couplage magnétoélectrique qui est généralement de très faible intensité. Là encore, le chimiste théoricien est susceptible d'apporter un éclairage sur les mécanismes qui sont à l'origine de ce couplage magnétoélectrique. Les travaux antérieurs de l'équipe ont conduit à l'étude de nombreux composés moléculaires anisotropes. Dans le travail présenté ici, nous avons souhaité élargir les sujets abordés précédemment dans plusieurs directions :

- L'exploration des systèmes mono-nucléaires présentant un couplage spin-orbite du premier ordre. Ce travail a été réalisé en collaboration avec des expérimentateurs du LCC.
- L'analyse et la rationalisation de l'impact du champ électrique sur les paramètres d'anisotropie axial et rhombique d'un composé mononucléaire. Le travail a fait l'objet d'une collaboration avec des expérimentateurs de l'ICMMO (Paris-Saclay).
- La détermination de l'origine physique du tenseur symétrique d'anisotropie d'échange dans les composés bi-nucléaires et ses effets d'interférence avec le terme dit de Dzyaloshinskii-Moriya. Ici encore, nous avons exploré les limites du régime du couplage spin-orbite du premier ordre et proposé un modèle plus sophistiqué permettant de traiter des systèmes qui relèvent de ce régime.
- Enfin, dans l'optique d'élargir nos objets d'études, nous avons extrait l'ensemble des interactions, isotropes et anisotropes, d'un matériau fortement corrélé : l'Herbertsmithite. Ce dernier travail a été réalisé en collaboration avec une collègue théoricienne en physique du solide.

Méthodologie de calcul

Le Hamiltonien électronique exact

Le point de départ de la description des propriétés électroniques des systèmes par calculs *ab initio* est le Hamiltonien électronique exact, formulé dans l'approximation de Born-Oppenheimer :

$$\hat{H}_{elec} = -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{|r_i - R_A|} + \sum_{i=1}^{N}\sum_{j<i}^{N}\frac{1}{|r_i - r_j|} \tag{0.0.2}$$

Les trois termes représentent dans l'ordre l'énergie cinétique des électrons, leur interaction avec les noyaux et la répulsion coulombienne entre électrons. Il contient l'ensemble des interactions nécessaires à la description de la structure électronique des systèmes étudiés. Cependant, sa résolution exacte est impossible pour des systèmes de taille chimique, d'où la nécessité d'introduire des approximations hiérarchisées permettant d'en dériver une description effective et interprétable.

Méthode basée sur la fonction d'onde

Dans les systèmes basés sur des métaux de transition, le magnétisme est une manifestation d'une couche d non remplie (exclusivement la couche 3d dans ce travail). En effet, les ions métalliques au cœur de ces études présentent des électrons célibataires, les composés sont donc à couches ouvertes. Une bonne description de ce type de système passe par l'utilisation de méthode de calcul multi-référenciel. La méthode Complete Active Space Self Consistent Field (CASSCF) [10] représente la méthode de choix pour les calculs que nous souhaitons réaliser. Elle propose de diviser les orbitales moléculaires en trois sous-espaces, les inactives qui resteront doublement occupées tout le long du calcul, les virtuelles qui resteront vides et finalement les orbitales actives dont le nombre d'occupation peut varier. Une interaction de configuration est alors calculée dans l'espace des orbitales actives. Cela permet d'obtenir une fonction d'onde qui s'exprime selon plusieurs déterminants de Slater :

$$|\Psi_{\mathrm{CASSCF}}\rangle = \sum_i c_i |\psi_i\rangle \tag{0.0.3}$$

où les déterminants de Slater $|\psi_i\rangle$ sont générés à partir de toutes les combinaisons électroniques possibles dans l'espace actif, et les coefficients c_i sont optimisés en parallèle avec les orbitales. Cette approche capture la corrélation statique, c'est-à-dire la corrélation entre les électrons à l'intérieur de l'espace actif. Cependant, la corrélation dynamique, issue des fluctuations instantanées d'électrons en dehors de l'espace actif, demeure absente à ce stade et doit être ajoutée par des méthodes nommées post-CASSCF.

Inclusion de la corrélation dynamique

L'extraction de valeurs précises des interactions que nous cherchons à déterminer requiert l'inclusion de la corrélation dynamique dans nos calculs. Cette corrélation est issues principalement des mono- et diexcitations apparaissant en dehors des configurations de l'espace actif. Il existe deux types d'approches pour récupérer ces effets de corrélation selon qu'elles sont basées sur des méthodes perturbatives ou des méthodes variationnelles.

Méthodes perturbatives de second ordre : CASPT2 et NEVPT2

Les approches **CASPT2** (*Complete Active Space Second-Order Perturbation Theory*) [11] et **NEVPT2** (*N-Electron Valence State Perturbation Theory*) [12, 13] permettent d'introduire la corrélation dynamique à partir de la fonction d'onde CASSCF de référence. Elles reposent sur un développement perturbatif du Hamiltonien électronique, dans lequel les effets des configurations extérieures à l'espace actif sont pris en compte jusqu'au second ordre. Ces deux méthodes conduisent à des résultats de qualités comparables mais chacune présente ses avantages et ses inconvénients. Ces approches perturbatives permettent d'accéder à une partie de la corrélation dynamique utilisée pour corriger le spectre énergétique des états calculés.

Méthode variationnelle : Difference Dedicated Configuration Interaction

La méthode **DDCI** (*Difference Dedicated Configuration Interaction*) [14] repose sur un formalisme variationnel, contrairement aux approches perturbatives. Elle consiste à diagonaliser explicitement le Hamiltonien électronique dans un espace de configuration restreint. Son caractère variationnel en fait une méthode très précise, mais au prix d'un coût computationnel élevé, limitant son application à des systèmes de taille réduite. Dans le but de réduire le temps de calcul nécessaire, trois variantes de cette méthode ont été développées, chacune prenant en compte des catégories d'excitation différentes. Ces variantes permettent d'apporter la corrélation dynamique, à la fois sur les énergies des états mais aussi en agissant sur la fonction d'onde de ces derniers, permettant la description *ab initio* la plus sophistiquée des couplages magnétiques au sein de complexes ou matériaux magnétiques.

Inclusion du couplage spin–orbite

Les effets relativistes, et en particulier le couplage spin-orbite (SO), sont à l'origine des propriétés d'anisotropie magnétique qui sont étudiées dans ce manuscrit. Dans ce travail, ces effets sont incorporés explicitement à partir d'une procédure en deux étapes. Dans un premier temps la fonction d'onde spin-orbite "free" est calculée au niveau CASSCF, l'ajout de la corrélation dynamique se fait par correction perturbative (CASPT2 ou NEVPT2) ou variationnelle (DDCI). Ensuite, la matrice d'interaction entre les différentes composantes M_S des états spin-orbite "free" générés dans la première étape, est construite en incluant les effets spin-orbite (et parfois spin-spin) à traver la méthode d'interaction d'état (SO-SI). Le Hamiltonien utilisé est :

$$\hat{H}_{\mathrm{SO}} = \sum_i \xi(r_i)\,\hat{\mathbf{l}}_i \cdot \hat{\mathbf{s}}_i \tag{0.0.4}$$

où $\xi(r_i)$ est le couplage radial dépendant de la distance à chaque noyau, $\hat{\mathbf{l}}$ et $\hat{\mathbf{s}}$ sont les opérateurs de moment angulaire et de spin. Une fois diagonalisée, cette matrice permet l'analyse du spectre énergétique SO ainsi que la fonction d'onde des états couplés. Les états propres résultant incluent ainsi les contributions relativistes dans la fonction d'onde. Cette inclusion cohérente du couplage spin-orbite est indispensable à la compréhension de l'anisotropie locale et des mécanismes d'échange anisotrope observés dans les systèmes étudiés.

Théorie des Hamiltoniens effectifs

Les états ainsi obtenus à partir de méthodes *ab initio* représentent la meilleur approximation des fonctions d'onde exactes. Cependant, les bases dans lesquelles ils sont développés peuvent être de très grande taille, rendant l'analyse compliquée. Afin de permettre la comparaison des résultats *ab initio* avec les modèles phénoménologiques utilisés par les expérimentateurs, une méthodologie basée sur la Théorie des Hamiltoniens effectifs [15] est appliquée. Elle permet, à partir de l'énergie des états du bas du spectre et de la projection de leur fonction d'onde sur un espace cible, une comparaison directe entre la matrice numérique ainsi obtenue et la matrice représentative du hamiltonien modèle supposé représenter les propriétés physico-chimiques du système étudié. Il est alors possible de valider ce modèle voire, si ce n'est pas le cas, d'identifier les configurations et/ou les opérateurs nécessaires pour l'améliorer.

Complexes mononucléaires

Dans un premier temps, notre intérêt se porte sur des complexes mononucléaires de métaux de transition synthétisés par des expérimentateurs avec qui nous avons collaboré afin d'identifier les ingrédients (nature et état d'oxydation du métal, nombre et position des ligands) permettant d'obtenir une anisotropie magnétique élevée. Le modèle de Zero-Field-Splitting, utilisé pour représenter les états de basse énergie des complexes étudiés et les interactions anisotropes qui leur ont donné naissance, s'écrit sous forme tensorielle de la sorte :

$$\hat{H}_{ZFS} = \hat{\mathbf{S}} \cdot \overline{\overline{D}} \cdot \hat{\mathbf{S}} \tag{0.0.5}$$

où $\overline{\overline{D}}$ est le tenseur symmétrique ZFS de rang deux. L'obtention de la valeur des opérateurs de ce tenseur se fait par la comparaison directe entre la matrice numérique obtenue par la théorie des Hamiltoniens effectifs et la matrice modèle développée dans la base des composantes M_S formant l'état spin-orbite "free" fondamental. Une fois obtenu, ce tenseur peut être diagonalisé et réduit à deux paramètres effectifs, D et E. Le premier est appelé paramètre d'anisotropie axiale, le second paramètre d'anisotropie rhombique. Le terme D quantifie la séparation énergétique entre les composantes M_S formant l'état fondamental, en l'absence de champ magnétique externe. Il traduit la préférence du moment magnétique pour une orientation parallèle ($D < 0$, anisotropie axiale facile) ou perpendiculaire ($D > 0$, anisotropie planaire) à l'axe principal du tenseur d'anisotropie. Le paramètre E, généralement plus faible, rend compte de la rhombicité du champ de ligands ; il traduit la levée supplémentaire de dégénérescence entre les directions x et y perpendiculaires à l'axe principal. Ainsi, les valeurs et le signe de D et E déterminent la nature et l'intensité de l'anisotropie magnétique locale, et constituent des grandeurs fondamentales pour la compréhension des propriétés magnétiques des complexes moléculaires.

L'étude de l'anisotropie locale présentée dans ce travail s'organise selon deux axes : (i) l'impact du couplage spin-orbite du premier-ordre sur les valeurs d'anisotropie ainsi que les conditions nécessaires à son observation ; (ii) l'impact d'une perturbation extérieure, telle qu'un champ électrique, sur les paramètres d'anisotropie et l'étude approfondie des sources de leur modulation.

Complexes de Fe(II)

L'étude des complexes pentacoordinés de Fe(II) s'inscrit dans le cadre plus large de la recherche de systèmes moléculaires où le couplage spin–orbite intervient au premier ordre. Dans les complexes de métaux de transition, les effets de couplage spin–orbite sont en général de second ordre, agissant par l'interaction de l'état fondamental avec des états excités bien séparés énergétiquement. Cependant, dans certains environnements de coordination présentant une quasi-dégénérescence entre orbitales d, le couplage spin–orbite peut agir directement entre deux états fondamentaux quasi-dégénérés. C'est cette situation particulière, génératrice d'anisotropies exceptionnellement fortes, qui a motivé la conception et la synthèse de la série de complexes Fe(II) étudiés ici.

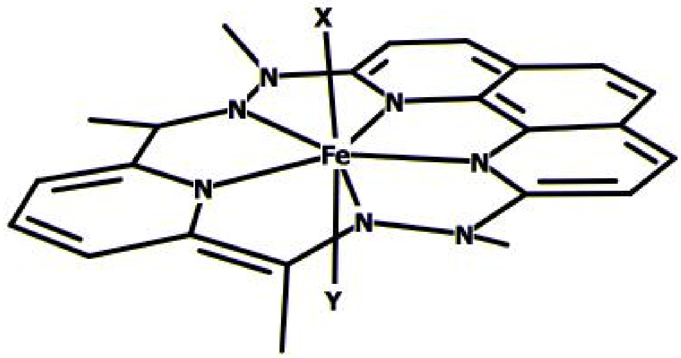

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FIGURE 1 – Représentation schématique des complexes de Fe(II) avec les atomes d'Hydrogènes cachés

Ces composés présentent un environnement bipyramidal pentagonal quasi idéal, dans lequel les orbitales d_{xz} et d_{yz} du métal central sont quasi-dégénérées. Cette configuration électronique permet au moment orbital de ne pas être complètement « quenché » par le champ de ligands, ouvrant la voie à un couplage spin–orbite de premier ordre. Dans ces conditions, le moment magnétique retrouve une contribution du moment angulaire L et ne peut plus plus être simplement considéré égal au moment de spin S. Dans ce régime, le Hamiltonien de champ nul (ZFS) n'est plus strictement valide, car les états magnétiques sont directement mélangés par l'interaction spin–orbite.

En complément de l'analyse *ab initio*, un modèle de champ de ligands, déjà appliqué dans des systèmes similaires [16, 17], a été utilisé afin de décrire la levée de dégénérescence entre les orbitales d_{xz} et d_{yz} par un Hamiltonien à deux paramètres :

$$\hat{H}_{LF} = \begin{array}{c} \begin{array}{cc} d_{xz} & d_{yz} \end{array} \\ \begin{pmatrix} -\delta_1 & \delta_2 \\ \delta_2 & \delta_1 \end{pmatrix} \end{array}$$

Le paramètre δ_1 quantifie la séparation énergétique entre les deux orbitales et δ_2 leur couplage hors-diagonal, lié à la distorsion de la géométrie locale. Ce modèle simple permet de dériver des expressions pour le spectre SO et établir un lien, quand autorisé, avec le Hamiltonien ZFS : dans le cas où δ_1 est faible, le mélange entre d_{xz} et d_{yz} devient important et le couplage spin–orbite agit au premier ordre. L'ajustement des paramètres du modèle aux calculs montre que, pour les complexes halogénés, la levée de dégénérescence entre états spin free est de l'ordre de quelques dizaines de cm^{-1}, confirmant un très fort couplage spin–orbite et la non-applicabilité du modèle ZFS classique. À l'inverse, le complexe avec ligand axial H_2O présente un écart beaucoup plus important (environ 400 cm^{-1}), et le couplage spin–orbite n'agit alors qu'au second ordre, ramenant le système dans le régime décrit par le Hamiltonien ZFS. Cette analyse met en évidence la transition progressive, au sein de la série étudiée, entre un régime de couplage spin–orbite de second ordre (modèle ZFS valable) et un régime de premier ordre. Elle montre ainsi qu'un contrôle fin de la symétrie locale et de la nature du ligand axial permet d'ajuster la proximité énergétique des orbitales d_{xz} et d_{yz}, et donc d'amplifier le couplage spin–orbite effectif. Cette stratégie constitue une voie prometteuse de conception de complexes à très forte anisotropie magnétique.

Complexe de Ni(II)

L'étude de ce complexe s'incrit dans la continuité d'une étude précédente qui a identifié une anisotropie locale forte dans un complexe de Ni(II) [17]. Dans cette géométrie, l'état fondamental est un triplet ($S = 1$), et l'anisotropie magnétique découle essentiellement du couplage spin–orbite de second ordre entre les composantes de ce multiplet et les états excités. Ce complexe présente une configuration électronique d^8 où les orbitales $d_{x^2-y^2}$ et d_{xy} sont quasi-dégénérées permettant de retrouver une contribution du couplage spin-orbite de premier ordre aux paramètres d'anisotropie locale. L'objectif de cette étude est d'identifier, à partir de calculs *ab initio* effectués sous l'effet d'un champ électrique homogène, les contributions électroniques responsables de la valeur du paramètre D et E et d'en comprendre les variations sous l'effet d'une perturbation extérieure.

Nous avons pu identifier le premier état excité comme étant celui qui contribue le plus à la valeur de D et E et, surtout, celui dont la contribution est le plus affectée par le champ électrique. Une analyse approfondie nous a alors permis de rationaliser ces observations à partir de l'évolution du champ de ligand en présence du champ électrique ainsi que celle du couplage spin-orbite.

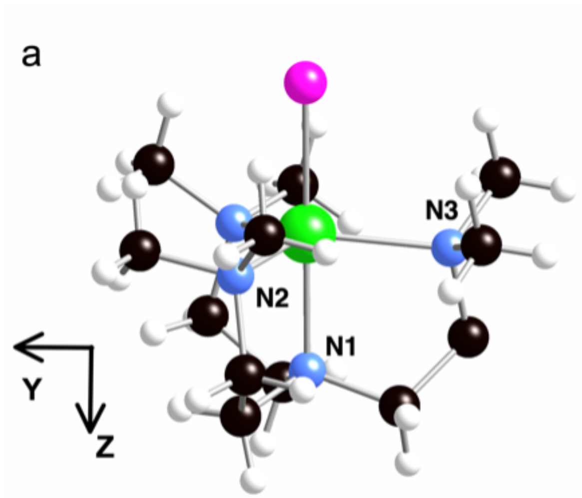

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FIGURE 2 – Représentation du complexe [Ni(Me_6trenCl)](ClO_4), Ni en vert, Cl en violet, N en bleu, C in noir and H in blanc

Nous avons pu montrer que la variation linéaire de D avec le champ électrique externe peut se décomposer en deux effets additifs, électroniques et géométriques. Nous avons aussi pu corréler l'évolution de E avec le champ électrique à la modification de l'asymétrie du champ de ligands dans le plan grâce aux variations du poids des déterminants impliquant une simple occupation des orbitales d_{xz} ou d_{yz}.

Ces résultats montrent la capacité du protocole CASSCF/NEVPT2/SO–SI à décomposer de manière quantitative les contributions de chaque excitation électronique au paramètre D et à relier ces grandeurs aux caractéristiques du champ de ligands local. L'étude du complexe de Ni(II) illustre ainsi comment les interactions de second ordre entre états de spin et de configuration peuvent être modulées par des variations de champ de ligands ou par l'application d'un champ électrique, offrant une compréhension détaillée des mécanismes à l'origine de l'anisotropie magnétique dans les complexes de métaux de transition.

Complexe binucléaire modèle

Le système étudié dans ce chapitre est un complexe de Cu(II) entouré de ligands Cl^-, choisi comme modèle représentatif pour la compréhension des interactions d'échange anisotrope dans des systèmes binucléaires. Ce composé a fait l'objet de plusieurs études théoriques [18-20] en raison de sa structure simple permettant l'application de méthode de calculs poussées et la dérivation de modèles analytiques simples.

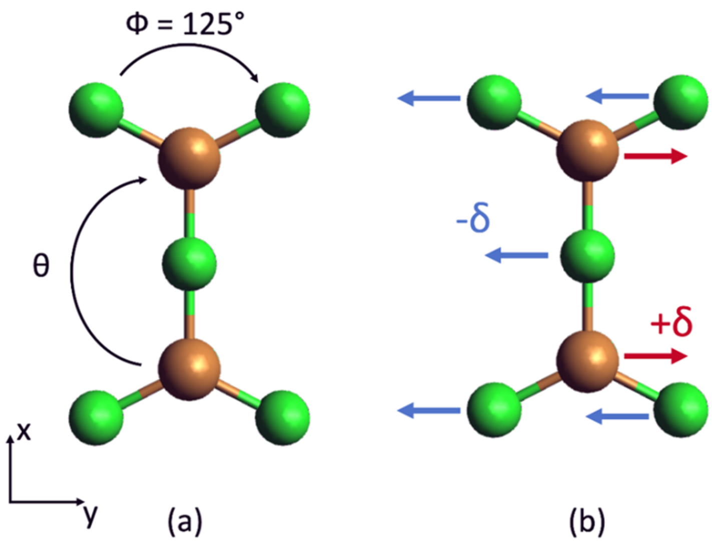

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FIGURE 3 – (a) Molécule modèle de Cu_2Cl_5 de symétrie D_{2h}, (b) Déplacements δ des atomes générés par l'application du champ électric selon la direction y.

Dans la géométrie considérée, les orbitales magnétiques des états de plus basse énergie sont de type $d_{x^2-y^2}$ et d_{xy}. Leur quasi-dégénérescence ouvre la possibilité à un couplage spin–orbite de premier ordre, capable de générer de grandes valeurs d'anisotropie. L'objectif central de cette étude est la détermination des origines physiques du tenseur symétrique d'échange, noté $\overline{\overline{D}}_{AB}$. Cette analyse repose sur le Hamiltonien modèle Multi-Spin, écrit sous la forme :

$$\hat{H}_{MS} = J_{AB}\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B + \hat{\mathbf{S}}_A \cdot \overline{\overline{D}}_{AB} \cdot \hat{\mathbf{S}}_B + \mathbf{d}_{AB} \cdot (\hat{\mathbf{S}}_A \times \hat{\mathbf{S}}_B) \tag{0.0.6}$$

où J_{AB} est le couplage isotrope, $\overline{\overline{D}}_{AB}$ le tenseur symétrique d'échange et $\mathbf{d}_{AB}$ le pseudo-vecteur de Dzyaloshinskii-Moriya. Afin d'isoler les mécanismes physiques responsables de l'apparition de D_{AB}, un modèle électronique simplifié a été développé, reposant sur six électrons répartis dans quatre orbitales (les orbitales $d_{x^2-y^2}$ et d_{xy} de chaque centre). Les calculs *ab initio* effectués permettent la détermination de la valeur des paramètres du hamiltonien modèle et une analyse détaillée des interactions électroniques et spin–orbite qui expliquent ces valeurs.

Le traitement théorique repose sur une méthode variationnelle développée dans le cadre de ce travail. Cette approche consiste à extraire, à partir des calculs *ab initio*, la matrice hamiltonienne complète incluant le couplage spin–orbite ainsi que les éléments de couplage électronique les plus pertinents, puis à la diagonaliser dans une base d'états judicieusement sélectionnée. Cette base est construite de manière à contenir l'état fondamental et les principaux états excités susceptibles de contribuer aux propriétés anisotropes. Ce formalisme permet ainsi de décomposer analytiquement les contributions de chaque état excité aux paramètres du Hamiltonien effectif, notamment à la partie symétrique du tenseur d'échange.

Dans un premier temps, la variation de l'angle avec les ligands a été étudiée afin de se rapprocher progressivement du régime de couplage spin-orbite du premier-ordre. Cette déformation géométrique influe sur l'écart énergétique entre les orbitales $d_{x^2-y^2}$ et d_{xy} des deux centres, et agit donc directement sur les valeurs d'anisotropie. Lorsque l'angle diminue, se rapprochant d'une symétrie C_3 autour de chaque centre magnétique, la séparation énergétique entre ces orbitales se réduit, et le couplage spin–orbite devient de plus en plus efficace, menant progressivement à un régime de premier ordre. L'analyse variationnelle met en évidence une augmentation nette des composantes du tenseur symétrique d'échange D_{AB} à mesure que l'on s'approche de ce régime. Cette évolution démontre que la distorsion géométrique peut être utilisée pour amplifier les effets du couplage spin–orbite et modifier la nature même de l'interaction magnétique entre les deux centres. Cette étude permet aussi de déterminer l'importance de la corrélation dynamique lors de l'extraction du tenseur symmétrie. Plusieurs schémas d'extraction ont été appliqués avec une inclusion graduelle de la corrélation dynamique au travers des énergies et de la fonction d'onde des états qui composent le modèle.

Dans un second temps, un champ électrique externe a été appliqué pour étudier la sensibilité du système aux perturbations extérieures. Lors des études précédentes, ce champ a été identifié comme source d'hybridation entre les orbitales d favorisant l'apparition d'une valeur non nulle du pseudo-vecteur de Dzyaloshinskii–Moriya ($\mathbf{d}_{AB}$) jusqu'alors absent dans le complexe sans champ (il présente un centre d'inversion). L'analyse des contributions des états excités a permis, dans le cas d'une certaine orientation du champ, d'identifier un phénomène d'interférence entre les deux types d'interaction, symétrique et antisymétrique.

L'ensemble de ces résultats met en évidence la capacité des méthodes de calcul employées à identifier les origines des composantes du hamiltonien modèle considéré. Il montre que la quasi-dégénérescence orbitale, combinée à une perturbation externe contrôlée, constitue le levier principal de l'apparition des composantes anisotropes d'échange, tant symétriques qu'antisymétriques. Cette approche offre ainsi un cadre cohérent pour comprendre, à partir d'un modèle simplifié, comment engendrer et moduler les interactions magnétiques anisotropes dans des systèmes binucléaires grâce à un champ électrique.

Herbertsmithite

Ce chapitre s'inscrit dans la continuité des travaux visant à étendre l'application des méthodes de chimie quantique à des systèmes d'intérêt pour la physique de la matière condensée, en particulier ceux relevant du magnétisme frustré. L'étude porte sur l'Herbertsmithite [21], $ZnCu_3(OH)_6Cl_2$, matériau emblématique du réseau kagomé, longtemps considéré comme l'une des meilleures réalisations expérimentales d'un liquide de spins quantique. Dans ce composé, les ions Cu^{2+} ($S = \frac{1}{2}$) forment un réseau de triangles équilatéraux partageant leurs sommets, conduisant à une frustration géométrique qui empêche l'établissement d'un ordre magnétique même à très basse température. De nombreux modèles ont été proposés pour décrire ce type de système, à commencer par un modèle d'Heisenberg de spins couplés entre plus proches voisins, mais d'importantes déviations aux prédictions de ce modèles ont été observées lors des études expérimentales. Plusieurs mécanismes sont soupçonnées d'être à la source de ces déviations. Le premier est l'existence de couplages isotrope au-delà des plus proches voisins au sein du plan Kagomé, entre ions cuivre séparés par de grande distances. Un autre est lié à la présence de défauts dans la structure correspondant à la substitution d'ions Zn^{2+} par des ions Cu^{2+} (ou vice-versa) permettant d'établir des couplages magnétiques entre plans kagomé sinon quasiment isolés magnétiquement. Une autre source de déviation est la présence d'interactions anisotropes, plus précisément du pseudo-vecteur de Dzyaloshinskii-Moriya.

La première partie de l'étude repose sur des calculs DFT appliqués à des fragments immergés du matériau judicieusement choisis pour évaluer les couplages magnétiques isotropes. Appliqués aux plans kagomé, ils montrent que l'interaction entre premiers voisins est antiferromagnétique ($J \approx 180$ K) et domine largement les couplages entre centres plus lointains dont le couplage est deux ordres de grandeur plus faible. Quand la substitution d'un ion Zn^{2+} par un ion Cu^{2+} est considérée, ces calculs mettent en évidence un fort couplage ferromagnétique (dont l'intensité est un tiers de celle des couplage premiers voisins au sein du réseau kagomé) entre les sites magnétiques du réseau kagomé et ces défauts, remettant fortement en question le caractère strictement bidimensionnel du matériau.

La seconde partie met en œuvre des méthodes multi-référence basées sur la fonction d'onde sur des fragments immergés afin d'extraire à la fois les interactions isotropes et anisotropes effectives. L'utilisation de la méthode DDCI, bien que très précise, conduit à une sous-estimation du couplage antiferromagnétique premiers voisins, rendant nécessaire une adaptation de la procédure. Celle-ci consiste à optimiser les orbitales des ligands pontants hydroxo responsables du couplage et à les inclure explicitement dans l'espace actif lors du traitement corrélé. Cette approche permet une description plus cohérente des mécanismes d'échange et conduit à des valeurs de couplage compatibles avec les observations expérimentales.

L'analyse des interactions anisotropes révèle que le tenseur symétrique d'échange est négligeable, tandis que l'interaction antisymétrique de Dzyaloshinskii–Moriya (DMI) présente une amplitude significative, dominée par une composante dans le plan du réseau. Cette composante, souvent négligée ou réduite à un terme perpendiculaire au plan kagomé dans les modèles usuels, doit pourtant, selon nos résultats, jouer un rôle essentiel dans la compréhension fine des propriétés magnétiques du matériau.

Ainsi, cette étude constitue la première application systématique de méthodes *ab initio* à l'extraction quantitative des interactions magnétiques dans l'Herbertsmithite. Elle met en évidence l'importance des défauts structuraux et des anisotropies d'échange dans l'interprétation des comportements expérimentaux et illustre la complémentarité entre chimie quantique et physique des matériaux fortement corrélés.

Conclusion

Plusieurs résultats remarquables ressortent de ce travail. Tout d'abord, au cours de l'étude de la série de complexes de fer, nous avons obtenu, en collaboration avec des expérimentateurs et pour la première fois, certains complexes qui relèvent du régime de spin-orbite au premier ordre. Ces résultats sont en parfait accord avec les observations expérimentales et ont permis de comprendre pourquoi les courbes d'aimantation et de susceptibilité magnétique ne pouvaient pas être correctement reproduites par les logiciels de fit des interactions D et E. En effet, la présence d'un moment angulaire non négligeable ne permet pas de définir le moment magnétique à partir du spin seul. L'étude de l'impact du champ électrique sur les paramètres D et E dans le complexe de Ni(II) a conduit à des résultats remarquables et non intuitifs. En effet, la réponse au champ électrique est considérable. La pente dD/dF est près de 100 fois supérieure à celle qui avait été calculée et mesurée pour un complexe de Mn(II) dont l'étude vient d'être très récemment publiée [22]. Par ailleurs, l'application du champ dans la direction Z, colinéaire avec l'axe magnétique de forte aimantation, induit une réponse sur D moins importante que celle dans la direction Y. Ces résultats ont été rationalisés et ils montrent que le facteur déterminant est la nature de l'excitation dans les états qui ont une forte contribution à l'anisotropie. Dans le cas considéré, nous avons vu que cette excitation qui en l'occurrence est entre les orbitales $d_{x^2-y^2}$ et d_{xy} se situe dans le plan (XY). C'est donc dans ce plan que l'application du champ électrique induira une réponse importante. Un autre résultat important est l'impact du champ sur la géométrie du complexe. En effet, sur le paramètre D des distorsions très faibles conduisent à des variations considérables de sa valeur. Le travail réalisé sur le complexe modèle Cu_2Cl_5 nous a permis de comprendre l'origine physique du tenseur symétrique d'anisotropie d'échange. Comme les paramètres de ce tenseur sont en général très faibles, nous avons considéré une molécule modèle dans une géométrie induisant un couplage spin-orbite proche du régime du premier ordre. Nous avons aussi étudié l'impact du champ électrique sur les paramètres d'anisotropie dans ce complexe bi-nucléaire, ce qui nous a permis de comprendre l'effet interférentiel entre les tenseurs symétrique et antisymétrique (Dzyaloshinskii-Moriya). La conclusion importante de ces précédents travaux est que tous les paramètres étudiés, i.e. le tenseur symétrique des complexes mono-nucléaires, le tenseur symétrique d'échange et le Dzyaloshinskii-Moriya qui apparaîssent dans les complexes poly-nucléaires contribuent au couplage magnétoélectrique. Ceci montre qu'il est possible de contrôler toutes ces propriétés magnétiques par le champ électrique. La partie la plus substantielle de cette thèse est l'étude du matériau Herbertsmithite. Mentionnons que, contrairement à plusieurs matériaux fortement corrélés étudiés auparavant (cuprates [23], nickélates [24], manganites [25], vanadates [26], etc.), celui-ci s'est avéré particulièrement difficile à décrire. La méthode DDCI qui est, à ce jour, la méthode la plus précise pour les calculs des couplages magnétiques, sous-estime de façon très importante le couplage antiferromagnétique entre premiers voisins. Ceci nous a contraint à utiliser une méthode alternative qui consiste à optimiser l'orbitale du ligand pontant qui médie le couplage et à l'introduire dans l'espace actif dans l'étape de calcul de la corrélation dynamique. Ici aussi plusieurs résultats physico-chimiques intéressants ont été obtenus :

— Le couplage avec les impuretés Cu^{2+} inter-plan issues d'un échange entre les Zn^{2+} inter-plan et les Cu^{2+} intra-plan est ferromagnétique et du même ordre de grandeur que le J premiers voisins. Ce résultat combiné avec une occurrence d'environ 15% de ces impuretés questionne la nature bi-dimensionnelle du matériau et pourrait expliquer l'impossibilité de reproduire ses propriétés collectives au moyen d'un modèle 2D.

— La présence de ces impuretés, qui occupent un site de symétrie C_3 et donnent des Cu^{2+} Jahn-Teller actifs, induit une distorsion locale du réseau. Ces distorsions sont à l'origine de l'apparition de deux couplages très différents entre les Cu^{2+} intra-plan premiers voisins.

— Le tenseur symétrique d'anisotropie et les interactions à plus longue portée sont d'amplitude négligeable. En revanche, le Dzyaloshinskii-Moriya est relativement important et sa plus forte composante est dans le plan. Une analyse fine des fonctions d'onde et des symétries a permis de rationaliser ce résultat. Il convient de noter que cette source d'anisotropie est très souvent négligée dans les études des propriétés collectives et que lorsqu'il est pris en compte, il est général supposé selon Z, *i.e.* perpendiculaire au plan du réseau Kagomé. Sa considération dans les modèles pourrait permettre d'améliorer la description de ce matériau.

De nombreuses perspectives se présentent aujourd'hui. L'une d'entre elles est d'ailleurs déjà abordée et pourrait donner lieu à une publication prochaine. Il s'agit d'une étude de tous les paramètres d'anisotropie dans un composé bi-nucléaire possédant un Cu^{2+} et un Ni^{2+}. Dans ce composé, le tenseur d'anisotropie locale s'ajoute aux autres tenseurs étudiés dans Cu_2Cl_5. Les dérivations analytiques d'un modèle du premier ordre montre des similitudes avec celle de l'étude précédente mais aussi quelques différences qualitatives. Par ailleurs, l'étude de l'impact du champ électrique dans un tel système pourrait faire apparaître d'éventuels effets interférentiels entre toutes les sources d'anisotropie du hamiltonien de l'Eq 0.0.1. Enfin, une perspective importante concerne les matériaux fortement corrélés. D'autres matériaux apparentés à l'Herbertsmithite comme la paratacamite [27] ou la claringbullite,[28] mériteraient d'être étudiés, d'une part pour tester les modèles couramment employés et éventuellement proposer des modèles alternatifs, d'autre part pour le défi que représente l'étude de ces systèmes si difficiles à décrire avec nos meilleures méthodes.

Introduction

Magnetic anisotropy is at the origins of single molecule magnet (SMM) behaviour, which has first been observed in the Mn_{12} molecule in 1993.[1] Since the first studies conducted on similar compounds,[2] this subject has evolved considerably and is now the focus of numerous studies in both the fundamental field of quantum mechanics (quantum tunnelling, quantum interference, etc.) and more applied fields (spintronics[3], quantum computing[4], high data storage, etc.). The ambition to create a qubit and control it in order to store and manipulate quantum information at the molecular level seemed excessive 30 years ago. Today, it seems achievable, and recent work has even succeeded in proposing logic gates [5] or transistors [6] at the single-molecule level. Magnetic anisotropy also leads to very interesting properties in strongly correlated materials. This is the case in multiferroic materials and in spin liquids. The work carried out here consists in determining isotropic and anisotropic interactions in magnetic systems of different types : model or real molecules, mono- or polynuclear, and a strongly correlated material. All these systems are highly correlated, and studying them using experimental techniques or theory is generally complex. For this reason, experimental chemists and physicists, as well as solid-state physics theorists, use Hamiltonians that are simpler than the exact electronic Hamiltonian (which only theoretical chemists manipulate), called model Hamiltonians. These Hamiltonians consider only magnetic sites, reducing them to their unpaired electrons, and often to the single degree of spin freedom, which considerably reduces the basis in which they are expressed. The interactions in these models are effective interactions and may correspond to complex mechanisms. The reduction of the basis, which is often sufficient for molecular materials where the molecules are of reasonable size (a few magnetic centres), is not sufficient when dealing with a strongly correlated material, for which the basis is infinite, and the use of methods for processing collective effects (such as tensor networks [7, 8] or machine learning [9]) using these models is then necessary. The latter are essentially the tools of theoretical physicists, and we do not presume to describe the collective properties of strongly correlated materials. However, determining models, validating them and quantifying their effective interactions is one of the challenges of theoretical chemistry in the field of magnetism and is at the heart of the work carried out in this thesis on both molecules and strongly correlated materials. Indeed, these models are often anticipated on the basis of physicochemical intuition and can be very complex, such as the following :

$$\hat{H}_{AB} = J(\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B) + \hat{\mathbf{S}}_A \cdot \overline{\overline{D}}_{AB} \cdot \hat{\mathbf{S}}_B + \mathbf{d}_{AB} \cdot (\hat{\mathbf{S}}_A \times \hat{\mathbf{S}}_B) \tag{0.0.7}$$

where J is the isotropic coupling constant, $\overline{\overline{D}}_{AB}$ the symmetric tensor of anisotropic exchange, and $\mathbf{d}_{AB}$ is the Dzyaloshinskii-Moriya pseudovector. The interactions appearing in Eq. 0.0.7 are numerous and are not observable. Specialists in the characterisation of magnetic systems perform fits of their values to best reproduce measurable properties (magnetic susceptibility, magnetisation curves, EPR spectrum, etc.). However, several sets of parameter values can sometimes reproduce the measured quantities with the same quality, and there is no way to ensure the validity of the model used. This is where the theoretical chemist comes in. Among their assets, the wave function of the exact electronic Hamiltonian, which only they can determine, plays a crucial role in validating the models used and proposing new models if necessary. The calculated wave functions are also essential for rationalising the properties studied. These rationalisations, which also use crystal field theory and/or orbital models, enable a dialogue to be established with experimentalists specialised in synthesis, which has often proved very useful and productive. They have been used on numerous occasions to guide syntheses in the design of new compounds with improved properties. Another major challenge of quantum computing is manipulating spin qubits using external perturbations. While magnetic fields appear to be the ideal perturbation for manipulating magnetic systems on a macroscopic scale, their use is highly problematic on a nanoscopic scale. Indeed, it is very difficult to focus on a single molecule and its application would involve manipulating several spin qubits simultaneously. For this reason, an electric field, which is easily focused, would be more desirable. However, it is not easy to manipulate magnetic systems, which often lack polarisation, using an electric field. Work on this subject, which has been extensively explored by specialists in multiferroic systems, has shown that hybrid materials (e.g. consisting of successive layers of magnetic and ferroelectric materials) that respond to both electric and magnetic fields rarely show a magnetic response to electric fields. For this to be the case, the same electrons must be involved in the responses to both fields, and therefore the system must exhibit a magnetoelectric coupling. Here again, theoretical chemists are likely to shed light on the mechanisms underlying this magnetoelectric coupling. The team's previous work has led to the study of numerous anisotropic molecular compounds. In the work presented here, we try to expand on the topics previously following four directions :

- The exploration of mononuclear complexes exhibiting first-order SOC in a collaboration with experimentalists from the LCC.
- Analysis and rationalisation of the impact of the electric field on the axial and rhombic anisotropy parameters of a mononuclear compound. The work was carried out in collaboration with experimenters from ICMMO (Paris-Saclay).
- Determining the physical origin of the symmetric exchange anisotropy tensor in binuclear compounds and its interference effects with the so-called Dzyaloshinskii-Moriya term. Here again, we explored the limits of the first-order SOC regime and proposed a more sophisticated model for dealing with systems that fall within this regime.
- Finally, we extracted all interactions, both isotropic and anisotropic, from a strongly correlated material : Herbertsmithite. This latest work was carried out in collaboration with a team of solid state physics theorists.

This thesis is organised around four chapters. The first is dedicated to the review of the theoretical models applied to molecular magnetic anisotropy by both experimentalist and theoricians. The model Hamiltonians and interactions will be presented and some insight on their origins will be given. The second part of this chapter is dedicated to the computational methodology that is applied throughout this work. It introduces the numerical methods used to obtain the wave-functions and energies that are crucial to make the connection to model Hamiltonians. Chapter 2 concerns the study of first-order spin-orbit coupling (SOC) and its implication on the models. Two types of mononuclear complexes were examined (Chapter 2), the first originates from a collaboration with experimentalists whose objective is the synthesis of complexes with large anisotropy induced by first-order SOC. Our work was complementary to the experimental characterisation of the complexes and to the evaluation of the validity of the usual models. The second complex has already been identified to exhibit large anisotropy values induced by a strong first-order SOC.[17] In a follow-up study, our interest lies in the evolution of the anisotropic parameters under the effect of an external electric field. Chapter 3 is the continuation of a study which had for purpose to determine the physical origins of the parameters of anisotropic exchange tensor shown in Eq 0.0.7 in a binuclear model molecule. Chapter 4 relates the work done on Herbertsmithite with a two-part study, first using Density Functional Theory for the determination of the isotropic couplings. And second, using wave-function based methods for the extraction of anisotropic parameters.

Chapitre 1

Theory and Methodology

1.1 Model Hamiltonian

1.1.1 Isotropic Hamiltonian

Heisenberg Hamiltonian

The Heisenberg-Dirac-van Vleck (HDvV) Hamiltonian is the most well-known and the most widely used Hamiltonian to model the behavior of system of interacting localized magnetic centers[29]. It writes :

$$\hat{H}_{HDvV} = -\sum_{i,j} J_{ij}\hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j \tag{1.1.1}$$

where $J_{i,j}$ is the coupling constant, $\hat{\mathbf{S}}_i$ and $\hat{\mathbf{S}}_j$ are the spin operators working on site i and j. Other conventions exist with a positive sign and/or a factor of 2 in front. With the convention of Eq. 1.1.1, a positive $J_{i,j}$ value indicates a ferromagnetic coupling between sites i and j (the coupling between these sites stabilizes the state with the largest total S value) while a negative value indicates an antiferromagnetic coupling (the state with the lowest S_{tot} value is stabilized). This Hamiltonian is regarded as a spin Hamiltonian since it acts only on the spin degrees of freedom of the system (*i.e.* the spatial part of all determinants is considered to be the same).

J is an effective parameter, that means that it considers, in the ideal case, all the mechanisms that couple $\hat{\mathbf{S}}_i$ and $\hat{\mathbf{S}}_j$. The most important ones are :

1- The direct exchange originating from the exchange integral K that has a ferromagnetic contribution.

2- The Anderson mechanism going through ionic determinants that has an antiferromagnetic contribution.

3- Super-exchange and spin polarization involving excitations of a diamagnetic bridging ligand.

In the case of two electrons in two orbitals a and b, the Anderson mechanism and its contribution to the effective integral value can be understood from the Hubbard Hamiltonian that considers both the neutral forms $|ab|$, $|\bar{a}\bar{b}|$, $|b\bar{a}|$, $|a\bar{b}|$ (*i.e.* those considered in the HDvV Hamiltonian) and the ionic forms, $|a\bar{a}|$ and $|b\bar{b}|$[30]. In the basis of M_S=0 determinants of the minimal valence space, the Hamiltonian is written :

$$
\begin{array}{cc}
 & \begin{array}{cccc} |a\bar{b}| & |b\bar{a}| & |a\bar{a}| & |b\bar{b}| \end{array} \\
\hat{H} = & \left(\begin{array}{cc|cc}
0 & K & t & t \\
K & 0 & t & t \\
\hline
t & t & U & K \\
t & t & K & U
\end{array}\right)
\end{array}
$$

$t = t_{ab}$ is the hopping integral between sites a and b and the on-site repulsion U is an effective parameter that describes the difference in energy between the ionic forms and the neutral ones. As long as U is large compared to t, neutral forms dominate the wave-function of the low energy singlet state (the triplet state is fully neutral) and it is possible to derive $J = J_{ab}$ from K, t and U. At the second-order of perturbation theory, one obtains the following expression for the effective integral J :

$$J = 2K - \frac{4t^2}{U_eff} \tag{1.1.2}$$

This expression shows the competition between a ferromagnetic component (direct exchange K>0) and an antiferromagnetic one (Anderson mechanism). Super-exchange contributions (involving the electrond and/or the orbitals of a bridging ligand) appear at higher order of perturbation and can be seen as correction to the value of the above parameters and can be expressed as :

$$J = 2K_{eff} - \frac{4t_{eff}^2}{U_{eff}} \tag{1.1.3}$$

where the parameters are now effective parameters taking into account the effect of mechanisms involving the bridging ligand electrons and orbitals. Stabilising the ionic forms compared to the neutral ones, by allowing excitation outside the minimal valence space, results in the screening of the U term. This leads to an increase of the antiferromagnetic contribution to the J coupling. For t_{eff} some mechanisms are illustrated in Figure 1.1. The hopping integral between a metal and the ligands of its coordination sphere being much larger than the metal to metal one, the t_{eff} may be much larger than the through space t_{ab} previously defined.

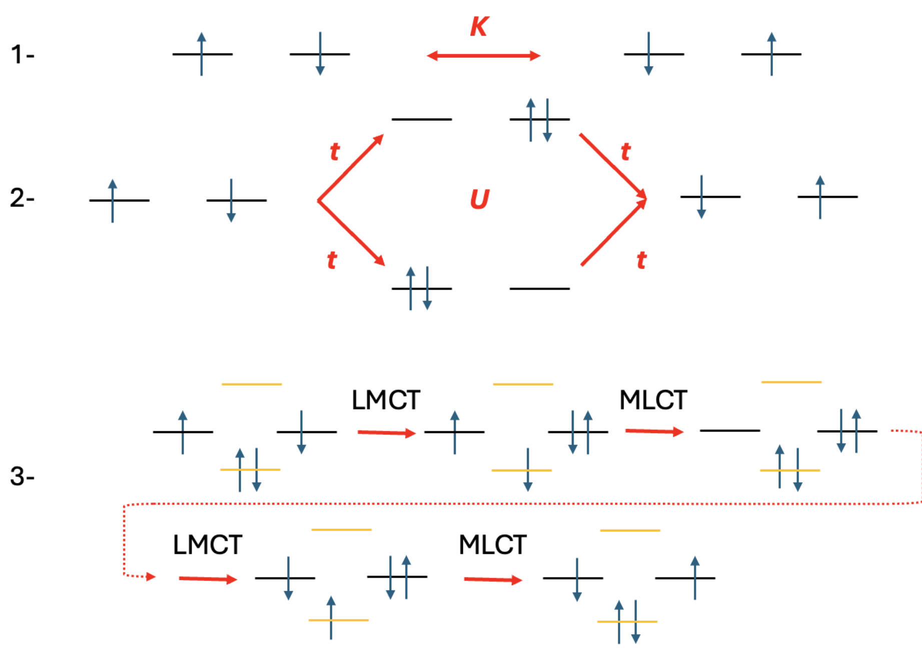

FIGURE 1.1 – Schematic representation of magnetic exchange mecanisms involving two electrons in two orbitals. 1- direct exchange, 2- kinetic exchange, and 3- super-exchange (one of the many pathways). In black, the magnetic orbitals and in orange the bridging ligand orbitals. MLCT (metal to ligand charge transfert) and LMCT (ligand to metal charge transfert) refer to charge transfert between the metals and ligand centres.

Ising Hamiltonian

The eigenfunctions of the Heisenberg Hamiltonian are eigenfunctions of the $\hat{\mathbf{S}}^2$ and $\hat{S}_z$ operators, making them intrinsically multi-determinantal (except for the maximum $|M_S|$ components), that requires Wave Function Theory (WFT) based calculation methods to be evaluated. Ising is a simpler model restricted to the components of $\mathbf{S}$ along the z axis (reducing it to a classical-like spin problem) whose states are eigenfunctions of $\hat{S}_z$ only :

$$H_{ising} = -\sum_{i,j} J_{ij}\hat{S}_{z,i}\hat{S}_{z,j} \tag{1.1.4}$$

The magnetic moment of each magnetic site are now considered to align with the z axis at all time, taking a $\pm S$ value. The main advantage of this approximation is that the eigenfunctions of this Hamiltonian can be assimilated to those of broken-symmetry DFT methods. It is then possible to fit J_{ij} values to DFT results. DFT avoiding the diagonalization of large matrices, it is much faster and applicable on larger systems than WFT methods.

1.1.2 Magnetic anisotropy in mononuclear complexes

Effects such as Spin-Orbit-Coupling (SOC) tend to create magnetic anisotropy that cannot be described by Eq.1.1.1. More specifically on mononuclear systems (only one magnetic centre) with a ground state of spin larger or equal to 1 and correct symmetry conditions, a lift of degeneracy between the different M_S components of a same S state can be observed even in the absence of magnetic field, this phenomenon is called Zero-Field-Splitting.

The associated model spin Hamiltonian[31] is written :

$$\hat{H}_{ZFS} = \hat{\mathbf{S}} \cdot \overline{\overline{D}} \cdot \hat{\mathbf{S}} \tag{1.1.5}$$

where $\hat{\mathbf{S}}$ is the spin vector operator of the ground state and $\overline{\overline{D}}$ is a two rank symmetric tensor. In an arbitrary axis frame (x, y, z), it is composed of six different parameters.

$$\overline{\overline{D}} = \begin{pmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{pmatrix} \tag{1.1.6}$$

This tensor can be reduced to three parameters by diagonalization, *i.e.* expressing them in the tensor principal axes that define the magnetic anisotropy axes. By convention, the Z axis is chosen by the diagonal value D_Z that differs the most from the others and X and Y so that $D_{XX} - D_{YY} > 0$. Going further as to work with traceless tensors allows us to use only two parameters, by convention the z-axis is taken as the main magnetic axis with D the axial parameter :

$$D = D_{ZZ} - \frac{1}{2}(D_{XX} + D_{YY}) = \frac{3}{2}D_{ZZ} \tag{1.1.7}$$

and the rhombic term :

$$E = \frac{1}{2}(D_{XX} - D_{YY}) \tag{1.1.8}$$

With $|D| \geq 3E \geq 0$ set as a convention.[32] The trace of the tensor is isotropic and does not affect the energy splitting of the M_S components. The ZFS Hamiltonian can then be written :

$$\hat{H}_{ZFS} = D(\hat{S}_z^2 - \frac{1}{3}\hat{S}^2) + E(\hat{S}_x^2 - \hat{S}_y^2) \tag{1.1.9}$$

In the case of a triplet S=1 ground state such as for a Ni (II) complex, in an arbitrary axes frame (x, y, z), the matrix representation of hamiltonian (1.1.5) writes :

H_{ZFS}	$\|1,-1\rangle$	$\|1,0\rangle$	$\|1,1\rangle$
$\langle 1,-1\|$	$\frac{1}{2}(D_{xx} - D_{yy}) + D_{zz}$	$-\frac{1}{\sqrt{2}}(D_{xz} + iD_{yz})$	$\frac{1}{2}(D_{xx} - D_{yy} + 2iD_{xy})$
$\langle 1,0\|$	$-\frac{1}{\sqrt{2}}(D_{xz} - iD_{yz})$	$D_{xx} + D_{yy}$	$\frac{1}{\sqrt{2}}(D_{xz} + iD_{yz})$
$\langle 1,1\|$	$\frac{1}{2}(D_{xx} - D_{yy} - 2iD_{xy})$	$\frac{1}{\sqrt{2}}(D_{xz} - iD_{yz})$	$\frac{1}{2}(D_{xx} + D_{yy}) + D_{zz}$

In the magnetic axes frame (X, Y, Z), this matrix becomes :

H_{ZFS}	$\|1,-1\rangle$	$\|1,0\rangle$	$\|1,1\rangle$
$\langle 1,-1\|$	$\frac{1}{2}(D_{XX}+D_{YY})+D_{ZZ}$	0	$\frac{1}{2}(D_{XX}-D_{YY})$
$\langle 1,0\|$	0	$D_{XX}+D_{YY}$	0
$\langle 1,1\|$	$\frac{1}{2}(D_{XX}-D_{YY})$	0	$\frac{1}{2}(D_{XX}+D_{YY})+D_{ZZ}$

Removing the trace from the tensor and applying the convention from Eq. 1.1.7 and Eq. 1.1.8 we get :

H_{ZFS}	$\|1,-1\rangle$	$\|1,0\rangle$	$\|1,1\rangle$
$\langle 1,-1\|$	$\frac{1}{3}D$	0	E
$\langle 1,0\|$	0	$\frac{2}{3}D$	0
$\langle 1,1\|$	E	0	$\frac{1}{3}D$

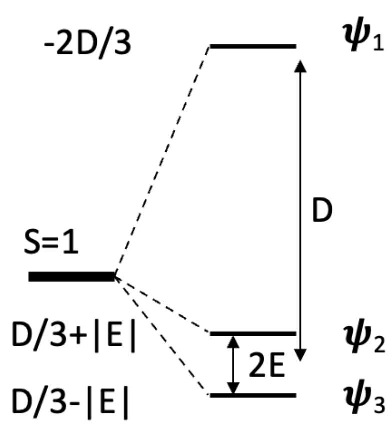

FIGURE 1.2 – Schematic representation of the three lowest SO-states undergoing ZFS in the case of triplet ground state.

Note that in this case (S=1) the D and E parameters can be estimated directly from the energy spectrum represented in Figure 1.2. Diagonalizing the model Hamiltonian matrix gives the three eigenvalues :

$$E_1 = -\frac{2}{3}D \tag{1.1.10}$$

$$E_2 = \frac{1}{3}D + E \tag{1.1.11}$$

$$E_3 = \frac{1}{3}D - E \tag{1.1.12}$$

with the eigenvectors :

$$\psi_1 = |1, 0\rangle \tag{1.1.13}$$
$$\psi_2 = \frac{1}{\sqrt{2}}(|1, -1\rangle + |1, 1\rangle) \tag{1.1.14}$$
$$\psi_3 = \frac{1}{\sqrt{2}}(|1, -1\rangle - |1, 1\rangle) \tag{1.1.15}$$

leading to the extraction of the D and E terms directly from the calculated or measured energy differences :

$$D = \frac{1}{2}(E_2 + E_3) - E_1 \tag{1.1.16}$$

and

$$E = \frac{1}{2}(E_2 - E_3) \tag{1.1.17}$$

In the case of an integer spin value S, a positive value of D indicates that the ground state is mainly composed of the M_S=0 component meaning that the projection of the spin moment along the z-axis is zero resulting in an easy-plane anisotropy. On the contrary if D<0, the ground state is formed from the M_S=$\pm M_{Smax}$ components with a maximum projection of spin moment along the z-axis resulting in an easy-axis anisotropy. For the study of system with a ground state of $S < 2$ these two parameters are enough to reproduce exactly the energy spectrum. In case of larger $S \geq 2$, one needs to introduce the equivalent Stevens operators[24, 33]. These operators were first derived to describe the lift of degeneracy induced by the crystal field potential studied in spectroscopy. The ZFS is now described by a more complete model Hamiltonian :

$$\hat{H}_{Stevens} = \sum_{k=0}^{2S} \sum_{n=0}^{k} B_k^n \hat{O}_k^n \tag{1.1.18}$$

where B_k^n are the Stevens parameter specific to each systems and $\hat{O}_k^n$ the Stevens Operators are spherical tensors of rank k. In case of mononuclear complexes, only the even rank k tensors contribute for symmetry reasons. The ZFS parameters are related to the second rank Stevens parameters :

$$D = 3B_2^0 \qquad\qquad E = B_2^2 \tag{1.1.19}$$

1.1.3 Magnetic anisotropy in polynuclear complexes

Opening the study to multiple magnetic centers introduces couplings that can be anisotropic between the magnetic moments of each centre. This coupling between two sites A and B with one unpaired electron each, is called anisotropic exchange interaction and is described by the following Giant Spin Hamiltonian[34].

$$\hat{H}_{AB} = \hat{\mathbf{S}}_A \cdot \overline{\overline{D}} \cdot \hat{\mathbf{S}}_B \tag{1.1.20}$$

The interaction tensor $\overline{\overline{D}}$ describes Zero-Field-Splitting the same way as before but now between two sites A and B, two parameters D and E can be defined similarly. This Hamiltonian is suitable for system with a ground state of spin larger or equal to one well separated in energy from other SO-states. This tensor can be split in two components, an isotropic term that relates to the HDvV Hamiltonian and an anisotropic interaction tensor $\overline{\overline{A}}$ that has no symmetry properties. This tensor can be further decomposed into a symmetric tensor $\overline{\overline{D}}_{ij}$ and an antisymmetric one $\overline{\overline{T}}_{ij}$. These tensors are defined as :

$$D_{ij} = \tfrac{1}{2}(A_{ij} + A_{ji}) \qquad\qquad T_{ij} = \tfrac{1}{2}(A_{ij} - A_{ji}) \tag{1.1.21}$$

The antisymmetric tensor $\overline{\overline{T}}$, is usually reported as a pseudovector known as the Dzyaloshinskii-Moriya interaction (DMI)[35], which only appears in the absence of an inversion centre between magnetic sites. It is then possible to study all the states of the configuration using the multi-spin Hamiltonian[36] :

$$\hat{H}_{AB} = J(\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B) + \hat{\mathbf{S}}_A \cdot \overline{\overline{D}}_{AB} \cdot \hat{\mathbf{S}}_B + \mathbf{d}_{AB} \cdot (\hat{\mathbf{S}}_A \times \hat{\mathbf{S}}_B) \tag{1.1.22}$$

where J is the isotropic coupling constant found in the HDvV Hamiltonian from Eq.1.1.1. The two following terms describe the anisotropic exchange with the symmetric tensor of exchange $\overline{\overline{D}}_{AB}$ and the DMI $\mathbf{d}_{AB}$ pseudo-vector.

The matrix representation for two interacting spin 1/2 systems is the following :

H_{MS}	$\lvert 1,-1\rangle$	$\lvert 1,0\rangle$	$\lvert 1,1\rangle$	$\lvert 0,0\rangle$
$\langle 1,-1\rvert$	$\frac{J}{4}+\frac{D_{zz}}{4}$	$\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$\frac{(D_{xx}-D_{zz}-2iD_{xy})}{4}$	$\frac{d_y+id_x}{2\sqrt{2}}$
$\langle 1,0\rvert$	$\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J}{4}-\frac{D_{zz}}{4}+\frac{(D_{xx}+D_{yy})}{4}$	$-\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$-\frac{id_z}{2}$
$\langle 1,1\rvert$	$\frac{(D_{xx}-D_{zz}+2iD_{xy})}{4}$	$-\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J}{4}+\frac{D_{zz}}{4}$	$\frac{d_y-id_x}{2\sqrt{2}}$
$\langle 0,0\rvert$	$\frac{d_y-id_x}{2\sqrt{2}}$	$\frac{id_z}{2}$	$\frac{d_y+id_x}{2\sqrt{2}}$	$-\frac{3J}{4}-\frac{D_{zz}}{4}-\frac{(D_{xx}+D_{yy})}{4}$

From this matrix, we notice that the Dzyaloshinskii-Moriya interaction creates a coupling between the singlet with the three M_S components of the triplet state, while the symmetric tensor of anisotropic exchange only couples the three components of the triplet. At the isotropic level, the difference between the triplet and singlet states is given by $\Delta E = J$, but with the inclusion of the anisotropic terms this becomes much more complex. It is impossible to obtain values for these interactions from the energy spectrum only, as such they will be extracted from the effective Hamiltonian theory. For couplings involving more than one unpaired electron per site, local tensors must be introduced :

$$\hat{H}_{AB} = J(\hat{\mathbf{S}}_A\cdot\hat{\mathbf{S}}_B)+\hat{\mathbf{S}}_A\cdot\overline{\overline{D}}_{AB}\cdot\hat{\mathbf{S}}_B+\mathbf{d}_{AB}\cdot(\hat{\mathbf{S}}_A\times\hat{\mathbf{S}}_B)+\hat{\mathbf{S}}_A\cdot\overline{\overline{D}}_A\cdot\hat{\mathbf{S}}_A+\hat{\mathbf{S}}_B\cdot\overline{\overline{D}}_B\cdot\hat{\mathbf{S}}_B \quad (1.1.23)$$

Note that this Hamiltonian 1.1.23 is not restricted to a ground state with $S \geq 1$.

1.2 Methodology

All types of calculations discussed here after have for purpose to solve, in some way, the time independant Schrödinger Equation :

$$\hat{H}\Psi_i = E_i\Psi_i \quad (1.2.1)$$

where $\hat{H}$ is a Hamiltonian used to describe the system, E_i is the energy associated to the wave-function Ψ_i. The solutions E_i of this problem are obtained pretty straightforwardly by diagonalizing the Hamiltonian matrix, except that in most cases, the Ψ_i vector are not known beforehand. The Hamiltonian treated during the calculations is called exact electronic Hamiltonian as it considers all electronic and nuclear interactions of the system, it is written as follow, in atomic units :

$$\hat{H} = -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 - \sum_{A=1}^{M}\frac{1}{2M_A}\nabla_A^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{|r_i - R_A|} + \sum_{i=1}^{N}\sum_{j<i}^{N}\frac{1}{r_{ij}} + \sum_{A=1}^{M}\sum_{B>A}^{M}\frac{Z_A Z_B}{|R_A - R_B|} \tag{1.2.2}$$

where r_i is the position vector of the i^{th} electron, R_A the position vector of the A^{th} nucleus with atomic number Z_A and mass M_A. This Hamiltonian can be simplified in the context of the Born-Oppenheimer approximation, in which the electrons are considered to adapt instantly to the movement of the nuclei. This allows to decouple the movement of the electrons from that of the nuclei, whose position can be fixed at the equilibrium (or any other geometry). Note that once the electronic energies and wave-function have been obtained for different geometries, it is also possibe to study the nuclei motion within this approximation. This approximation is justified as long as the electrons are much faster than the nuclei (since electrons are much lighter).

$$\hat{H}_{elec} = -\frac{1}{2}\sum_{i=1}^{N}\nabla_i^2 - \sum_{i=1}^{N}\sum_{A=1}^{M}\frac{Z_A}{|r_i - R_A|} + \sum_{i=1}^{N}\sum_{j<i}^{N}\frac{1}{|r_i - r_j|} \tag{1.2.3}$$

This Hamiltonian describes the electronic problem while the nuclei contribution, last term from eq 1.2.2 is a constant, having for only effect a shift in the overall energy spectrum. While this greatly simplifies the equations, it is still not solvable analytically and several approximations were developped to tackle this problem. All the methods used in this work rely on the construction of molecular orbitals (**MO**) as expansions of atomic orbitals (**AO**).

$$\psi_k = \sum_i c_{ik}\phi_i \tag{1.2.4}$$

where ψ_k is the k^{th} MO built from the AOs ϕ_i with coefficient c_{ik}. This expansion relies on an infinite number of AO but in real case application this is not achievable and will thus be restricted to a finite number of basis functions. As the electronic Hamiltonian does not include any information about spin, it will be included within a so called spin orbital with the introduction of two orthonormal functions $\alpha(\omega)$ and $\beta(\omega)$, *i.e.* spin up or down function, with ω a dummy variable. From each spatial molecular orbital, two spin orbitals can be created such that :

$$\chi_i = \begin{cases} \psi_i(r)\alpha(\omega) \\ \quad or \\ \psi_i(r)\beta(\omega) \end{cases} \tag{1.2.5}$$

This definition for the spin orbital is well adapted for close-shell systems where all molecular orbitals are doubly occupied, the spatial part for any spin orbital is the same for both spin function defining the restricted formalism. A commonly used notation in this formalism is to refer to a spin orbital by its spatial component and indicate the spin function with an overline, *i.e.* $\chi_i = \psi_i(r)\alpha(\omega) = \psi_i(r)$ or $\chi_i = \psi_i(r)\beta(\omega) = \overline{\psi}_i(r)$. Such definition changes when working with open shell system, *i.e.* some orbitals are singly occupied necessiting the application of unrestricted formalism.

1.2.1 Hartree-Fock Method

The cornerstone of *ab initio* calculation in quantum chemistry is the Hartree-Fock (HF) method which is usually the initial step for computing a first approximation of the wave-function in molecules. It is a variational approach that aims to treat the N-electron problems as a problem of N non interacting electrons in the presence of an average potential replicating interactions between them, it is in this sense a mean-field theory. This wave-function is constructed on the optimisation of a single Slater determinant Ψ :[37]

$$|\Psi(x_1, x_2, \ldots, x_N)| = \frac{1}{\sqrt{N!}} \begin{vmatrix} \chi_i(x_1) & \chi_j(x_1) & \cdots & \chi_k(x_1) \\ \chi_i(x_2) & \chi_j(x_2) & \cdots & \chi_k(x_2) \\ \vdots & \vdots & & \vdots \\ \chi_i(x_N) & \chi_j(x_N) & \cdots & \chi_k(x_N) \end{vmatrix} \tag{1.2.6}$$

where χ_i are spin orbitals and the variable $x_i = \{r_i, \omega_i\}$, it involves all combination of all N electrons in all k spin orbitals. We introduce the shorthand notation for such determinant $|\Psi(x_1, x_2, \ldots, x_N)| = |\chi_1(x_1)\chi_2(x_2)\ldots\chi_N(x_N)|$ showing only the diagonal elements of the determinant.

The way to obtain the best adapted Hartree-Fock wave-function, *i.e.* the molecular orbitals, comes through the resolution of the Roothan's equation :[38]

$$FC = SC\epsilon \tag{1.2.7}$$

S is the overlap matrix of the basis function, ϵ the matrix of orbital energies and C is the matrix of the trial vector. The Fock operator F is defined as :

$$\hat{F}(i) = \hat{h}(i) + \sum_{j}^{N/2}[2\hat{J}_j(i) - \hat{K}_j(i)] \tag{1.2.8}$$

where $\hat{h}(i)$ is a mono-electronic operator that contains the kinetic energy operator of the i^{th} and the coulomb attraction with the fixed nuclei. The two operators $\hat{J}_j(i)$ and $\hat{K}_j(i)$ describe the coulomb repulsion and exchange mechanism between the i^{th} electron and all the other electrons. These operators carry a direct dependance on the trial vectors from Eq. 1.2.7 as such, the Roothan's equation are non-linear and will be solved iteratively following a $Self\ Consistent\ Field$ (SCF) procedure. At convergence, one has optimised the determinant that minimizes the energy, *i.e.* the orbitals are optimised during the SCF procedure. One should note that as all bielectronic integrals are calculated exactly, the method allows one to calculate the expectation value of the exact Hamiltonian on this determinant.

1.2.2 Configuration Interaction

A single Hartree-Fock determinant usually provides a good starting point for closed-shell systems (since their wave function is usually dominated by this HF determinant) but this is not the case for magnetic systems (one or more unpaired electrons) that are open-shell systems and then intrinsically multi-determinantal. The most straight-

forward way to compute such a wave-function is to perform a full Configuration Interaction (full CI) calculation. This method relies on the diagonalisation of the N-electron Hamiltonian in a basis of N-electron functions (Slater Determinants). In other word, each configurations of N-electrons in the k-spin orbitals is considered and added to the space in which is spanned the Hamiltonian. With an infinite basis, the energy obtained would be exact in the Born-Oppenheimer approximation for a discrete set of basis function (a continuous basis would be necessary to solve the Schrödinger equation exactly). However, this is not applicable due to the finite nature of our calculation as we cannot handle infinite basis set. Still, full CI calculations are possible on small systems and provide a good lower bound to the exact energy.
The determinant $|\psi_0\rangle$ is a Slater determinant describing the case where electrons occupy the lowest orbitals obtained by a Hartree-Fock calculation. The determinant $|\psi_a^r\rangle$ describes the configuration where an electron from orbital a is excited to orbital r. In the same way, all the excitations are taken into account and the generated determinants are used as the basis for the many-electron wave-function $|\Psi_0\rangle$

$$|\Psi_0\rangle = c_0|\psi_0\rangle + \sum_{ar} c_a^r|\psi_a^r\rangle + \sum_{\substack{a<b \\ r<s}} c_{ab}^{rs}|\psi_{ab}^{rs}\rangle + \dots \tag{1.2.9}$$

As the number of determinants grows factorially with the number of ways of arranging the electrons amongst the given orbitals, this is very costly calculation wise. This method is thus only applicable on small systems where the number of electrons and the basis sets used are small. This is not the case for the systems studied in this work, so the need to decrease the computational time is of importance. One can expect from the wave-function computed, that a significant number of determinants do not play an important role in the physics of the wave-function. The ones with high excitations should appear with little weight unless high excited states are computed, therefore they can be left out of the wave-function without removing too much of physical relevance. The main focus of this is on the characterisation of the ground and lowest excited states in 3d-transition metals complexes, whose wave-functions are mainly expressed on determinants which differ by the occupation of the d-orbitals of the metal ion. It becomes

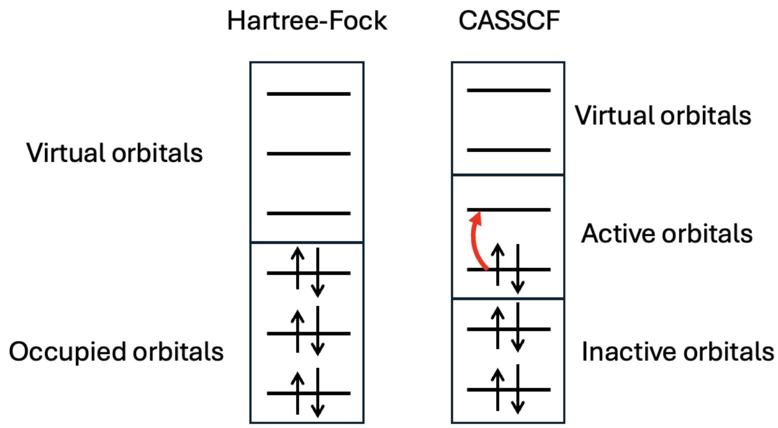

FIGURE 1.3 – Difference between the orbitals space in Hartree-Fock and CASSCF Theory

clear that in these systems, most of the determinants taken into account in a full CI calculation are not relevant to their physics.

1.2.3 Complete Active Space Self Consistent Field

One of the most used method to introduce multiple references in the wave-function is called *Complete Active Space Self Consistent Field* (*CASSCF*)[10]. Compared to the Hartree-Fock theory where two types of orbitals (occupied and unoccupied) are considered, in CASSCF theory orbitals are separated in three sub-spaces. The inactive orbitals are doubly occupied throughout the calculation, virtual orbitals stay empty while the active orbitals have a variable occupancy ranging from zero to two electrons. These active orbitals (and the active electrons they bear) define the active space into which a configuration interaction will be realised considering all the possible determinants within this sub-space. As the number of determinants grows significantly with the number of active electrons and orbitals, this method is still costly and its size becomes a limiting factor for the calculations. The new wave-function is now written as an expansion of Slater determinants which is obtained via a SCF procedure where both the expansion coefficients and the orbitals are optimised as to minimize the ground state energy or of several states in an "average" way. This dual optimisation scheme can render the convergence troublesome, as such the definition of the active space and the choice of the starting orbitals become crucial. One usually starts from a set of orbitals previously obtained via a Hartree-Fock calculation. For computation of magnetic properties in transition metal systems, the minimum active space should consist of the magnetic orbitals, *i.e.* the singly occupied d-orbitals of the metallic centers, this can later be extended to all of the 3-d orbitals of the metal as well as some orbitals of the ligands. As a result of the CASSCF method we obtain the energies of all states included in the calculation and a new set of orbitals into which they are expressed. Note that depending on the computational code used, the sets of orbitals may be *specific* to a single state or *averaged* to lower the average energy of all of the states computed. This calculation takes into account the correlation between the electrons inside the active space in the mean field created by the other electrons. While this method provides the non-dynamic correlation, it fails to capture the correlation with the inactive electron and their reponse to excitations, called dynamic correlation which plays a crucial work in the physics of magnetic systems.

1.2.4 Perturbation Theory

Perturbative treatment can be applied to extract dynamic correlation, mainly brought by mono and diexcitations on top of CASSCF configurations. For such calculation, one can use the CASPT (Complete Active Space Perturbation theory) or the NEVPT (N-electron valence state perturbation theory). Both of these methods rely on the assumption that the Hamiltonian can be partitioned into a zeroth-order term and a perturbation $\hat{V}$ with the parameter λ :

$$\hat{H} = \hat{H}^{(0)} + \lambda\hat{V} \tag{1.2.10}$$

The wave-function and energy are expanded in a similar way and the zeroth order term $\Psi^0_{(0)}$ is chosen to be the CASSCF wave-function. The effect of the configurations outside the active space on the energy and the wave-function is estimated though perturbation theory and the expansions of the equations allows one to obtain the corrected energies, usually at second-order of perturbation with CASPT2[11] or NEVPT2[12, 13]. These two methods mainly differ from the choice of the zeroth-order Hamiltonian, CASPT2 relies on a monoelectronic Hamiltonian built from a one electron Fock operator that falls back to the Moller-Plesset Hamiltonian in single reference case. This treatment may lead to what is called "intruder states" coming from divergence in the denominator of the corrections term where a difference of orbital energy is taken. Several approaches have been adapted such as the introduction of *level shifts*, real or imaginary, as to fix this divergence. In case of NEVPT2, the zeroth-order Hamiltonian is a bi-electronic (Dyall) Hamiltoninan which in itself includes a shift in energy between state from inside or outside the active space preventing the appearance of intruder states. These methods, in their simple scheme, are contracted, *i.e.* they do not change the ratio between CASSCF determinant coefficients. One should note that the ORCA package provides both partially and fully contracted scheme for NEVPT2.

1.2.5 Difference dedicated configurational interaction

Dynamic correlation brought by single and double excitations can also be retrieved through a truncated to single and double excitations Configuration Interaction calculation (SDCI). This way, wave-functions are fully decontracted, *i.e.* coefficients of the CASSCF configurations (as well as all other configurations) are optimized by the diagonalization of the MRCI matrix. While very promising, the size of the system remains a limiting factor since the MRCI matrix size increases rapidely with the number of electrons and orbitals (occupied, active and virtual). It is then compulsory to truncate the SDCI space as done in the DDCI method. Excitations can be classified into eight categories depending on the number of hole/particle created, as represented in Figure 1.4. A hole is an excitation from an inactive orbital to an active or virtual orbital, while a particle is an excitation from an active or inactive orbital to a virtual one.

The most numerous, and thus the most troublesome, are 2h-2p excitations. Nevertheless, it can be shown that at second-order of perturbation, as long as a common set of MO is used, they do not contribute to the energy difference between states but

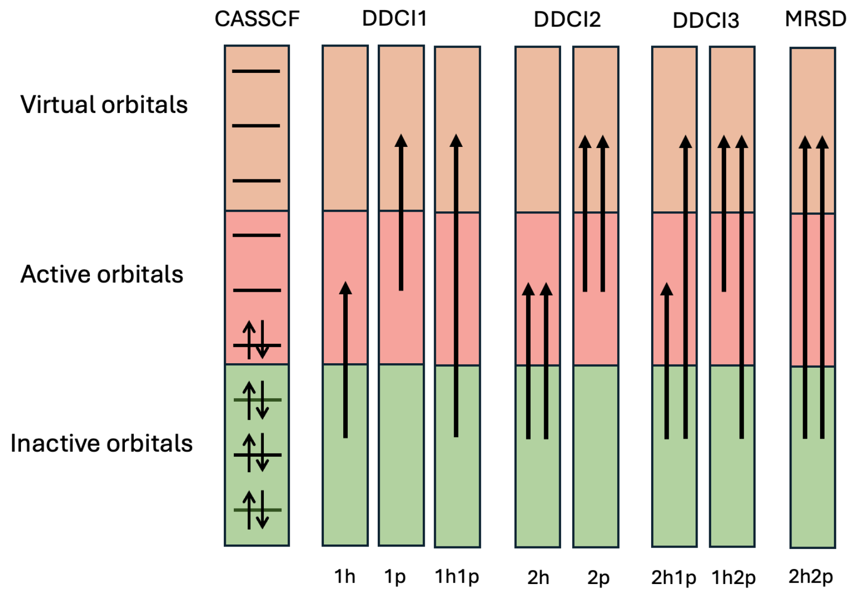

FIGURE 1.4 – Classes of excitations and associated DDCI method

only act as a shift of all energies. As such they can be neglected giving rise to a variant of MRCI called Difference-Dedicated-Configurational-Interaction (DDCI). Three sub-variant of DDCI exists taking into account different classes of excitations as pictured on Figure 1.4 It has been established that the extraction of magnetic coupling involving bridging ligand requires the use of the DDCI3 method to provide good estimates. One should note that any truncated CI suffers from size-consistency errors. While the Coupled Cluster method enables one to correct these erros in the case of a single reference, generalisations of this method to multireference approaches are quite complicated and lead to very costly calculations. They will not be used in this thesis.

1.2.6 Density functional theory

Another way to get a description of the electronic structure is through density functional theory (DFT). The main interest of such a method is the description of the ground state properties through the determination of the ground state energy E_0 following the variational theorem :

$$E_0 = \min \langle\Psi| \hat{H} |\Psi\rangle \tag{1.2.11}$$

Here, the electronic wave-function of the molecule Ψ is approximated to a single Slater determinant obtained from the determination of the electronic density $\rho(\mathbf{r})$:

$$\rho(\mathbf{r}) = N \int |\Psi(\mathbf{x}_1 \ldots \mathbf{x}_N)|^2 ds_1 dx_2 \ldots dx_N \tag{1.2.12}$$

The Hamiltonian1.2.2 can be written :

$$\hat{H} = \hat{T} + \hat{V}_{ee} + \hat{V}_{ne} \tag{1.2.13}$$

with $\hat{T}$ the kinetic energy operator, $\hat{V}_{ee}$ the electron-electron interaction operator and $\hat{V}_{ne}$ the nuclei-electron interaction operator. By replacing $v_{ne}(\mathbf{r})$ with a known external potential $v(\mathbf{r})$, one can obtain the ground state wave-function Ψ by solving the Schrödinger equation and the electron density follows. The first Hohenberg-Kohn theorem states that this external potential is a unique functional of the electron density.[39] As such, knowing the electron density allows to determine the properties of the ground states. The second Hohenberg-Kohn gives, in case of non degenerate ground state, that the wave-function Ψ is itself a functional of $\rho(\mathbf{r})$ which allows to define the total energy :

$$E[\rho] = F[\rho] + \int \rho(\mathbf{r}) v(\mathbf{r}) dr \tag{1.2.14}$$

with $F[\rho]$ a universal density functional which contains the kinetic and potential contributions. The ground state energy E_0 is the minimum of eq 1.2.14 which is reached when the electron density $\rho_(\mathbf{r})$ is that of the ground state $\rho_0(\mathbf{r})$. In theory the knowledge of the electron density allows to determine E_0, however the density dependance expression of $F[\rho]$ is not known.

$$F[\rho] = T[\rho] + V_{ee}[\rho] \tag{1.2.15}$$

Kohn and Sham introduced new definition of this functional[40] by replacing the interacting system with a fictious system of N non interacting electrons.The functional becomes :

$$F[\rho] = T_s[\rho] + J[\rho] + E_{xc}[\rho] \tag{1.2.16}$$

where $T_s[\rho]$ is the non-interacting kinetic energy functional of density ρ expressed in the basis of $\phi_i(\mathbf{r})$ that are built to reproduce the ground state density $\rho_0(\mathbf{r})$.

$$\rho_0(\mathbf{r}) = \sum_i^N |\phi_i(\mathbf{r})|^2 \tag{1.2.17}$$

$$T_s[\rho] = \sum_i^n \langle\phi_i| - \frac{1}{2}\nabla^2 |\phi_i\rangle \tag{1.2.18}$$

The Hartree-potential $J[\rho]$ gives the coulomb repulsion between electrons pair :

$$J[\rho] = \frac{1}{2}\int\int \frac{\rho(\mathbf{r}_1)\rho(\mathbf{r}_2)}{|\mathbf{r}_2 - \mathbf{r}_1|} d\mathbf{r}_1 d\mathbf{r}_2 \tag{1.2.19}$$

The energy expression Eq. 1.2.14 finally becomes :

$$E[\rho] = \int v(\mathbf{r})\rho(\mathbf{r})d\mathbf{r} + T_s[\rho] + J[\rho] + E_{xc}[\rho] \tag{1.2.20}$$

This definition gives the exact energy given that the expression of the Exchange-Correlation E_{xc} is known, unfortunately this is not the case. Over the years multiple attempts have been made to find a universal expression of E_{xc} in terms of ρ and its derivative. In the most basic approach, called Local Density Approximation (LDA), the system is taken to behave as a uniform electron gas where E_{xc} depends only in the density ρ. This model was later adapted to study different spin configuration with the Local Spin Density Approximation (LSDA). Other attemps tried to improve the functional by incorporating a dependance in the derivative of ρ with the Generalized Gradient Approximation (GGA) or a small portion of the Hartree-Fock exact exchange in what is called hybrid functionals.

1.2.7 Inclusion of Spin-Orbit effects

As the goal of this thesis is the characterisation of the magnetic anisotropy, which is a direct consequence of relativistic mechanisms, it is crucial to include the most important ones in our calculation, namely spin-orbit coupling (spin-spin coupling is less important but also included in all our calculations). Following the already established procedure,

a two-step approach is chosen for this work. It relies on the computation of the spin-orbit free states (*i.e.* independant from spin-orbit), using CASSCF or post-CASSCF method as the first step. The spin-orbit is then included *a posteriori* following the State-Interaction method (SO-SI). This method calculates the spin-orbit matrix, in the basis of the different M_S components of all the spin-orbit free states included in the calculation. This matrix is then diagonalised to obtain the "spin-orbit" states and their corresponding energies. In pratice, the Douglas-Kroll-Hess Hamiltonian is used, which allows the separation of the spin-independant part and spin-orbit dependant part. To greatly reduce computation time, the bi-eletronic terms are treated as a screening term within the mono-electronic terms. This allows to work with only mono-electronic terms in what is called Atomic Mean Field Integrals (AMFI).
In order to rationalise our results we will use the following operator :

$$\hat{H}^{SO} = \sum_i \xi_i(\mathbf{r}_i)\hat{l}_i \cdot \hat{s}_i \tag{1.2.21}$$

where ξ_i is the effective spin-orbit constant, $\hat{l}_i$ is the angular momentum operator and $\hat{s}_i$ the spin momentum operator. The effective spin-orbit constant ξ should yield different value depending on which electron is treated but for the rationalisation, a unique tabulated value of ξ will be used for all electrons. In practice, we use that of the free ion. One may note that it is usually lower in the molecules due to covalency between metal and ligands.

1.2.8 Embedded Cluster Method

In the case of molecular materials, the study of magnetic anisotropy is well described by single molecule calculations one does not usually need to describe its environment. The restricted size of the molecule generally allows for an explicit treatment of all the atoms involved. The same cannot be said about highly correlated materials which are ionic crystals of infinite size and for which the Madelung Field plays a crucial role. We have to work on smaller systems called "clusters" or "fragments". In theory one would want the fragment to be as large as possible to reduce the error created by cutting off the fragment from its environment, unfortunately to reduce computational cost we have to limit ourselves with small sized fragments. It then becomes crucial to choose correctly the fragment based on our knowledge of the physics we want to reproduce. The properties of the cluster alone differ from the one inside its natural environment. To reproduce it accurately it is immersed inside an embedding. This embedding is composed of point charges that aim to replicate the electrostatic field of the crystal, the Madelung Field, inside the cluster region. To keep the symmetries of the crystal, the point charges are positionned at the lattice sites in a sphere of radius R_c. Another

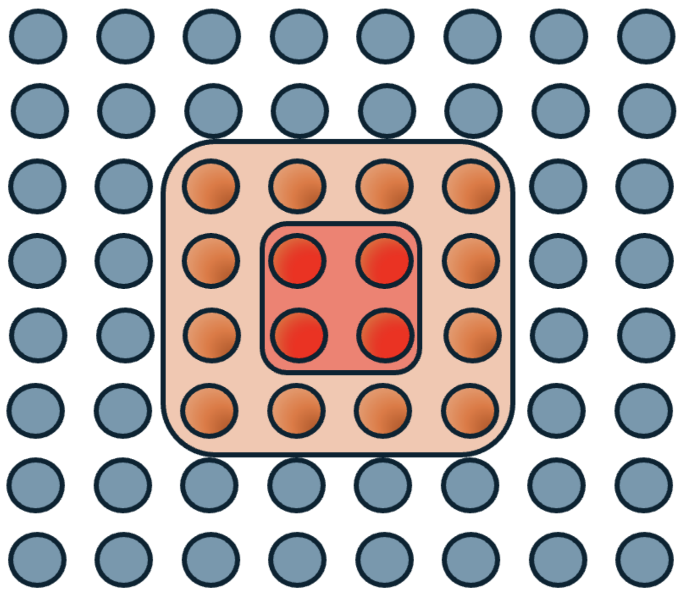

FIGURE 1.5 – Schematic representation of the three regions composing the embedding. red : cluster, orange : pseudo-potentials, blue : point charges

problem in selecting fragments in an ionic crystal is the overall charge of the fragment. The centre of the study is the metallic ions, charged positively, forming bonds with negatively charged ligands. A correct description of these magnetic centres requires to include all closest neighbors which often makes the overall fragment negatively charged. The outer ions closest to the edge of the fragment, replaced by point charges, are then positive. This induces an artificial displacement of the electron density of the outermost ions of the fragment. To avoid the electrons escaping the fragment, pseudo-potentials, placed around the fragment at the lattice sites, act as walls which prevent electrons from approaching these positive charges as if they were real atoms. These potentials are either *ab initio* model potential (AIMP) or Effective Core Potential (ECP) depending on which Quantum Chemistry software is used.

Two procedures were explored to create such embedding :

(1) Formal charges were used in a sphere with very large R_c (around 50 Å).

(2) Optimised point charges were obtained using Ewald Summation from formal charges, this allows to reduce the R_c to a few angstrom around the fragment.

All of this is done to ensure that the properties extracted from the fragment calculations, *i.e.* for a finite system, are a good representation of the crystal properties, *i.e.* for an infinite system.

1.2.9 Effective Hamiltonian Theory

Having ways to compute the *ab initio* multi-reference wave-functions usually leads to lenghty expressions where the wave-function is spanned along a large number of Slater determinants making its analysis complicated. On the other hand, the model Hamiltonians works in a much smaller space called the model space where a small number of electrons is considered to introduce effective interactions. A way to connect the two Hamiltonians is to apply a theory developped by Bloch called Effective Hamiltonian theory[15]. Its purpose is to build an effective Hamiltonian whose eigenvalues and eigenvectors reproduce the energy spectrum and the projected wave-functions onto the model space of the *ab initio* Hamiltonian but using a much smaller number of eigenfunctions.

Starting from the Schrödinger equation with the Born-Oppenheimer Hamiltonian $\hat{H}$:

$$\hat{H} |\Psi_m\rangle = E_m |\Psi_m\rangle \tag{1.2.22}$$

where Ψ_m are the eigenvectors with the associated eigenvalue E_m forming the space S of size N. We define a smaller space S_0 called model space with N_0 states $|\phi_i\rangle$, which is usually composed of the low lying states where the model Hamiltonian will be developped. An effective Hamiltonian H_{eff} is built to reproduce the energy spectrum of

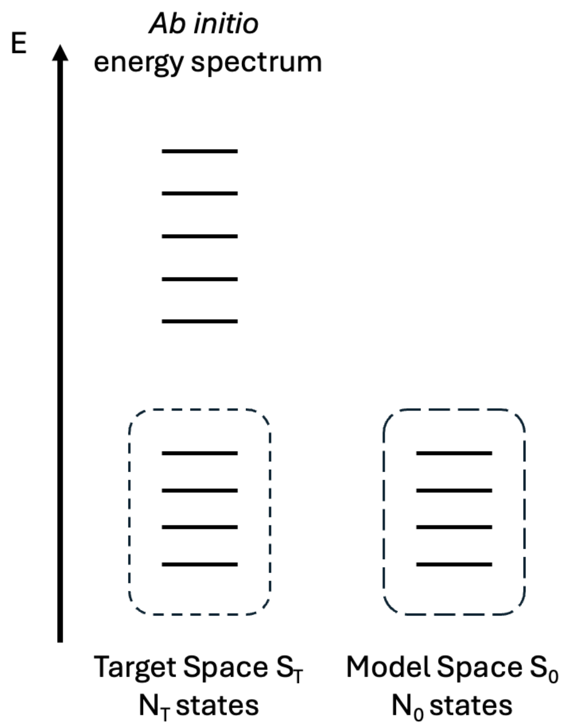

FIGURE 1.6 – Schematic representation of the two spaces involved in effective Hamiltonian theory

the exact Hamiltonian in the target space S_T using a small number of low-lying states $\tilde{\Psi}_m$ such that :

$$\hat{H}_{eff} |\tilde{\Psi}_m\rangle = E_m |\tilde{\Psi}_m\rangle \tag{1.2.23}$$

This target space S_T is chosen to be of the same size as the model space S_0 allowing for direct comparison.

The first step is to project the eigenfunctions Ψ_m onto the model space using the projector :

$$\hat{P}_0 = \sum_m^{N_0} |\tilde{\phi}_m\rangle \langle\tilde{\phi}_m| \tag{1.2.24}$$

$$|\tilde{\Psi}_m\rangle = \hat{P}_0 |\Psi_m\rangle \tag{1.2.25}$$

These eigenvectors are not necessarily orthogonal leading to a non-Hermitian Hamiltonian, this is problematic when comparing to the model Hamiltonians which are Hermitians. In this work, we apply the formalism proposed by Des Cloizeaux where the vectors are symmetrically orthogonalized[41] from the overlap matrix S :

$$|\tilde{\Psi}_m^{\perp}\rangle = S^{-1/2}\,|\tilde{\Psi}_m\rangle \tag{1.2.26}$$

At this point, it is possible to check the quality of the model space by calculating the norm of the projected vectors present in the S matrix. If too small, the interactions of the system are not captured correctly by the model space making it inadequate.

When these projections are close to one, we can consider the model space adequate, the effective Hamiltonian is then built :

$$H_{eff} = \sum_m^N |\tilde{\Psi}_m^{\perp}\rangle\, E_m\, \langle\tilde{\Psi}_m^{\perp}| \tag{1.2.27}$$

Once this is done, this numerical matrix H_{eff} constructed from the *ab initio* exact Hamiltonian are compared to the model Hamiltonian representation matrix. This allows for the extraction of values for the model Hamiltonian parameters as well as a test of validity. If there is too much deviation between the two matrices, it is possible that there is some missing interaction in the model. In such a case, analytical derivation allows one to derive a new model Hamiltonian from a more sophisticated Hamiltonian.

Chapitre 2

Study of the magnetic anisotropy in mononuclear complexes

The field of mononuclear complexes is a perfect area for the study of magnetic anisotropy. The finite size of such complexes as well as the localised character of the magnetic entities allow for the explicit treatment of all atoms at play with reasonnable computational cost. This combined with the large amount of experimental data available enable for the validation of models and computational methodology. Numerous studies have been dedicated to the ability of the ZFS Hamiltonian to model the behaviour of mononuclear complexes based on fourth period transition metals. It has established that the value of the ZFS parameters are highly dependent on the metallic ion nature and charge, its d-shell configuration, and its environment which impact the energy difference of the d-orbitals. The ligands around the metallic centre becomes a crucial choice that dictates some of the complex properties as we will see in this chapter.

State of the art methodology for such studies is to generate a wave-function starting from a CASSCF calculation. This should includes all electronic states generated by an active space composed of all the d-orbitals of the metallic ion and the electrons occupying them. The energies of these states are then computed using the NEVPT2 method to include the dynamical correlation. Spin orbit coupling between M_S components of all states are accounted for using Spin-Orbit-State-Interaction (SO-SI) method with NEVPT2 correlated energies as the diagonal elements of the SO matrix. Finally, the ZFS parameters are extracted using the Effective Hamiltonian Theory in a procedure implemented in the ORCA package. To go beyond the simple extraction of the parameters, a more thorough study requires the analysis of the wave-functions, energy differences and contributions to ZFS parameters.

2.1 First-Order Spin-Orbit Coupling

The search for large magnetic anisotropy in molecular systems remains a central objective in the design of next-generation single-molecule magnets (SMMs). In 3d transition-metal complexes, the challenge arises from the fact that usually SOC is weak compared to the ligand field, the orbital angular momentum is then almost quenched. SOC can be considered as a perturbation of spin-orbit free states (situation called second-order spin-orbit coupling) and magnetic anisotropy properties are correctly described by the zero-field splitting (ZFS) Hamiltonian, in which the axial D and rhombic E parameters quantify the splitting of the M_S components in the ground spin state. While this formalism has proved successful in identifying SMMs candidates, it fundamentally limits the achievable anisotropy, since second-order processes rely on large orbital energy gaps. One way to increase the anisotropy is to try and retrieve a contribution of first-order SOC, where near-degenerate orbitals allow the spin and orbital angular momenta to couple directly. This situation is well known for lanthanide ions, where unquenched orbital moments naturally give rise to extreme anisotropy, but it is rarely accessible for 3d ions, except in very low coordination numbers or highly symmetric geometries. The theoretical appeal of such systems is clear : when orbital quasi-degeneracy is preserved, the ZFS model breaks down, and anisotropy must instead be understood in terms of the mixing of spin–orbit states at the *ab initio* level. With this in mind, a series of five Iron(II) complexes were synthesized with the purpose of retrieving a first-order SOC. Through a joint study by experimentalists and theoricians, this work aims to determine the properties of the complexes and identify the contribution of first-order SOC. The use of a pentadentate ligand in the equatorial plane enforces a pentagonal bipyramidal geometry around the metallic center. Combined with halides or H_2O ligands in the apical positions, this allows the conditions of quasidegeneracy of the d_{xz} and d_{yz} orbitals to be met.

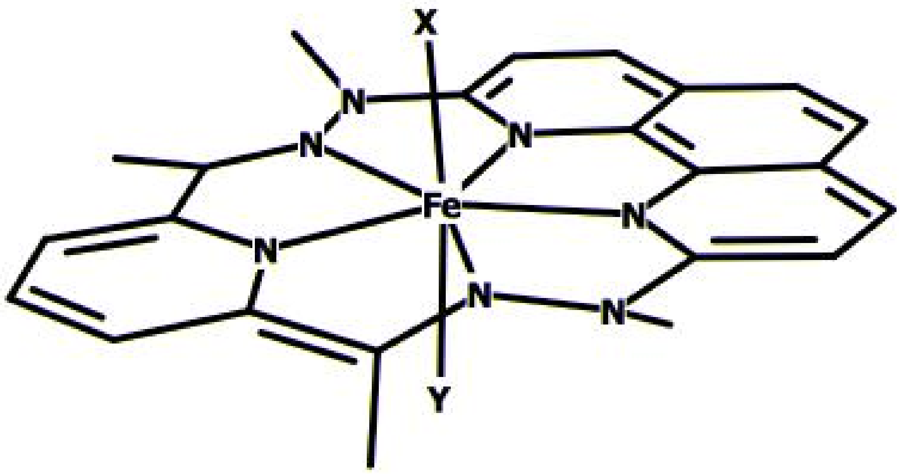

FIGURE 2.1 – Schematic representation of the Fe(II) complex with H atoms attached to carbon hidden

The electronic structure of the complexe were computed using CAS(6,5)SCF calculations, which allow for the evaluation of the energy splitting of the d-orbitals. First-order SOC involves the two lowest quintuplets Q_1 and Q_2, which are essentially monoconfigurational with one electron occupying either the d_{xz} or d_{yz} orbital. The incorporation of the SOC was performed using the SO-SI method generating the full manifold of SO-states for the extraction of anisotropic parameter of the ZFS Hamiltonian. In parallel, a first-order SOC model was adopted, previously developed for low-symmetry iron[16] and nickel complexes[17]. In this picture, the ligand field Hamiltonian is represented in a basis of the nearly degenerate orbitals d_{xz} and d_{yz}. This lift of degeneracy is induced by distortion of the fully symmetric geometry and is described by two parameters :

— δ_1 the splitting between the two orbitals, reflecting the extent to which degeneracy is broken.

— δ_2 the off-diagonal mixing between the orbitals, which couples them in the ligand field Hamiltonian.

The ligand field Hamiltonian is then written :

$$\hat{H}_{LF} = \begin{array}{c} \begin{array}{cc} d_{xz} & d_{yz} \end{array} \\ \begin{pmatrix} -\delta_1 & \delta_2 \\ \delta_2 & \delta_1 \end{pmatrix} \end{array}$$

An expression of the SO-states energy spectrum can be derived and directly compared to the *ab initio* one to obtain values for the δ_i terms. It also provides a way to estimate the D and E parameter from the ZFS formalism if applicable. The study indicates that the ZFS formalism is only applicable in the case of the H_2O ligand with a lift of orbital degeneracy of 412 cm^{-1} between the d_{xz} and d_{yz}. The results indicates a significant axial anisotropy and rhombicity that is slightly underestimated compared to experimental determination with a S=2 model. The two models, ZFS and first-order SOC, provide values that are in very good agreement. In the case of other complexes, results indicate that they behave like SMMs with large anisotropy. However, the energy splitting between the d_{xz} and d_{yz} orbitals is very small, which leads to a strong mixing of the two lowest quintuplet states in the SO-ground state. Only a few cm^{-1} separate the five lowest SO-states from the other, rendering the ZFS hamiltonian inappropriate. Overall, this study highlights that first-order SOC, traditionally associated with low-coordinate 3d complexes, can also be engineered in higher coordination environments by careful ligand design. This not only advances the understanding of magnetic anisotropy in transition metals, but also provides new opportunities for designing molecular materials with enhanced anisotropic interactions.

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Cite this: *Inorg. Chem. Front.*, 2025, **12**, 3456

Engineering first-order spin–orbit coupling in a pentagonal bipyramidal Fe(II) complex and subsequent SMM behavior†

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Received 18th December 2024,
Accepted 1st March 2025

DOI: 10.1039/d4qi03255a

rsc.li/frontiers-inorganic

Pentagonal bipyramidal (PBP) complexes with a first-order spin–orbit coupling contribution can be readily obtained, mainly through chemical design optimization ensuring minimum structural distortion and a more symmetrical ligand field. This conclusion follows from the investigation of a series of five Fe(II) complexes: $[FeL^{N5}(H_2O)Cl]Cl\cdot 4.5H_2O$, **1**; $[FeL^{N5}Cl_2]\cdot 3H_2O$, **2**; $[FeL^{N5}Br_2]$, **3**; $[FeL^{N5}I_2]$, **4**; and $[Fe_{0.12}Zn_{0.88}L^{N5}I_2]$, **5** ($L^{N5}$ stands for the pentadentate macrocyclic ligand formed by the condensation of 2,6-diacetylpyridine and 2,9-di(α-methylhydrazino)-1,10-phenanthroline). Theoretical calculations revealed quasi-degeneracy of the d_{xz} and d_{yz} orbitals for the complexes with halide ligands at the apical positions (ΔE = 91, 134, and 142 cm^{-1}, respectively, for **2–4**). This small energy gap leads to SO states with very strong mixing of the ground and first excited quintet states. Therefore, the ZFS Hamiltonian is not suitable for modelling the magnetic properties of complexes **2–5**. This does not apply for **1** with ΔE = 412 cm^{-1}. The recorded magnetic behaviors indicated strong magnetic anisotropy; for **1** D = −24 cm^{-1} was obtained. The Br and I derivatives were found to behave as SMMs (with a U/k_B of about 90 K), the latter even in the absence of a static field.

10th anniversary statement

Inorganic Chemistry Frontiers is undoubtedly the benchmark journal for inorganic chemistry. From the beginning, it has been the policy of the journal that quality should prevail over quantity, and this policy has enabled it to quickly make a difference. I know that I am going to find in the journal the latest remarkable developments in my field. This is important to me, not only as a reader, but also as an author.

Introduction

The paradigm of single-molecule and single-chain magnets (*i.e.* SMM and SCM)[1,2] has triggered efforts to achieve large magnetic anisotropy in molecular compounds.[3–8] One way of achieving this is to be close to the first-order spin–orbit regime. However, while the orbital angular momentum, L, plays a dominant role in lanthanide ions, it is generally quenched by the ligand field in transition metal ions.[9,10] In 3d ion complexes, the coordination sphere leads to an energy distribution of the d orbitals with differences of a few hundreds to a few thousands cm^{-1} between them. The magnetic anisotropy exhibited by the metal center stems from the interaction between the fundamental and excited levels resulting from the promotion of an electron to a higher-energy orbital, a process known as second-order spin–orbit coupling (SOC). This magnetic anisotropy is described by the zero-field splitting (ZFS) model,[11] and quantified by the axial parameter, D,

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† Electronic supplementary information (ESI) available: Additional experimental information; crystal structure solution for **2**, crystallographic information, related plots, and bond lengths and angles for all the compounds, PXRD, Mössbauer spectra, EDX analysis of **5**, additional magnetic behaviors, and data from theoretical calculations. CCDC 2352840–2352845. For ESI and crystallographic data in CIF or other electronic format see DOI: **https://doi.org/10.1039/d4qi03255a**

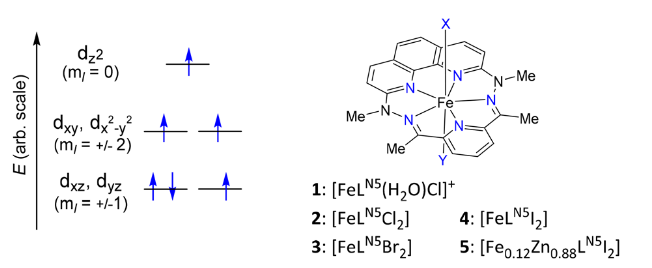
and the rhombic term, E. The sign of D, which is positive for in-plane anisotropy and negative for axial anisotropy, is regulated by the coordination polyhedron and the electronic configuration of the metal center,[12] and its strength is inversely proportional to the energy difference of the ground and excited states. However, under certain conditions, the splitting of the orbital levels can lead to quasi-degenerate orbitals giving rise to a certain amount of first-order spin–orbit contribution. This results in much larger anisotropy and, in the case of an axial-symmetry ligand field, axial (*i.e.* Ising-type) anisotropy is observed. The use of low coordination number complexes is an effective approach to generate first-order SOC. This has been illustrated for Fe derivatives with two-coordinate linear[13–16] or trigonal-pyramid[17,18] coordination spheres, most exhibiting slow relaxation of magnetization (*i.e.* SMM behavior). Similar results have been reported for Co(II)[19,20] and Ni(I/II)[21–23] derivatives. However, the slightest deformation of the coordination sphere or the symmetry of the ligand field leads to an increase in the energy difference between the d orbitals of the metal ion and lifts the quasi-degeneracy of the spin–orbit free states, resulting in the suppression of first-order SOC.[15,24–26]

While the contribution of the first-order SOC is clearly essential for enhancing the magnetic anisotropy in 3d ions, achieving this objective by chemical design remains highly challenging. And this becomes even more difficult if such complexes are to form the building blocks of polynuclear systems such as SCMs; a crucial requirement will be that the suitable geometry of the coordination polyhedron is structurally robust.

Energy diagrams with degenerate orbitals are also observed for geometries with higher coordination numbers.[12] This is the case for heptacoordinated species with pentagonal bipyramidal (PBP) arrangements for which two sets of two orbitals with the same energy levels are expected in the ideal geometry (Scheme 1). In real compounds, structural distortions and low symmetry of the ligand field lead to the lifting of degeneracies of the energy levels of these orbitals (from a few hundreds to thousands cm^{-1}). However, a significant ZFS effect has been demonstrated for such complexes, with D parameters of the order of 30 cm^{-1} for Co(II) to −30 cm^{-1} for Fe(II) and Ni(II).[27] For a given metal ion, the value of D depends on the actual energy difference between these orbitals.[28] For instance, in Fe(II) derivatives, the main contribution to the negative D value arises from the transfer of an electron between d_{xz} and d_{yz} orbitals; therefore their energy separation significantly affects the overall value of D. Theoretical calculations have revealed that D will not exceed −10 cm^{-1} for an energy difference above 400 cm^{-1}, but that D value may reach −20 to −30 cm^{-1} when the energy difference is reduced to about 200 cm^{-1}.[29–32] The same applies to Ni(II) in PBP coordination, but for the d^8 configuration it is the d_{xy} and $d_{x^2-y^2}$ orbitals that are relevant.[28] Thus, to achieve large magnetic anisotropy, it is essential to minimize the energy gap between d orbitals. The optimum situation would be to achieve quasi-degeneracy, which would *de facto* induce a substantial contribution from first-order SOC.

We show here that PBP complexes with a first-order SOC contribution can be readily obtained, mainly through chemical design optimization ensuring minimum structural distortion and a more symmetrical ligand field.

The PBP coordination polyhedron is typically induced by a pentadentate ligand, which occupies the equatorial positions of the complex and confers remarkable structural robustness. The phenanthroline-based macrocyclic ligand L^{N5} (Scheme 1) gained our preference because it is planar, obviously stiff, and symmetrical with five coordinating sp^2-N atoms. A suitable metal ion is Fe(II). Its high-spin d^6 electronic configuration in the PBP geometry leads to three electrons occupying the two d_{xz} and d_{yz} orbitals. As these orbitals are ideally degenerate and are linear combinations of $m_l = \pm 1$ spherical harmonics, 1st order SOC can take place. However, to maintain the (quasi-) degeneracy of these orbitals, the apical ligands of the complex must have the same interaction with each of these orbitals, and any dissymmetry would induce an energy difference between them.[29] Halides (Cl^-, Br^-, and I^-) seemed best suited for this purpose. The corresponding complexes are reported here. Theoretical calculations revealed quasi-degeneracy of the d_{xz} and d_{yz} orbitals (ΔE < 200 cm^{-1}) for all those with halide ligands in the apical positions. Their magnetic behavior indicated strong magnetic anisotropy, and the bromide and iodide derivatives were found to behave as SMMs, the latter even in the absence of a static field.



Scheme 1 Relative energy diagram of the d orbitals in an ideal pentagonal bipyramidal geometry (D_{5h}), the electron filling is for high-spin Fe(II), and sketch of the molecular complexes **1–5**.

Results and discussion

Syntheses

The PBP complexes $[FeL^{N5}(H_2O)Cl]Cl\cdot 4.5H_2O$, **1**; $[FeL^{N5}Cl_2]\cdot 3H_2O$, **2**; $[FeL^{N5}Br_2]$, **3**; $[FeL^{N5}I_2]$, **4**; and $[Fe_{0.12}Zn_{0.88}L^{N5}I_2]$, **5**, were obtained by adapting a reported procedure.[33] The Fe(II) complex $[FeL^{N5}(H_2O)Cl]Cl\cdot 4.5H_2O$, **1**, was prepared by reacting $FeCl_2\cdot 4H_2O$, 2,6-diacetylpyridine, and 2,9-di(α-methylhydrazino)-1,10-phenanthroline in H_2O in the presence of HCl. When excess NaCl was added to the reaction mixture, co-crystallization of **1** and $[FeL^{N5}Cl_2]\cdot 5H_2O$, **2** occurred. However, the complexes $[FeL^{N5}Br_2]$, **3** and $[FeL^{N5}I_2]$, **4** were isolated as single products by the same procedure (*i.e.* in the presence of an excess amount of either NaBr or NaI, and the corresponding HX acid). The reactions were performed under strictly oxygen-free conditions, although the addition of ascorbic acid was required to avoid traces of oxidation. This was systematically confirmed by Mössbauer spectroscopy (see below). All the complexes were isolated as crystallized solids and the same batch was used for the characterization studies.

In order to investigate these Fe complexes diluted in a diamagnetic matrix, the corresponding Zn(II) complexes were synthesized to verify structural concordance. Only the iodide derivative [$ZnL^{N5}I_2$] proved isostructural to the Fe homologue. With Br^-, a pentagonal pyramidal complex, [$ZnL^{N5}Br$]$Br\cdot0.5H_2O$, with a single axial ligand was obtained (Fig. S6†), and the Cl derivative could not be crystallized. Thus, only mixed-metal derivative [$Fe_{0.12}Zn_{0.88}L^{N5}I_2$], **5**, was prepared. Information on these Zn complexes are provided in the ESI.†

Crystal structures

The crystal structures for all the complexes have been determined by single-crystal X-ray diffraction analyses; the crystallographic data are given in Table S1 (ESI).† The metal–ligand bond lengths and axial bond angles for the first coordination sphere of **1–4** are gathered in Table 1, and additional information can be found in the ESI (Fig. S1–S6†).

Complexes **1–4** and [$ZnL^{N5}I_2$] are mononuclear and have a chemical organization similar to known PBP complexes of 3d metal ions with ligand L^{N5}.[33–38] The crystal arrangement of **2** was found to be incommensurable and composed of two superimposed inversed configurations, more information can be found in the ESI.† For each molecular complex, the M(II) center is heptacoordinated and sits in a slightly distorted pentagonal bipyramid environment (Fig. 1 and ESI†). The equatorial plane is formed by five N atoms from pentadentate ligand L^{N5}, while axial positions are occupied by different donor groups: H_2O and Cl^- for **1**, Cl^- for **2**, Br^- for **3**, and I^- for **4** and [$ZnL^{N5}I_2$]. The Fe–N bond distances in the equatorial sites are found between 2.07 and 2.28 Å, the shortest is with the N of the pyridyl moiety and the longer with the imine groups (Table 1). These bond distances are very similar to those reported for [$FeL^{N5}(H_2O)_2$]$^{2+}$.[33] The bonds between the metal center and the halogen atoms in the apical sites are longer, extending from 2.25 to 2.92 Å for Cl^- to I^-, in line with the increasing van der Waals radii of these atoms. The equatorial coordination arrangement is perfectly planar, the five nitrogen atoms and the iron atom (as well as Zn in [$ZnL^{N5}I_2$]) lie in the same plane (defined by the 5 N, see Fig. S1–S5†), and only for **1** the Fe atom is very marginally outside the plane by 0.022 Å. The apical arrangement very slightly deviates from normal to the equatorial plane, and the X–Fe–X links form an angle of around 170° (Table 1) in all the complexes. Thus, the coordination polyhedron of all these complexes is best described by a pentagonal bipyramidal geometry.

To evaluate the degree of deviation from the ideal PBP geometry, continuous shape measures analysis of the coordination polyhedron was performed with SHAPE software.[39,40] The CShM parameter values related to the deviation of the actual shape from ideal PBP for **1–4** are 0.499, 1.060, 1.326, and 2.214, respectively. The divergence, which increases from **1** to **4**, can be ascribed to the increasing size of the halogen atoms. Indeed, the distortion of the equatorial plane from the pentagonal geometry is quite small and similar for all complexes (Table S2†). The increasing deviation from the ideal PBP geometry is therefore mainly due to the lengthening of the $Fe–X_{axial}$ bonds from **1** to **4** (Table 1) due to the larger size of the halogen atoms which results in an elongated pentagonal bipyramidal environment.

For [$ZnL^{N5}Br$]Br, the molecular complex exhibits a pentagonal pyramidal coordination sphere with the five N atoms of L^{N5} bonded to Zn forming the pentagonal base and a coordinated Br located on the apex of the pyramid (Fig. S6†). Zn(II) is located approximately 0.441 Å above the basal plane with N–Zn bond distances ranging from 2.134 Å to 2.350 Å (Fig. S6†), which are slightly longer than those in [$ZnL^{N5}I_2$]. The charge of this cationic complex is compensated by a bromide anion. Although unusual, the formation of a pentagonal pyramidal complex between L^{N5} and a 3d metal ion is not unprecedented, and it was reported for an Mn(II) derivative with Cl as the apical ligand.[41]

For all the compounds the crystal packing shows that the complexes are stacked in parallel (Fig. S1–S4†). For **1** and **2** the complexes are organized in layers, with uncoordinated Cl and/or H_2O solvates in between (Fig. S1 and S2†), whereas the packing is more compact for the solvent-free **3** and **4**. The closest distances between the Fe centers are similar, with values of 8.352(1) to 8.7229(3) Å from **1** to **4**. For the Br- and I-derivatives, some rather close separations between the halogen atoms and the hydrogen of adjacent complexes, especially with phenanthroline and pyridyl moieties, become evident when the van der Waals radii of the halogens are considered (about 1.85 and 2.0 Å, respectively). These weak hydrogen bonds (H⋯X: 2.8062(4)–2.9585(4) Å in **3** and 3.0592(2)–

Table 1 Metal–ligand bond lengths (Å) and axial bond angles (°) for the first coordination sphere for **1–4**

1		**2**				**3**		**4**	
Fe1–N1	2.1112(2)	Fe1–N3	2.13(1)	Fe2–N6	2.07(1)	Fe1–N1	2.128(5)	Fe1–N1	(2.1053)
Fe1–N2	2.273(2)	Fe1–N4	2.248(7)	Fe2–N5	2.252(9)	Fe1–N2	2.304(3)	Fe1–N2	2.280(2)
Fe1–N7	2.270(2)	Fe1–N7	2.117(9)	Fe2–N2	2.124(9)	Fe1–N4	2.114(3)	Fe1–N4	2.123(2)
Fe1–N4	2.137(2)	Fe1–Cl1	2.575(6)	Fe2–Cl1	2.544(2)	Fe1–Br1	2.7106(5)	Fe1–I1	2.9231(2)
Fe1–N5	2.134(2)	Fe1–Cl2	2.555(6)	Fe2–Cl2	2.591(6)	Br–Fe–Br	171.08(4)	I–Fe–I	170.556(4)
Fe1–Cl	2.2549(7)	Cl–Fe1–Cl	171.5(1)	Cl–Fe2–Cl	170.0(1)				
Fe1–O1	2.193(2)								
Cl–Fe–O	174.26(5)								

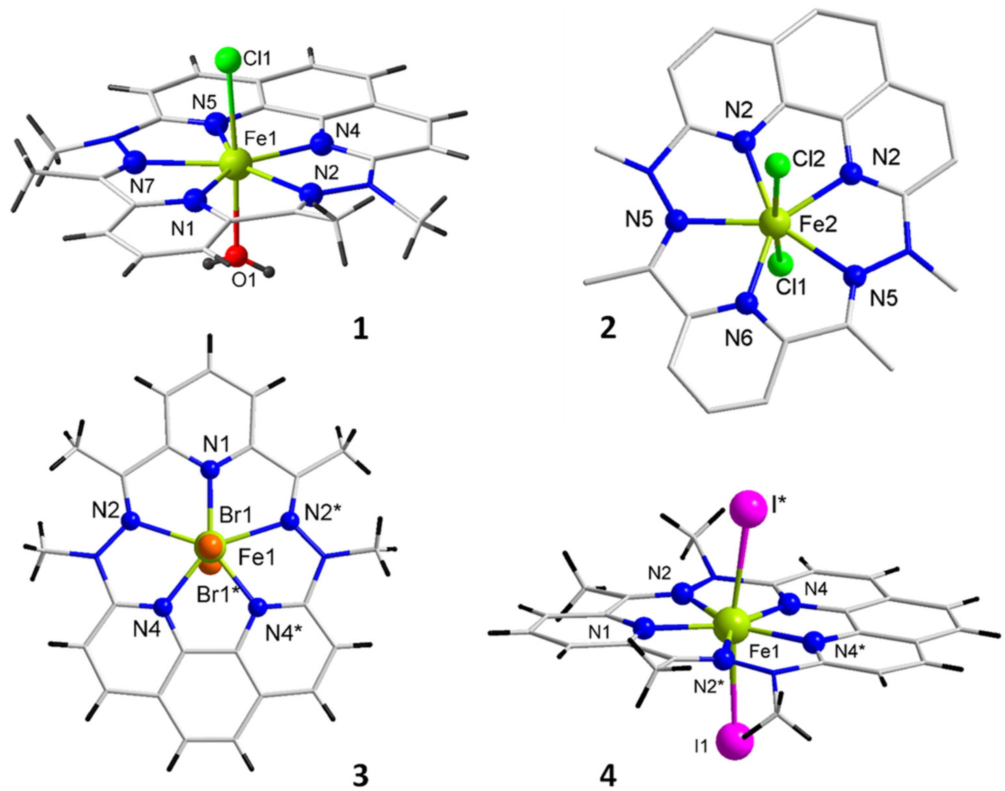


Fig. 1 Molecular structures of the Fe complexes **1–4**. One of the two independent complexes of **2** is shown; counter ion and lattice solvent molecules are not depicted; * refers to the 1 − *x*, *y*, 1,5 − *z* symmetry operation.

3.1730(2) in **4**) are likely to inter-connect all the molecules in the solid (Fig. S4†) and propagate magnetic interactions (*vide infra*).

With the exception of compound **2**, the phase concordance of the bulk samples with the crystal structures was confirmed by powder X-ray diffraction (Fig. S7†).

Mössbauer spectroscopy

^{57}Fe Mössbauer spectroscopy was carried out to confirm the oxidation state of the Fe centers and the purity of samples **1**, **3**, and **4** for magnetic studies. The spectra recorded at 80 K show a unique doublet (Fig. S8†) with a chemical shift near δ = 1.1 mm s^{-1} and a quadrupole splitting of about Δ = 2.1 mm s^{-1} (Table 2). These characteristics are comparable to those reported (*i.e.* 1.1 and 2.74 mm s^{-1}, respectively) for a homologous high-spin Fe(II) complex with H_2O as ligands in the apical positions.[33] Moreover, Mössbauer data have been reported for several other Fe(II) complexes in the PPB surrounding featuring either a macrocyclic ligand[42–44] or an open pentadentate ligand[29,45–49] in their equatorial coordination sphere. In all cases the chemical shift is found between 1.0 and 1.2 mm s^{-1} while their quadrupole splitting is more versatile as it varies with the ligands located in the apical positions.[29] Interestingly, Fe(III) in the same PBP environment exhibits a decrease in the chemical shift to about δ = 0.5 mm s^{-1} with smaller splitting,[50,51] which makes it very distinguishable and easy to identify in the Mössbauer spectrum of a Fe(II) derivative. The spectra obtained for **1**, **3** and **4** are typical of high spin PBP Fe(II) complexes with no detectable traces of oxidation products.

Table 2 Experimental Mössbauer parameters and magnetic characteristics for **1**, **3–5**

	1	**3**	**4**	**5**
Mössbauer doublet (80 K)				
δ (mm s^{-1})	1.10	1.09	1.05	—
Δ (mm s^{-1})	2.12	2.14	2.03	—
Magnetic characteristics				
$\chi_M T$ at 300/2 K (cm^3 mol^{-1} K)	3.16/2.21	3.90/2.11	3.27/0.58	—
M (μ_B) for 50 kOe at 2 K	2.52	2.73	2.14	2.28
U/k_B (K)	—	89 ± 5	99 ± 8	88 ± 9
τ_0 (s)	—	7×10^{-12}	3×10^{-11}	2×10^{-10}

Theoretical calculations

Wavefunction based calculations (SO-CAS(6/5)SCF + NEVPT2) (see the Experimental section for computational information) have been performed on complexes **1** to **4** followed by the procedure of anisotropic parameter extraction proposed by Maurice *et al.*[52] The atom coordinates from the X-ray structures were considered except for the positions of the H atoms that were optimized using DFT calculations.

Table 3 (left part) Energy differences (cm^{-1}) between the ground state Q_1 spin–orbit free and the 4 excited quintuplet states (Q_2 is the 1st excited one, …) in complexes **1–4** (the apical ligands is indicated in brackets), obtained at the NEVPT2 level of calculations, and (right part) AILFT 3d MO energies obtained at the NEVPT2 level of calculations

Complex	Q_2	Q_3	Q_4	Q_5	d_{yz}	d_{xz}	$d_{x^2-y^2}$	d_{xy}	d_{z2}
1 (Cl H_2O)	450	6661	7951	9228	0	412	5275	6278	7197
2 (Cl)	92	6804	7756	9060	0	91	5323	6122	7048
3 (Br)	42	5734	7999	8070	134	0	4597	6415	6470
4 (I)	37	6314	7962	8739	142	0	5020	6570	7035

The x, y, and z axes are those presented in Fig. S15.†

The NEVPT2 energy differences between the spin–orbit free quintuplet ground state and the first excited quintuplet state resulting from d–d transitions are given in Table 3, together with the energies of the AILFT 3d MOs. It should be specified that the lowest Q_1 and Q_2 states essentially have a mono-configurational wavefunction with a double occupancy either in the d_{xz} or the d_{yz} orbital (depending on the complex).

Atanasov *et al.*[26] have studied a family of trigonal pyramidal iron(II) complexes that exhibit very similar spectra to those calculated here with almost degenerate d_{xz} and d_{yz} orbitals. They proposed a model that accounts for the first-order SOC and the lift of degeneracy arising from geometrical distortions from the ideal C_3 symmetry. A related approach has also been developed to characterize a Ni(II) complex.[21] This model is applicable to the complexes studied here, for which the ideal symmetry would contain a C_5 symmetry axis. In the case of strictly degenerate d_{xz} and d_{yz} orbitals, the ground state is degenerate and consists of a 1 : 1 mixture of two quintuplet states (Q_1 and Q_2) in which either the d_{xz} or the d_{yz} orbital contains a single electron. Distortions of the ideal C_5 symmetry will lead to a lift of the degeneracy ($2\delta_1$) of these orbitals and to their mixing (δ_2) so that the ligand field Hamiltonian matrix can be written as:

$$\begin{array}{cc} d_{xz} & d_{yz} \end{array}$$
$$\begin{pmatrix} -\delta_1 & \delta_2 \\ \delta_2 & \delta_1 \end{pmatrix}. \tag{1}$$

In the complexes studied here, the states Q_1 and Q_2 are almost mono-configurational, hence the δ_2 parameter is close to zero. As the d_{xz} and d_{yz} orbitals are linear combinations of the $M_l = \pm 1$ spherical harmonics, the two spin–orbit free quintuplet states Q_1 and Q_2 can only be coupled through the $\zeta l_z s_z$ part of the spin–orbit Hamiltonian:

$$\hat{H}^{\mathrm{SOC}} = \zeta \sum_i \left(l_i^Z s_i^Z + (l_i^+ s_i^- + l_i^- s_i^+)/2\right) \tag{2}$$

where ζ is the covalently reduced effective spin–orbit constant.[53]

SOCs will generate 10 spin–orbit states, which are the eigenfunctions of the spin–orbit and ligand field Hamiltonian, the representative matrix of which is provided in the ESI (Table S6†). In the absence of SOC with the other excited states, the analytical expressions of the energy of the SO states resulting from this first-order SOC would be:

$$\begin{aligned} E(\mathrm{SO_1, SO_2/SO_9, SO_{10}}) &= \pm \frac{1}{2}\sqrt{4\delta^2 + \zeta^2} \\ E(\mathrm{SO_3, SO_4/SO_7, SO_8}) &= \pm \frac{1}{4}\sqrt{16\delta^2 + \zeta^2} \\ E(\mathrm{SO_5/SO_6}) &= \pm \delta \end{aligned} \tag{3}$$

where $\delta = \sqrt{\delta_1{}^2 + \delta_2{}^2}$.

The *ab initio* energy of these SO states is given in Table 4 as well as their wavefunction decomposition on the basis of the two spin–orbit free quintuplet states Q_1 and Q_2. Of course, the SOC with other excited states is non-zero and is treated *ab initio*, indicating that the reported energies do not strictly follow these spacings.

In complex **1**, the ground state is mainly based on a configuration with a doubly occupied d_{yz} MO (and a single occupation of all other MOs), while Q_2, Q_3, Q_4, and Q_5 exhibit double occupancy of the d_{xz}, $d_{x^2-y^2}$, d_{xy} and d_{z2} MOs, respectively. The spectrum presents an energy gap of 450 cm^{-1} between the ground and first excited states originating from the energy gap between the d_{yz} and d_{xz} MOs. Therefore, the 5 lowest SO states SO_1–SO_5 are mainly based on the M_s components of Q_1 (more than 86%) and are low in energy compared to SO_6–SO_{10} that are mainly based on Q_2 components. In this case, the anisotropic ZFS spin Hamiltonian applies (eqn (4)); it is extended with a fourth-order tensor operator represented by B_{40}, as recommended.[26]

$$\begin{aligned} \hat{H}_{\mathrm{ZFS}} = D[\hat{S}_Z{}^2 - S(S+1)/3] + \frac{E[\hat{S}_+{}^2 + \hat{S}_-{}^2]}{2} \\ + B_{40}[35\hat{S}_Z{}^4 - (30S(S+1) - 25)\hat{S}_Z{}^2 + 3S(S+1)(S(S+1)-2)] \end{aligned} \tag{4}$$

Parameters B_{40}, D and E, reported in Table 5, have been extracted for complex **1** using the following formulas which result from the diagonalization of the Hamiltonian $\hat{H}_{\mathrm{ZFS}}$ matrix (given in Table S7†):

$$\begin{aligned} D &= -\frac{1}{7}[(\mathrm{En}_{\pm 2} - \mathrm{En}_{\pm 1}) + (\mathrm{En}_{\pm 2} - \mathrm{En}_0)] \\ B_{40} &= \frac{1}{140}\left[(\mathrm{En}_{\pm 2} - \mathrm{En}_0) - \frac{4}{3}(\mathrm{En}_{\pm 2} - \mathrm{En}_{\pm 1})\right]. \end{aligned} \tag{5}$$

In eqn (5), $\mathrm{En}_{\pm 2}$, $\mathrm{En}_{\pm 1}$ and En_0 are the mean energy values of the SO states essentially carried by the $M_s = \pm 2$, $M_s = \pm 1$, and

Table 4 Energy (cm^{-1}) and decomposition of the ground and first excited spin–orbit states on the basis of Q_1 and Q_2

Complex		SO_1	SO_2	SO_3	SO_4	SO_5	SO_6	SO_7	SO_8	SO_9	SO_{10}
1	Energies	0.0	1.0	34	48	58	507	521	548	601	603
	Weights	86/12	87/12	96/3	95/5	98/0	0/96	3/95	5/94	12/87	12/85
2	Energies	0.0	0.0	81	82	141	228	280	320	408	410
	Weights	61/39	61/39	71/28	69/31	99/0	0/99	28/71	31/69	39/60	39/60
3	Energies	0.0	0.0	84	86	161	207	275	315	411	411
	Weights	55/44	55/44	62/38	59/40	99/0	0/99	37/62	41/59	44/55	44/55
4	Energies	0.0	0.0	87	88	169	208	284	320	421	421
	Weights	54/45	54/45	60/40	58/42	99/0	0/99	40/60	42/58	45/54	45/54

Weights (in %) are given for Q_1/Q_2. SO states SO_1, SO_2, SO_9 and SO_{10} are based on the $M_s = \pm 2$ of Q_1 and Q_2 respectively, SO_3, SO_4, SO_7 and SO_8 on the $M_s = \pm 1$ components, and SO_5 and SO_6 on the $M_s = 0$ components.

Table 5 Parameters extracted from the formulas (see text) and provided by the ORCA code[59]

Complex	D (eqn (5))	D (ORCA)	D (exp)	E (eqn (5))	E (ORCA)	E (Exp)	B_{40}	δ	ζ (ORCA)
1	−14.00	−13.64	−24.9 ± 0.4	2.33	2.24	4.06 ± 0.04	0.09	225	411
2	—	—	—	—	—	—	—	43	409
3	—	—	—	—	—	—	—	23	404
4	—	—	—	—	—	—	—	20	396

Axes are those of the **D** tensor for complex **1** (Fig. S15†). z is almost aligned with the bond between the metal and the apical ligands and x and y are in the plane of the pentadentate ligand. x is almost along the Fe–N(pyridine) direction in **1** (N1).

$M_s = 0$ spin components, respectively. E is 1/6 of the splitting between the two SO states essentially carried by the $M_s = \pm 1$ spin components. The values of the ZFS parameters D and E, and of the spin–orbit constant calculated in ORCA are also reported (Table 5). The decrease of the spin–orbit constant in complexes (in comparison with the free metal ion) is due to the covalent interaction between the metal ion and the ligand and is called the relativistic nephelauxetic effect. As expected, the obtained values follow the spectrochemical series, and therefore, they decrease as the nephelauxetic effect increases, *i.e.* $H_2O \ll Cl^- < Br^- < I^-$. This effect is well documented in the literature.[54–58] As expected from the nature of the excitation from Q_1 to Q_2–Q_5, the contribution of Q_2 to D is negative (−18.9 cm^{-1}) while those of Q_3 and Q_4 are positive and of much smaller amplitude (+1.97 and 0.72 cm^{-1}, respectively) since these states are much higher in energy, and the contribution of Q_5 is almost zero.

The case of complexes **2**, **3** and **4** is more complex since the energy gap between Q_1 and Q_2 is much smaller than in **1**, as d_{yz} and d_{xz} are almost degenerate. This small energy gap leads to SO states with a very strong mixing of Q_1 and Q_2 spin–orbit free states (except for $M_s = 0$ components that do not interact through SOC) and no clear gap is observed between the 5 lowest and 5 highest energy states. Therefore, the ZFS Hamiltonian is not suited to model magnetic properties of complexes **2–4**,[60] and only their δ and ζ parameters have been reported in Table 5. Note that parameters D and E are provided by the ORCA code and can be calculated using the formulas given above, but the magnetic moment, which now contains a non-negligible contribution from the angular momentum, can no longer be equated with spin momentum alone. For this reason, an evaluation of the D and E parameters from the magnetic data using spin S = 2 is irrelevant for these complexes.

The calculated g-factors show the anticipated trend with $g_x \approx g_y \ll g_z$, values are tabulated in Table S8.† In addition, the magnetization *versus* field has been calculated at various temperatures and will be compared below to the measured magnetizations.

Compared with previous works on PBP Fe(II) complexes,[29,48] the first order SOC contribution in complexes **2** to **4** is much more important. Indeed, it should be noted that the energy differences between the ground state Q_1 spin–orbit free quintuplet and the first excited state (Q_2) are particularly small here. For instance, in the earlier reports these values vary between 170 and 410 cm^{-1}, compared to the 37–92 cm^{-1} found for **2–4** (Table 3). Complex **1**, with H_2O in the apical position, has an energy difference of 450 cm^{-1} between Q_1 and Q_2, and can therefore be described using a ZFS model. In **1**, this energy difference is governed by the energy difference between the d_{yz} and d_{xz} orbitals, which in turn is governed by the difference of interactions in the x and y axes. In this respect, the orientation of the π-doublets in the apical positions plays a crucial role in these complexes.[48] In **1**, the π doublet of H_2O is oriented along the x axis (Fig. S15†) and therefore introduces a dissymmetry between x and y axes, which is responsible for the lift of degeneracy between the two orbitals. On the other hand, for halogen atoms as in **2–4**, 2 π doublets are involved with identical interactions with d_{yz} and d_{xz}, leading to quasi-degeneracy of the orbitals.

Magnetic properties

The temperature dependence of the molar magnetic susceptibility, χ_M, in the 2 to 300 K range, and the field dependence of the magnetization at different temperatures between 2 and 8 K were recorded for complexes **1**, **3**, and **4** (Fig. 2 and S10†).

All the derivatives exhibit a paramagnetic behavior characterized by a constant $\chi_M T$ value between 300 and 100 K, followed by an increasingly steep decline as T approaches 2 K. The characteristic values found at 300 and 2 K are listed in Table 2. As can be seen in Fig. 2, the decline is clearly more pronounced for **4**. For this compound the *M versus H* at 2 K also exhibits an S-shape behavior (Fig. 2b). These behaviors indicate antiferromagnetic interactions between the complexes in the solid state.[61] This was anticipated from the crystal lattice arrangement of **4**, which shows short contacts existing between the Fe complexes by means of I atoms and phenanthroline moieties (*vide supra*). The *M versus H* behavior for **5** confirms this hypothesis (Fig. 2b), as such intermolecular magnetic interactions no longer occur for the diluted derivative.

The field dependences of the magnetizations for **1**, **3**, and **4** between 2 and 8 K are depicted in Fig. S10.† For all complexes,

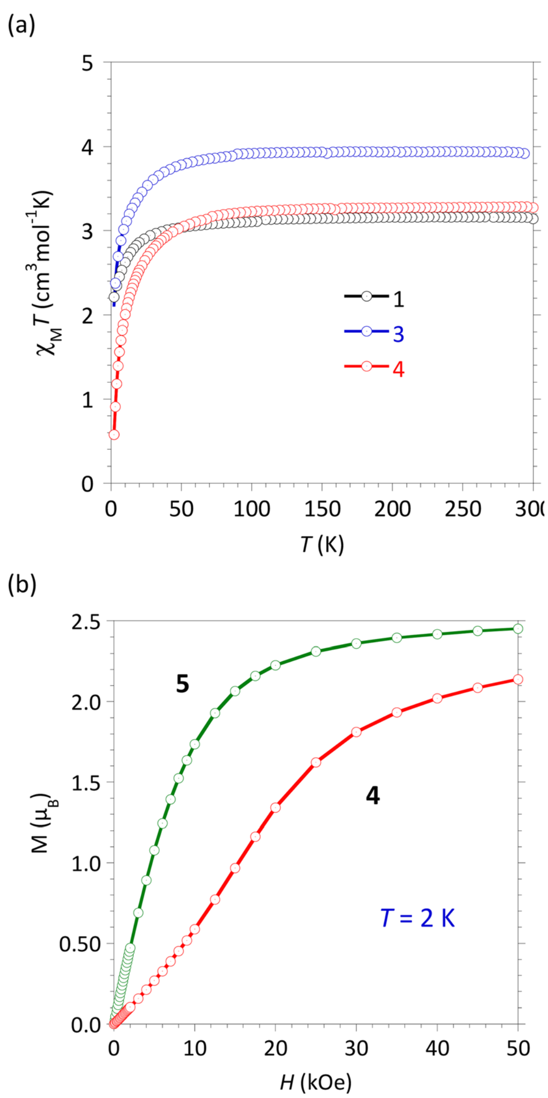


Fig. 2 (a) $\chi_M T$ *versus T* for **1**, **3**, and **4** and (b) *M versus H* at 2 K for **4** and **5**.

the magnetization achieved for a field of 50 kOe at 2 K is around 2.5μ_B (Table 2), a value lower than the expected contribution of spin alone for a center with S = 2, confirming the existence of substantial magnetic anisotropy. The theoretical calculations revealed that only derivative **1** shows no mixing between ground and excited states, and so its magnetic behaviors were likely to be analyzed with a ZFS-based model. Indeed, a good adjustment of the experimental *M versus H* behaviors was obtained with a model for an S = 2 with magnetic anisotropy accounted by ZFS, yielding $D = -24.9 \pm 0.4$ cm^{-1} and $E = 4.06 \pm 0.04$ cm^{-1} for $g = 2.34$ (Fig. S10a†). The negative value for D is in agreement with the anticipated axial magnetic anisotropy for a high-spin d^6 ion in PBP geometry,[27] and with the calculated value. A similar analysis proved impossible for **3** (let alone **4**), which is in line with the conclusion of the theoretical studies. However, for these complexes the experimental *M versus H* behaviors are well reproduced by the calculated behaviors as described above (Fig. S10†).

The existence of a slow magnetization relaxation phenomenon was examined by AC-mode magnetic susceptibility measurements in zero field and in the presence of an applied magnetic field. For **1**, just the onset of an out-of-phase component of the susceptibility, χ''_M, is observed in the applied field (Fig. S11†), a behavior indicative of fast relaxation above 2 K. For **3**, a slow relaxation is revealed when a small field is applied to cancel the QTM (quantum tunneling of the magnetization), while the χ''_M *versus T* for **4** in zero field shows a frequency dependent peak (Fig. S12a and S13a,† respectively).

For **3**, the optimum applied DC field, *i.e.* for which the peak signal of χ''_M is the largest and at the lowest frequency, turned out to be 5 kOe (Fig. S12b†). Detailed AC studies were carried out with this applied field to reveal frequency and temperature dependencies of the χ''_M characteristic for an SMM (Fig. 3 and S12†). The relaxation times, τ, between 2 and 5.6 K were obtained from the analyses of χ''_M *versus* ν behaviors using a generalized Debye model.[62] The temperature dependency of τ is well modeled by an expression combining an Orbach process (Arrhenius law) and a direct mechanism to account for the lower temperature behavior (eqn (6), respectively first and second term).[1,5] The best fit gave $U/k_B = 89 \pm 5$ K, $\tau_0 = 7 \times 10^{-12}$ s, $DH^n = 14.6 \pm 0.1$ K^{-1} s^{-1}.

$$\tau^{-1} = \tau_0^{-1} \exp(-U/k_B T) + DH^2 T + QTM^{-1} \qquad (6)$$

For **4**, the temperature dependency of χ''_M in the absence of a static applied magnetic field exhibits a well-defined peak shape with the maximum shifts from about 6 K to lower T with decreasing AC frequencies from 1500 to 1 Hz (Fig. S13b†). However, χ''_M does not tend to return to zero after reaching the peak when the temperature is lowered. The behavior at lower temperatures (<3 K) is characteristic of QTM, but a second, smaller peak is also observed at around 3.5 K. This second component could be the signature of magnetic ordering, which is also suggested by the meta-magnetic behavior shown by **4** (Fig. 2b). Indeed, the second (T-independent) signal is no longer found for **5** (*vide infra*). Moreover, the signal at 3.5 K is

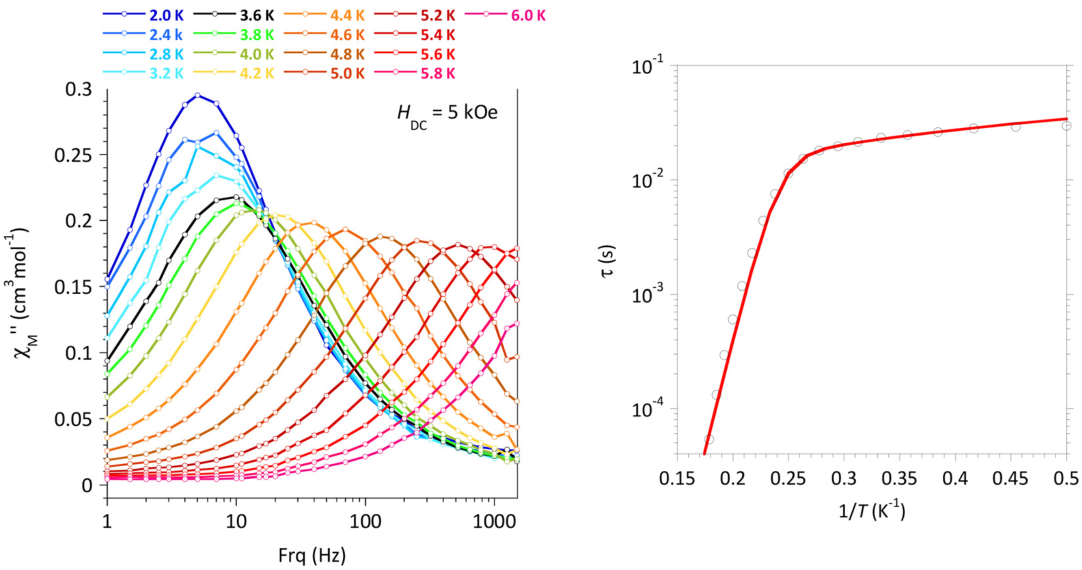


Fig. 3 [$FeL^{N5}Br_2$], **3**: (left panel) χ''_M *versus* ν, in the 2–6 K temperature range with H_{DC} = 5 kOe; (right panel) experimental (O) and calculated (—) relaxation time, τ, *versus* 1/*T* with best-fit associated with eqn (6) (see the main text).

dampened by the application of a DC field (Fig. S13e and f†). In the field, the maximum of χ''_M at 2 K steadily shifts to a lower frequency until about 5 kOe, suggesting QTM cancellation, but it also continuously increases in intensity with DC fields up to 10 kOe (Fig. S13e†). The latter trend can be attributed to a gradual decoupling of intermolecular interactions by the applied field.

The relaxation times, τ, for **4** in the absence of an applied field could be obtained between 2 and 6.4 K from the analyses of χ''_M *versus* ν behaviors by a generalized Debye model (Fig. S13c and Table S4†). The plot of τ *versus* 1/*T* (Fig. S13d†) shows three domains characteristic of temperature activated relaxation (6.4–5.8 K), *T*-independent relaxation (5.6 to about 3.5 K), and for lower *T* a steady increase of the relation time likely due to the intermolecular interactions taking place at these temperatures. The temperature dependence of τ between 6.4 and 3.0 K is well modeled by an expression combining an Orbach process and QTM (first and third terms in eqn (6)). Best fit gave U/k_B = 76 ± 6 K, $\tau_0 = 8 \times 10^{-10}$ s, and QTM = 3.41 × 10^{-4} s (Fig. S13d†), confirming the SMM behavior in zero-field for **4**.

In order to avoid the contribution of the QTM, an investigation of the SMM characteristics was performed with an applied field of 5 kOe (Fig. S13g–l†). Under these conditions, the temperature dependence of the relaxation of the relaxation time follows an Orbach behavior in the higher T-domain and a Direct process below 5 K, characterized by U/k_B = 99 ± 8 K, $\tau_0 = 3 \times 10^{-11}$ s, and DH^n = 11.47 ± 0.09 $K^{-1}\ s^{-1}$. It can be noticed

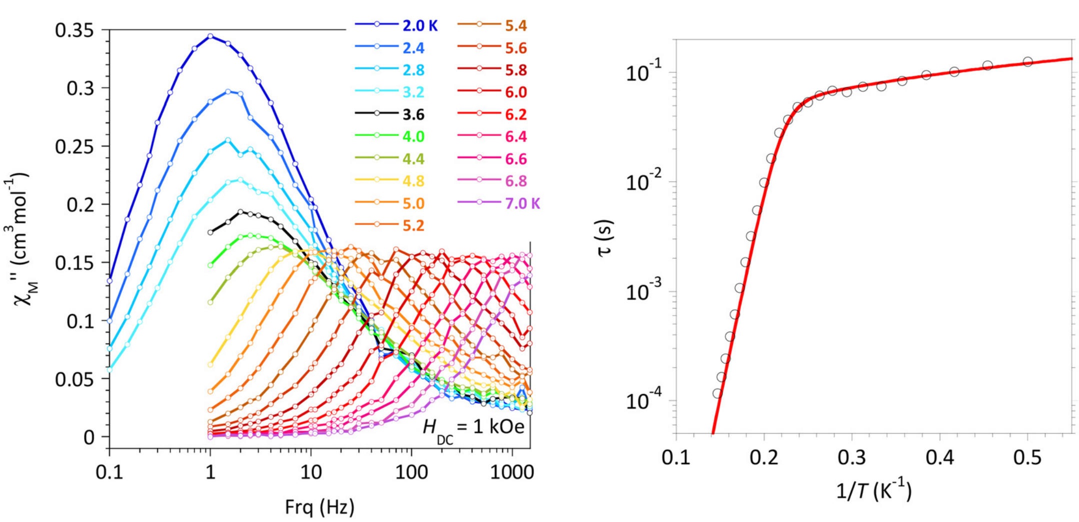


Fig. 4 [$Fe_{0.12}Zn_{0.88}L^{N5}I_2$], **5**: (left panel) χ''_M *versus* ν with H_{DC} = 1 kOe, in the 2–7 K temperature range; (right panel) experimental (O) and calculated (—) relaxation time τ *versus* 1/*T* with best-fit associated with eqn (6) (see the main text).

that these parameters are very similar to those of the Br-derivative, **3**.

For the solid solution of $[FeL^{N5}I_2]$ in $[ZnL^{N5}I_2]$, **5**, AC magnetic susceptibility recorded in the absence of an applied static magnetic field confirms the emergence of a χ''_M signal below 9 K for a test frequency of 997 Hz (Fig. S14a†). However, the profile of the behavior, with a poorly resolved peak and a continuously increasing contribution with decreasing temperature, is a characteristic of rapid relaxation processes such as QTM. The latter is quenched when a static field is applied; for **5**, 1 kOe was found to be optimal (Fig. S14b†) and was applied for detailed AC studies. The χ''_M *versus* ν behavior at different temperatures between 2 and 7 K and the related temperature dependence of τ are depicted in Fig. 4 (additional plots are shown in Fig. S14†). And here again, τ *versus* $1/T$ of $[FeL^{N5}I_2]$ in **5** is perfectly modeled using an Orbach and a Direct process (eqn (6)) with $U/k_B = 88 \pm 9$ K, $\tau_0 = 2 \times 10^{-10}$ s, and $DH^n = 4.13 \pm 0.04$ K^{-1} s^{-1}. The energy barrier for magnetization reversal is very similar to that obtained for the bulk $[FeL^{N5}I_2]$, confirming the molecular origin of the low T magnetization freezing. No evidence for magnetic hysteresis was found in the M *versus* H behaviors recorded at 2 and 5 K (Fig. S10†).

An SMM-type behavior with a magnetization reversal regulated mainly by a thermal activation is clearly evidenced for **2–5**. For the I-derivative, slow magnetization relaxation is even observed in zero-field, a behavior hardly found in mononuclear 3d ion SMMs. For all, the effective energy barriers, U/k_B, are much larger than those usually reported for PBP complexes of Fe(II),[27] and reach that described for linear two-coordinated Fe(II) with first-order SOC.[15]

Concluding remarks

Unlike lanthanide ions, 1st-order spin–orbit coupling is cancelled by the ligand field for transition metal ions, and achieving high magnetic anisotropy is generally considered the Holy Grail for these ions. The results gathered herein show that the contribution from first-order SOC is not confined to low-coordination number complexes but can be implemented in a pentagonal bipyramidal Fe(II) complex by appropriate chemical design.

The rigid pentadentate ligand, L^{N5}, imposes an effective equatorial coordination sphere close to a pentagonal and symmetrical ligand field, and apical ligands (halogens in the present case), which exhibit a very similar overlap with the d_{xz} and d_{yz} orbitals, lead to a quasi-degenerate state for these orbitals sharing 3 electrons. The mixing of the M_s levels, revealed by the theoretical calculations, confirms the existence of the 1st-order SOC and rules out the possibility of describing the magnetic behavior within the ZFS formalism.

Further evidence is provided by the experimental behavior, which is consistent with the existence of large magnetic anisotropy, the most salient feature being the SMM behavior with large energy barriers for magnetization reversal. The observation of such a behavior without an applied static magnetic field is quite unique for a mononuclear 3d complex.

Finally, such complexes with quasi-degenerated magnetic orbitals are also desirable building units for the design of exchange-coupled polynuclear compounds. Some recent studies have shown that close to a first-order SOC regime, anisotropic interactions such as the Dzyaloshinskii–Moriya[63,64] or the symmetric exchange tensor of anisotropy may also reach high values.[65,66]

Experimental section

Materials and methods

All the chemicals used were commercially available and were employed without further purification. 2,9-Di(α-methylhydrazino)-1,10-phenanthroline·2HCl, denoted as $Phen^{MeNH2}$·2HCl, was synthesized from methylhydrazine and 2,9-dichloro-1,10-phenanthroline[67] according to a literature procedure.[68] All the syntheses of Fe^{II} complexes were carried out under N_2 using Schlenk techniques and degassed solvents (diethyl ether was purified using an Innovative Technology Solvent Purification® system while the alcohol and water solvents were distilled under N_2 prior to use). The solid samples for the characterization of the Fe^{II} complexes were prepared in a glovebox. Technical information can be found in the ESI.†

Synthetic procedures

$[FeL^{N5}(H_2O)Cl]Cl·4.5H_2O$ 1. $FeCl_2·4H_2O$ (100.0 mg; 0.54 mmol) was solubilized in 70 mL of water, followed by the addition of ascorbic acid (100.0 mg; 0.57 mmol). $Phen^{MeNH2}$·2HCl (170.0 mg; 0.54 mmol) was then added to the solution that turned yellow. The addition of 2,6-diacetylpyridine (81.0 mg; 0.54 mmol) followed by 3 drops of 37% aqueous HCl solution turned the solution color to olive green. The reaction mixture was heated to reflux for 1 h and then concentrated to 10 mL before being stored overnight at 5 °C. Dark green-brown crystals suitable for X-ray diffraction were collected by filtration and washed with cold methanol, shortly dried under vacuum and stored under a nitrogen atmosphere (164.0 mg, 49%). Elemental analysis: calcd for $C_{23}H_{29}N_7Cl_2FeO_{4.5}$ (with the loss of 1 H_2O molecule of the solvent) C, 45.79; H, 5.01; N, 16.25. Found C, 45.96; H, 4.93; N, 16.22. IR (ν_{max}/cm^{-1}): 3317br, 3170br, 2989w, 1589vs, 1575vs, 1553s, 1486vsbr, 1419s, 1363s, 1300s, 1186s, 1148vs, 1090vs, 1045vs, 858vs, 736w, 686w, 664w, 624vw.

$[FeL^{N5}Br_2]$ 3 and $[FeL^{N5}I_2]$ 4. **3** and **4** were synthesized following the same procedure as that for **1**. 48% HBr and 47% HI aqueous solutions were used instead of HCl to acidify the reaction mixture after the addition of 2,6-diacetylpyridine. The addition of NaBr (500.0 mg; 4.85 mmol) or NaI (800.0 mg; 5.33 mmol) at the end of the reflux time formed dark green crystals of **3** (121 mg, 41%) or **4** (176 mg, 50%), respectively, and these were isolated after cooling the solution to 5 °C for several days. For **3**, elemental analysis: calcd for $C_{23}H_{21}N_7Br_2Fe$ C, 45.20; H, 3.46; N, 16.04. Found C, 45.35; H, 3.47; N, 16.20. IR (ν_{max}/cm^{-1}): 3039w, 2951w, 1590vs, 1575vs, 1545s, 1495vs, 1473s, 1427s, 1409s, 1359s, 1302s, 1256s, 1187s, 1150vs, 1094vs, 1027vs, 1009s, 953s, 873vs, 812s, 795s, 742s, 658w. For **4**, elemental analysis: calcd for $C_{23}H_{21}N_7FeI_2$ C, 39.18; H, 3.00; N, 13.90. Found C, 39.08; H, 2.74; N, 13.66. IR (ν_{max}/cm^{-1}): 3037w, 3000w, 1589vs, 1573vs, 1542s, 1494vs, 1471s, 1451s, 1423s, 1405s, 1354s, 1302s, 1255s, 1180s, 1148vs, 1091vs, 1026vs, 867vs, 798s, 740s, 695w, 658w.

[$Fe_{0.12}Zn_{0.88}L^{N5}I_2$] 5. $FeCl_2\cdot 4H_2O$ (8.0 mg; 0.040 mmol) and $Zn(NO_3)_2\cdot 6H_2O$ (118.0 mg; 0.40 mmol) were solubilized in 40 mL of H_2O. Ascorbic acid was added (20.0 mg; 0.11 mmol) followed by Phen$^{MeNH2}\cdot$2HCl (113.0 mg; 0.36 mmol), 2,6-diacetylpyridine (69.0 mg; 0.42 mmol), and then 3 drops of 47% HI aqueous solution. The solution turned yellowish green and it was heated to reflux for 1 h. A brown-yellow precipitate started to appear during the heating time. NaI (380.0 mg; 2.53 mmol) was added before allowing the suspension to cool down to room temperature. The solid was then isolated by filtration through a cannula and filter paper. It was washed with acetone and dried under vacuum (120.0 mg, 47%). Elemental analysis: calcd for $C_{23}H_{21}N_7Fe_{0.12}I_2Zn_{0.88}$ C, 38.72; H, 2.97; N, 13.74. Found C, 38.70; H, 2.95; N, 13.75. EDX analysis (Fig. S9†) was performed on several crystals that confirmed the presence of both Fe (12 mol%) and Zn (88 mol%). IR (ν_{max}/cm^{-1}): 3035w, 3010w, 1592vs, 1571vs, 1542s, 1494vs, 1470s, 1450s, 1427s, 1408s, 1355s, 1302s, 1255s, 1179s, 1148vs, 1091vs, 1030vs, 866vs, 798s, 739s, 693w, 658w.

Single-crystal X-ray diffraction

Intensity data were collected at low temperature on an Apex2 Bruker diffractometer equipped with a 30 W air-cooled microfocus source (λ = 0.71073 Å). Structures for **1**, **3**, **4**, **[$ZnL^{N5}Br$]·Br** and **$ZnL^{N5}I_2$** were solved using SUPERFLIP[69] or ShelxT[70] and refined by means of least squares procedures using the PC version of the CRYSTALS program.[71] Atomic scattering factors were taken from the international tables for X-ray crystallography.[72] All non-hydrogen atoms were refined anisotropically. Structure for **2** was found to be modulated with a modulation vector of 0.6754*a* + 0.1052*c*. The structure was solved by charge flipping methods using the SUPERFLIP program[69] using the super-space group *C*2/*m*(a0g)0s and refined using Jana2006.[73] The structure is composed of two superimposed inversed configurations of the complex. The incommensurately modulated structure of this complex was successfully solved and refined in (3 + 1) dimensions, evidencing an ordered-disorder along the fourth dimension as well as a significant atomic displacement from the average positions. Additional information can be found in the ESI.†

Computational information

CASSCF and NEVPT2 calculations were performed with Orca5.0 package[59] considering 6 active electrons and 5 active MOs, *i.e.* those with a major Fe 3d character. Extended DKH-def2 atomic basis sets were used for all atoms except I: QZVPP for Fe atoms (14s10p5d4f2 g), TZVP for C (6s3p2d1f), N (6s3p2d1f), O (6s3p2d1f), Cl (8s4p2d1f), Br (10s8p4d1f), SVP for H atoms (2s1p) and a 15s11p5d1f basis set for I atoms. Optimization of the positions of the H atoms was performed by DFT calculations using def2-SVP atomic basis sets and the PBE functional.

Data availability

Crystallographic data have been deposited at the CCDC under 2352840 (**1**), 2352845 (**2**), 2352841 (**3**), 2352842 (**4**), 2352843 ([$ZnL^{N5}Br$]·Br), and 2352844 ($ZnL^{N5}I_2$) and can be obtained from **https://www.ccdc.cam.ac.uk/structures/**.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

The authors are grateful to M. J.-F. Meunier (LCC) and M. V. Collière (LCC) for technical assistance in magnetic, Mössbauer, and EDX data collection.

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2.2 Impact of the Electric field on the ZFS parameters

This section follows on from a previous study of a Ni(II) complex exhibiting strong unaxial anisotropy.[17] This complex has a tetradentate ligand (hexamethyl-2,2',2"-triamino-triethylamine) in which three nitrogen atoms form bonds in the plane with the metal ion, while the fourth nitrogen atom forms bonds in the perpendicular plane, as shown in Figure 2.2. Opposite this fourth nitrogen atom is an apical halide ligand, either chlorine or bromine.

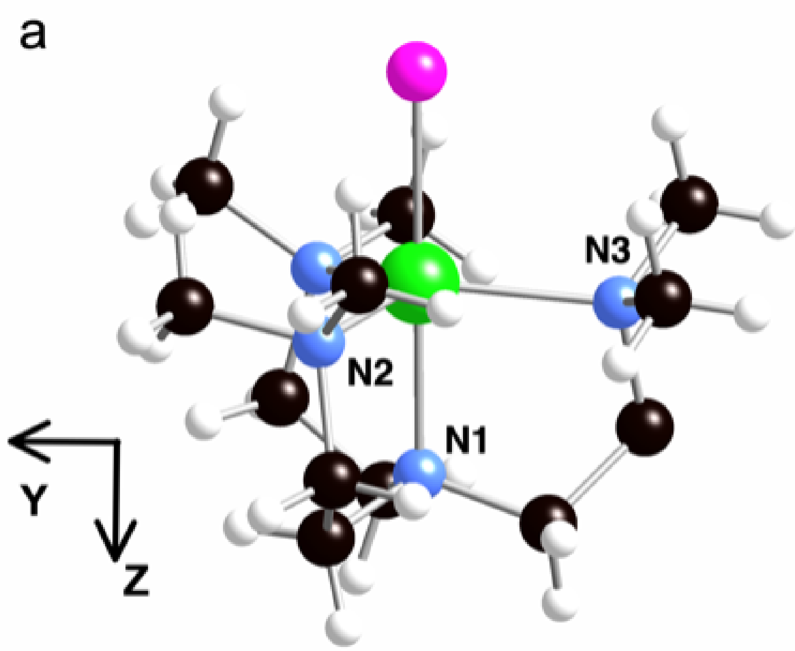


FIGURE 2.2 – Representation of the complex [Ni(Me_6trenCl)](ClO_4), Ni in green, Cl in purple, N in blue, C in black and H in white

Cristallographic data indicates that this complex crystallises in the trigonal space R_{3c} with a C_3 axis of symmetry along the Ni-Cl axis. This coordination induces a degeneracy of the d-orbitals that share the same angular component m_l=±2(±1), *i.e.* d_{xy} and $d_{x^2-y^2}$ (d_{xz} and d_{yz}). The conditions for 1^{st} order SOC are met between two degenerate ground states carried by the two d^8 configurations depicted on Figure 2.3 (left). This would result in a large splitting of the M_S components of the ground state with a large value for the D term as strong as the spin orbit coupling constant of the nickel ion (ξ_{Ni}=644cm^{-1}).[42] At this point, a competing effect appears, the Jahn-

Teller effect, which breaks the orbital degeneracy by distorting the complex (mainly removing the Ni atom from the barycenter of the N2, N3 and N4 atoms), reducing the axial anisotropy and gives rise to non-zero rhombic term ($E \neq 0$). This distortion is confirmed by both electron paramagnetic resonance (EPR) studies and a theoretical study, which revokes of 1^{st} order SOC, leaving only 2^{nd} order SOC as the source of the anisotropy.

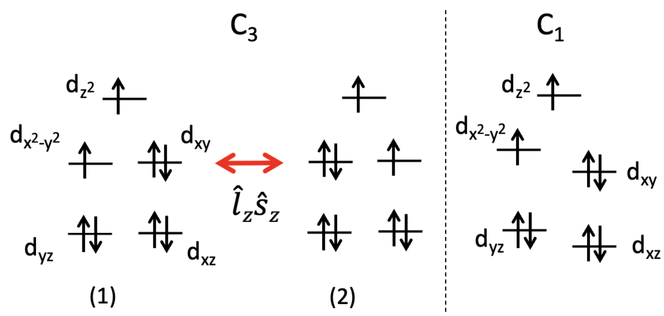


FIGURE 2.3 – Energy diagrams of the d-orbitals in case of three-fold symmetry C_3 (left) and in presence of Jahn-Teller distortion (right) with the induced lift of degeneracy of the d-orbitals.

The ground state is now mainly carried by the configuration illustrated in Figure 2.3 (right) where $d_{x^2-y^2}$ is simply occupied while d_{xy} remains doubly occupied. State of the art methodology was applied to this complex as to extract the ZFS parameters, the concluding remark was that this complex exhibits strong axial anisotropy with $D \approx -104cm^{-1}$ and a significant rhombic term E=1.6cm^{-1}. Although the Jahn-Teller effect has a significant impact on the value of the D parameter, it does not completely negate it, which shows that the ligand rigidity is an important factor to reach large values of anisotropy.

The focus of the present study lies in the interest of controlling the anisotropy via an external stimulus, such as an electric field. While the effect of a magnetic field on the different M_S components is well understood in terms of the Zeeman effect, it is practically impossible to focus a magnetic field at the nanoscale. This makes it hard to conceptualise any real life application relying on magnetic fields for systems of this scale. Electric fields, however, allow very accurate control at this scale and can affect the magnetic properties of a material via magneto-electric couplings. *Ab initio* calculations allow for the appreciation of the Jahn-Teller distortion as well as the determination of the anisotropic parameters D and E as functions of the applied electric field. Such a mechanism can be rationalized by analysing the Spin-Orbit coupled wave-function and the contribution of the excited states to the ZFS, both of which are provided by the ORCA package. To identify, discriminate and quantify the mechanisms at the origin of the magneto-electric effect, series of calculations were performed for different orientations and intensity of the electric field. It is then possible to evaluate the effect of :

(a) electronic structure changes, ZFS calculation under an applied field on the zero-field geometry.

(b) geometric structure changes, ZFS calculation under no applied field on the field-optimised geometry.

(c) electronic and geometric changes, both geometry optimisation and ZFS calculation performed under an applied electric field.

In order to explore the response to the electric field for different orientations, the field $\mathbf{F}$ was initially applied along the Ni-Cl axis, which corresponds to the easy axis of magnetisation. This orientation includes the maximum component of the dipole moment and is referred to as $\mathbf{F} \parallel Z$. The electric field was then applied perpendicular to this axis ($\mathbf{F} \parallel Y$) in the YZ plane containing the Ni, N1 and N3 atoms (see to Figure 2.2 for atom numbering). The YZ plane was chosen because the Ni-N3 bond length is the shortest of the three Ni-N bonds in the XY plane.

Analysis of the contributions

The main contribution to the D term comes from the SOC with the first excited triplet state (T_1), which is predominantly expressed in the determinant formed by exciting an electron from the d_{xy} to the $d_{x^2-y^2}$ orbitals. The Hamiltonian used to rationalise the SOC is given by $\hat{H}^{SO} = \sum_i \zeta(\hat{l}_x^i \hat{s}_x^i + \hat{l}_y^i \hat{s}_y^i + \hat{l}_z^i \hat{s}_z^i)$ where i runs over all the electrons in the d^8 configuration. The two lowest triplets mainly present singly occupied orbitals that are combinations of m_l=$\pm$2, the resulting SOC originates only from the part of the

Hamiltonian involving the m_s=±1 of each triplet, *i.e.* from the $\hat{l}_z^i \hat{s}_z^i$ term. This coupling stabilises the m_s=±1 of the T_0 ground state without affecting the m_s=0 component, which gives a negative contribution to the D term. This contribution can be estimated using second-order perturbation theory :

$$C(D)^{(2)} = -\frac{|\langle T_0^{\pm 1}| \hat{H}^{SO} |T_1^{\pm 1}\rangle|^2}{\Delta E} \tag{2.2.1}$$

with $\Delta E = E(T_1) - E(T_0)$ the energy difference between the two states. As this energy difference is relatively small, a better treatment of the contributions can be obtained by diagonalising the 2×2 SOC matrices in the basis of $T_0^{\pm 1}$ and $T_1^{\pm 1}$ using a variational approach :

$$C(D)_{VAR} = \frac{\Delta E - \sqrt{\Delta E^2 + 4|SOC|^2}}{2} \tag{2.2.2}$$

where $|SOC|^2 = |\langle T_0^{\pm 1}| \hat{H}^{SO} |T_1^{\pm 1}\rangle|^2$ is the squared value of the off-diagonal term in the SO Hamiltonian between the ground and excited components. Other states may contribute positively or negatively to the overall value of D, their contributions can be obtained in the same way. The ORCA program also provides these contributions and will be used if possible. To approach the final value of D, one needs to sum all these contributions. However, the variation of the electric field has almost no impact on the contributions of the excited states except for $T_1^{\pm 1}$, so they will be excluded from the rationalization. The rhombic term E will be rationalised, but as the sum of the contributions provided by the program does not match the overall variation of the E term, the rationalisation will be performed based on the analysis of the wave-functions.

2.2.1 Computational Informations

Wave-function based calculations have been performed using the ORCA package [43]. In a first place, CASSCF calculations have been performed to account for non-dynamic correlations. The CAS(8,5) active space contains eight electrons in the five essentially 3d orbitals of the Ni(II) ion, and one set of orbitals have been optimized using an average procedure over all triplet and all singlet states of the configuration. To account for dynamic correlations at the 2^{nd} order of perturbation, NEVPT2 calculations have been carried out for all states. The SO-SI method was then used to account for SOCs between all Ms components of all spin states using CASSCF wave-functions and NEVPT2 correlated energies (*i.e.* as diagonal elements of the SO matrix). ZFS parameters were extracted using a procedure based on the effective Hamiltonian theory which proved to provide accurate values of the anisotropic parameters. The QZVPP extended basis sets [44] have been used for Ni atoms (14s10p5d4f2g), TZVP for other atoms (6s3p2d1f for N, 8s4p2d1f for Cl and 6s3p2d1f for C) and SVP for H (2s1p). As the field may change the geometrical structure, geometry optimizations were performed for different values and orientations of the field. We used the Hay-Wadt LanL2TZ(f) basis set for Ni and its corresponding relativistic effective core, Pople triple-ζ basis set (6-311G) for N [44] and Cl atoms [45], a Pople double-ζ plus polarization basis set (6-31G*) [46] for C atoms, and a Pople double-ζ basis set (6-31G) for H atoms [46] and the B3LYP functional[47]. Small variations in the geometries have proved to have a large impact on the ZFS parameters, as such the convergence thresholds were set as VERYTIGHT.

The field is applied in the two opposite directions referred as positive +F and negative –F with an amplitude of $|\mathbf{F}|= 1.028.10^9$ V.m^{-1}. The positive direction for the field in the Z direction induces an elongation of the Ni^{2+}-Cl^- bond, *i.e.*, the positive region of the electric potential is above Cl^- in Figure 1. The positive direction for the field in the Y direction induces an elongation of the Ni^{2+}-N3, *i.e.*, the positive region of the electric potential is to the right of N3 in Figure 2.2.

2.2.2 Application of the electric field along the Z axis

First, we consider the $\mathbf{F} \parallel Z$ situation since the important component of the electric dipole moment is along Z, coinciding with the Ni-Cl bond while its components along X and Y are small. Table 2.1 shows the three components of the dipole moment and its magnitude in absence of electric field.

Figure 2.4 shows the ground state energy as a function of the electric field for the three cases of calculation. As expected from the Stark effect, the variation of case (a) is linear with respect to the field $\mathbf{F}$, which tends to stabilise the ground state when the

	μ_X	μ_Y	μ_Z	$|\mu|$
F=0	-0.02	-0.08	-8.11	8.11

TABLE 2.1 – Dipole moment components in Debye at F=0 V.m^{-1}

applied field and dipolar moment share the same orientation. In case (b) the energy follows a parabolic curve as the electric field disrupts the equilibrium geometry found at $F = 0V.m^{-1}$. The curvature is much smaller compared to the variation in case (a) and the overall behaviour in case (c) follows the electronic effects.

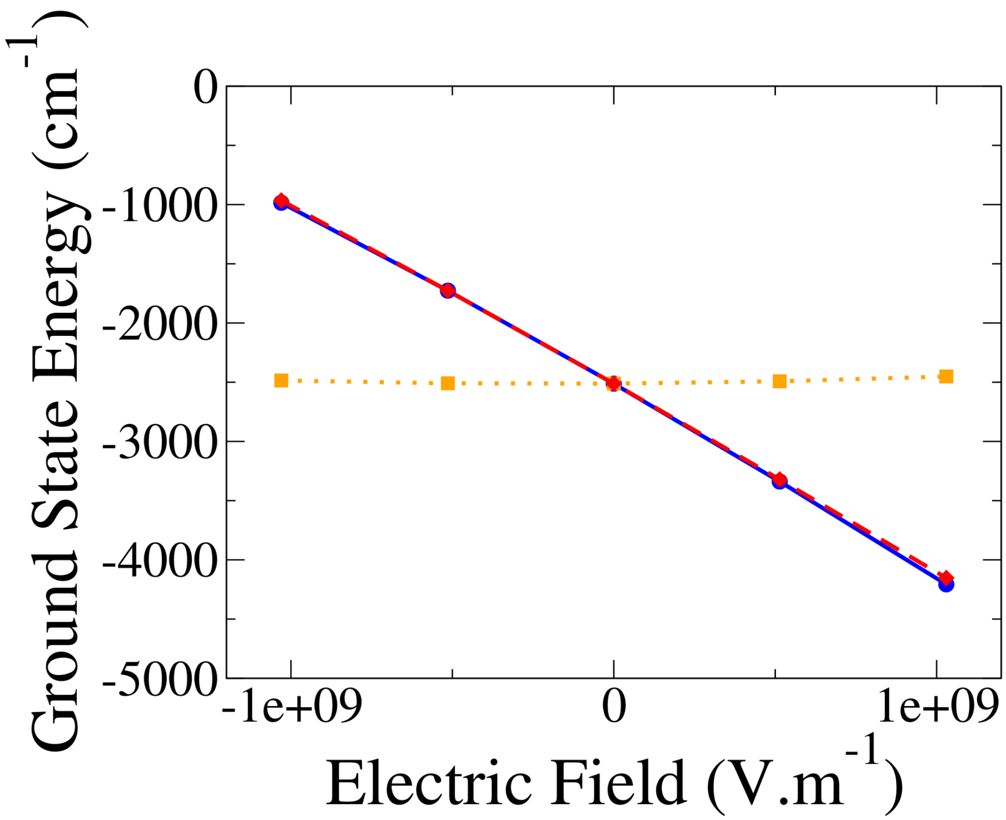


FIGURE 2.4 – Energies in cm^{-1} (shifted by 58.35795.10^7 cm^{-1}) of the ground state obtained for the cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line) as functions of the field, $\mathbf{F} \parallel Z$.

Figure 2.5 shows how the D parameter evolves as a function of the applied field $\mathbf{F}$. In all three cases the variation (dD/dF) is linear, with a small negative slope in case (a) and a large positive slope in cases (b) and (c). This demonstrates that the geometric effects (b) have a stronger impact on the variation of the D value than electronic effects (a) as the overall tendency of case (c) follows that of case (b). Figure 2.6 shows the evolution of the E parameter. Although some deviation is observed for the geometric case (b), mainly due to the lack of precision in the optimised geometry, the overall variation is linear with the field. However, the variation in case (b) is very small (<0.05 cm^{-1}) and the behaviour of case (c) is now governed by the electronic effects with a negative slope.

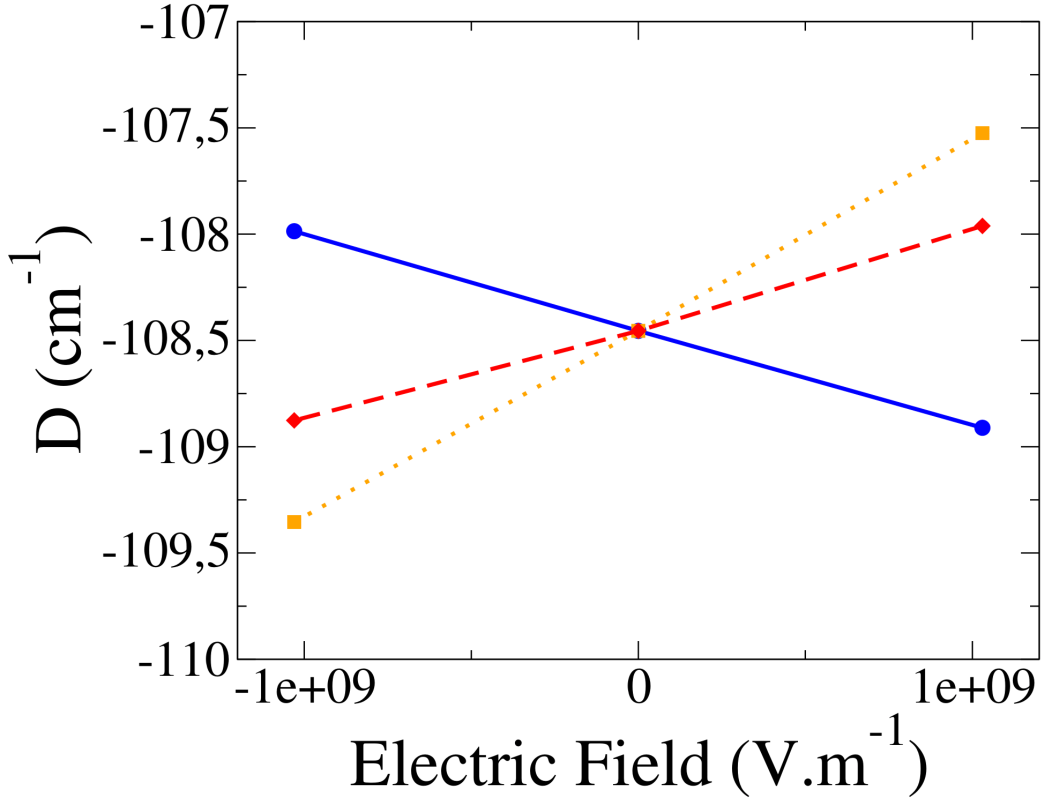


FIGURE 2.5 – Evolution of the axial D parameter (in cm^{-1}) as a function of the field applied in cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line), $\mathbf{F} \parallel Z$

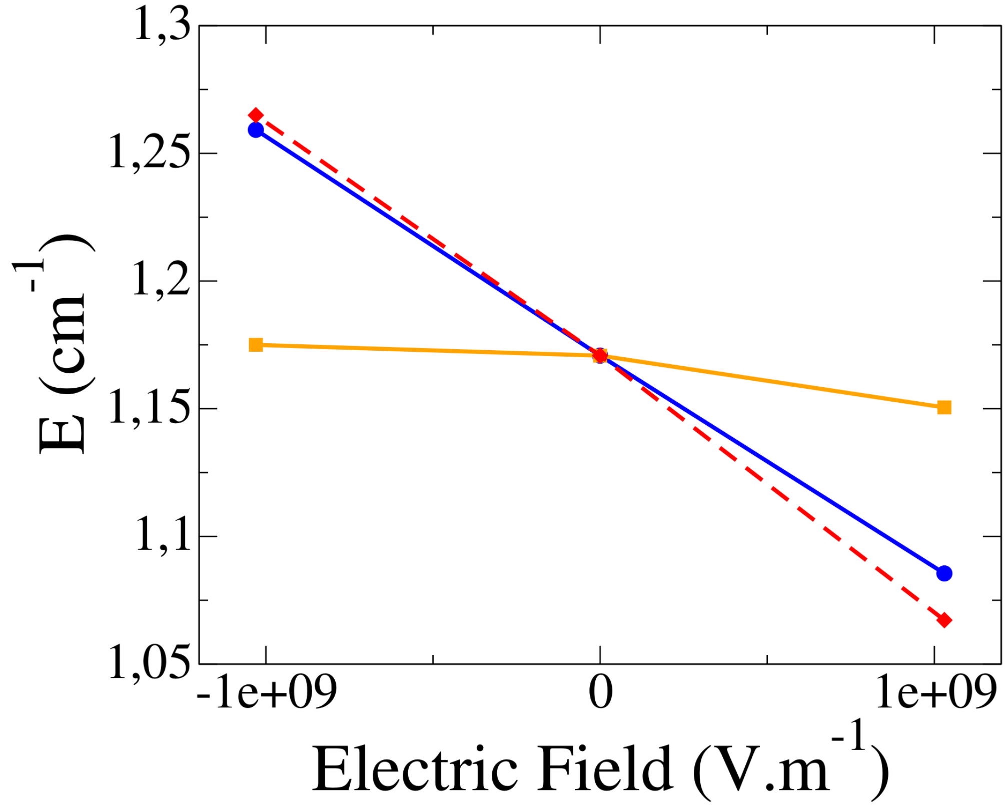


FIGURE 2.6 – Evolution of the rhombic E parameter (in cm^{-1}) as a function of the field applied in cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line), $\mathbf{F} \parallel Z$

The slopes dD/dF and dE/dF for all three cases of calculations and their individual sums are reported in Table 2.2. Note that these slopes are additive with good precision for both the D and E parameters.

$Hz/(V.m^{-1})$	(a) Electronic structure	(b) Geometric structure	(c) Both	Sum of (a) and (b)
dD/dF	-13.52	+26.66	+13.15	+13.14
dE/dF	-2.53	-0.36	-2.88	-2.89

TABLE 2.2 – Slopes of the straight lines in cases (a), (b) and (c) for the D and E parameter for $\vec{\mathbf{F}} \parallel Z$, Fig. 2.5 and 2.6

To rationalise these results, the most significant contributions to the D parameter C(D) were examined as detailed in Table 2.3. As mentionned previously, the main contribution comes from the first excited triplet, which is also the most affected by the field **F**. While other states may have large contributions, the variation from $-F$ to $+F$ is minimal. Given that these contributions are positive (while the total D is negative) it is clear that the first excited triplet is the main source of the substantial axial D parameter and its variation in this system.

	Excited state	C(D) for -F	C(D) for +F	$\Delta C(D)$ ORCA	$\Delta C(D)$ VAR.	$\Delta C(D)$ PERT.	$\Delta\|\mathrm{SOC}\|^2$	$\Delta(\Delta E)$
(a)	T_1	-179.18	-180.06	-0.88	-0.82	-0.96	1722.78	-1.60
	T_2	23.84	23.72	-0.12				
	T_3	19.61	19.68	0.07				
	S_3	20.48	20.57	0.09				
(b)	T_1	-180.95	-178.25	2.70	2.70	3.42	-209.90	36.2
	T_2	23.82	23.72	-0.10				
	T_3	19.71	19.58	-0.13				
	S_3	20.55	20.51	-0.04				
(c)	T_1	-180.49	-178.66	1.83	1.90	2.47	1509.15	34.7
	T_2	23.88	23.66	-0.22				
	T_3	19.67	19.61	-0.06				
	S_3	20.51	20.56	0.05				

TABLE 2.3 – Contributions to the axial parameter D (in cm^{-1}) of the most contributing excited states and their variations ΔC(D) between the -F and $+F$ in the Z direction either provided by ORCA, calculated variationally and perturbatively. $\Delta|\mathrm{SOC}|^2$ is the variation of the square of the module of the SOC between the ground and the first excited triplet states while $\Delta(\Delta \mathrm{E})$ is the variation between the -F and $+F$ of their energy difference (in cm^{-1}) obtained at the NEVPT2 level.

The variation of the contribution of the first excited triplet $\Delta C(D)_{VAR}$ matches well with that provided by ORCA, indicating that the variation of the SOC, as well as the variation of the difference in energy $\Delta(\Delta E)$, can be used for the analysis. The perturbative estimation of this contribution yields a slightly larger value ($-193cm^{-1}$ at $F = 0$), as the two lowest states are rather close in energy ($\approx 2166cm^{-1}$), which takes them outside the perturbation regime. Four observations can be made :

— In case (a), the energy difference between the two lowest triplet states remains roughly the same ($\Delta(\Delta E) = -1.6cm^{-1}$), so the variation of the SOC is the leading factor in the change of $|D|$. As the SOC increases, so does $|C(D)|$ of T_1 and then $|D|$.

- In case (b), the SOC decreases and the energy difference increases ($\Delta|\text{SOC}|^2 = -210cm^{-1}$ and $\Delta(\Delta E) = 36cm^{-1}$), inducing a decrease of $|D|$ (Eq. 2.2.1).
- In case (c), both the SOC and energy difference increase. Since $|D|$ decreases, we can conclude that the dominant factor is the increase of the energy difference.
- Since case (c) and (b) have a similar trend, we can also conclude that geometrical structure effects dominate electronic ones.

Ligands	$-F$	$+F$	$\Delta(distance)$
Cl	2.31391	2.32709	$131.8.10^{-4}$
N_1 (apical)	2.15016	2.14727	$-28.9.10^{-4}$
N2	2.22739	2.22043	$-69.6.10^{-4}$
N3	2.14628	2.14139	$-48.9.10^{-4}$
N4	2.22688	2.22023	$-6.65.10^{-4}$

TABLE 2.4 – Variation of the distances in Angstrom between the Ni(II) ion and the atoms of its coordination sphere between $-F$ and $+F$ with $\vec{\mathbf{F}} \parallel Z$.

This last conclusion is supported by the changes in geometry under the application of the electric field, as reported in Table 2.4, with the variation of the distances between the Ni(II) ion and its coordination sphere between $+F$ and $-F$. The Ni-Cl bond length increases in a positive field $+F$, as this bond carries most of the dipole moment and is therefore expected to lengthen in order to maximise its interaction with the field. However, the others atoms tend to move closer to the Ni(II) ion, increasing the ligand field in the XY plane. This destabilises both the $d_{x^2-y^2}$ and d_{xy} orbitals, the former, which points directly towards N3 is more affected. This leads to an increase in the energy difference between these orbitals and thus between the ground and excited states. As can be seen from Tables 2.3 and 2.5, the variation of the two energy differences follows the same pattern in each case of calculation.

Variation of the orbital energy differences	(a)	(b)	(c)
$\Delta(\Delta\epsilon(d_{x^2-y^2} - d_{xy}))$	6.58	258.98	267.76

TABLE 2.5 – Variation between $-F$ and $+F$ applied in the Z direction of the orbital energy difference $\Delta(\Delta\epsilon(d_{x^2-y^2} - d_{xy}))$ in cm^{-1}

The rhombic term E arises from the difference between the X and Y directions. For its rationalisation, the same analysis based on the contributions provided by ORCA should be used. Unfortunately, these contributions cannot be used because their sum increases with the field while the E values extracted via effective Hamiltonian theory decrease as shown in Figure 2.6. Nevertheless it is possible to rationalise the E term by identifying the determinants that distinguish the X direction from the Y direction. In the case of an applied field along the Z axis, the $d_{x^2-y^2}$, d_{xy} and d_{z^2} orbitals are symmetric in X and Y. As such the only determinants that treat these two directions differently are those with a single occupation in the d_{xz} or d_{yz} orbitals. There are six of these determinants with M_S=1, two of them have the largest coefficients in the wave-function of the T_2 and T_3 excited states, resulting from single excitation from either of these orbitals to the $d_{x^2-y^2}$ or d_{z^2} which are singly occupied in the ground state T_0. If there is no difference between the two directions, *i.e.* no rhombicity E=0, the weights of these determinants should be the same in both states leading to two contributions that cancel each other out. To estimate the evolution of the rhombicity, one can look at the variation of these weights with respect to the field. We can define $\Delta w(X-Y)$ which is the difference between the weight w(d_{xz}) and w(d_{yz}), and its overall variation with the field $\Delta(\Delta w(X-Y))$ as reported in Table 2.6.

		$-F$	$-F$	$\Delta(\Delta w(X-Y))$
(a)	$w(d_{xz})$	96.200	96.741	-0.528
	$w(d_{yz})$	97.210	97.224	
	$\Delta w(X-Y)$	1.010	0.482	
(b)	$w(d_{xz})$	96.543	96.494	0.088
	$w(d_{yz})$	97.282	97.145	
	$\Delta w(X-Y)$	0.739	0.651	
(c)	$w(d_{xz})$	96.271	96.762	-0.621
	$w(d_{yz})$	97.282	97.152	
	$\Delta w(X-Y)$	1.011	0.390	

TABLE 2.6 – Weights w(d_{xz}) and w(d_{yz}) of the determinants with single occupation in the respective orbitals, their difference $\Delta w(X-Y)$ and their overall variation $\Delta(\Delta w(X-Y))$ in all three case of calculation for $\mathbf{F} \parallel Z$.

First, we can observe the small variation in case (b) which matches well the variation of E for this case as can be seen in Figure 2.6. The variation $\Delta(\Delta w(X-Y))$ is very similar between case (a) and case (c), which confirms that the overall variation of the E term is dictated by the changes in the electronic structure induced by the application of the electric field.

2.2.3 Application of the electric field along the Y axis

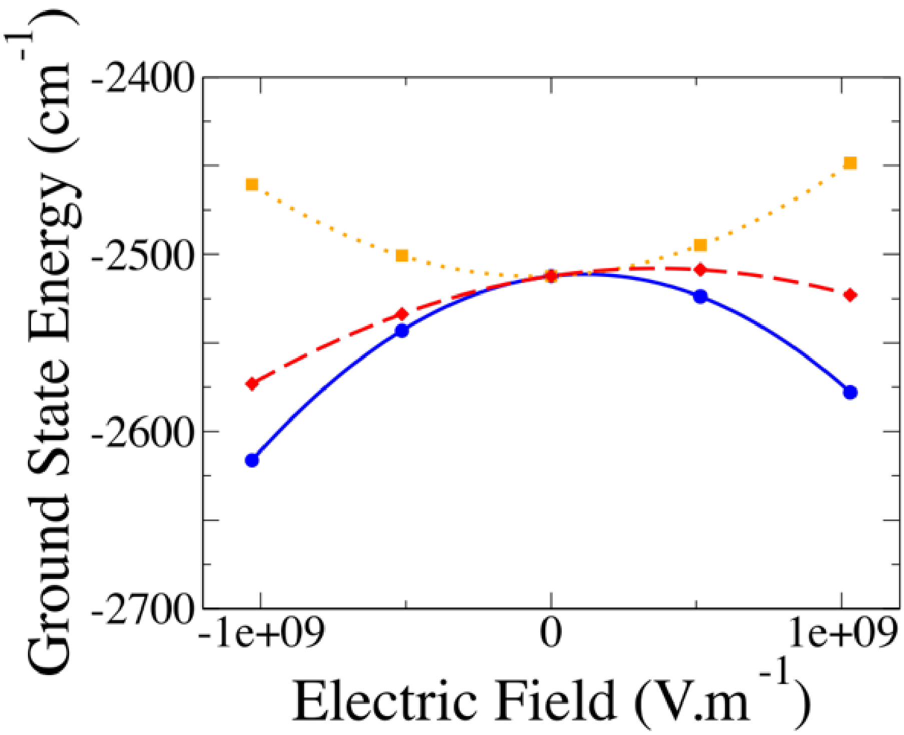


FIGURE 2.7 – Energies in cm^{-1} (shifted by 58.35795.10^7 cm^{-1}) of the ground state obtained for the cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line) as functions of the field applied in the Y direction.

Now, let us consider the applied field in the Y direction $\vec{\mathbf{F}} \parallel Y$. The field is now perpendicular to the main component of the dipole moment (μ_Z). Note that the field is not applied directly along the Ni-N3 bond, which is slightly out of the XY plane, but rather along the Y axis which is perpendicular to the Z axis. Figure 2.7 shows the evolution of the ground state energy when **F**//Y. Case (b) follows the same parabolic trend as when **F**//Z, the argument remains the same. The situation differs from that of $\mathbf{F} \parallel Z$ in case (a), the evolution is now quadradic rather than linear. The applied field induces a change in the sign of the dipole moment component μ_y which is very small, see Table 2.1. The Stark effect then acts to stabilise the ground state for both signs of the electric field **F**.

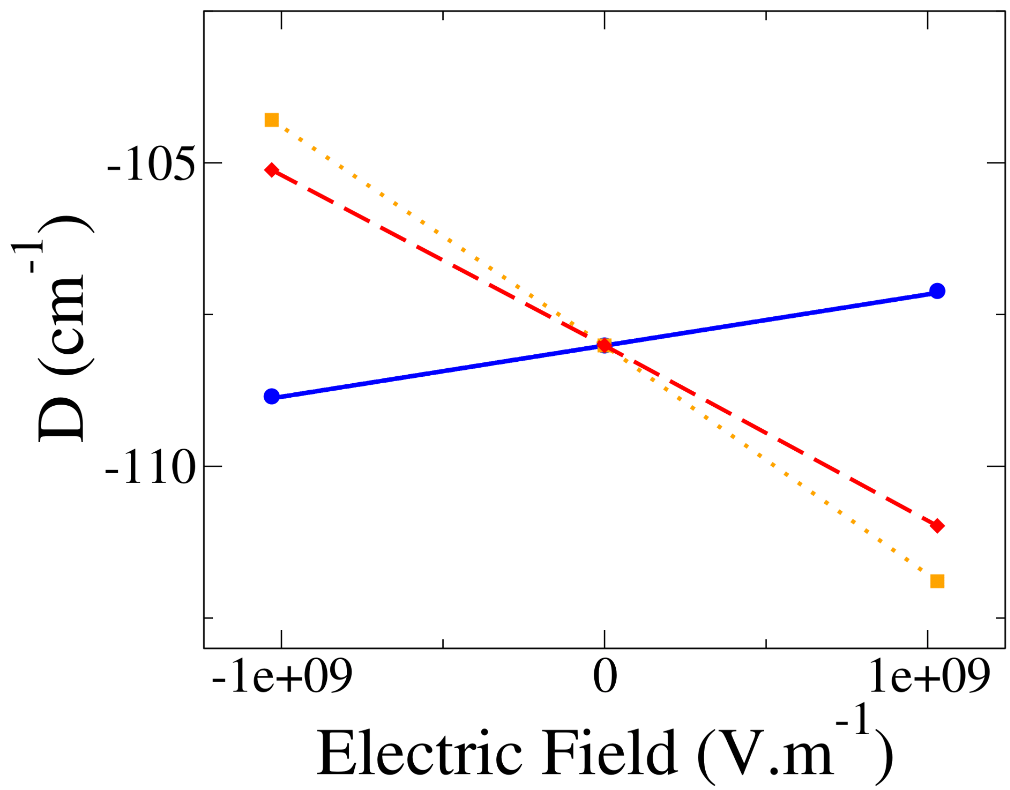


FIGURE 2.8 – Evolution of the axial D parameter (in cm^{-1}) as a function of the field $\vec{\mathbf{F}} \parallel Y$ in cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line)

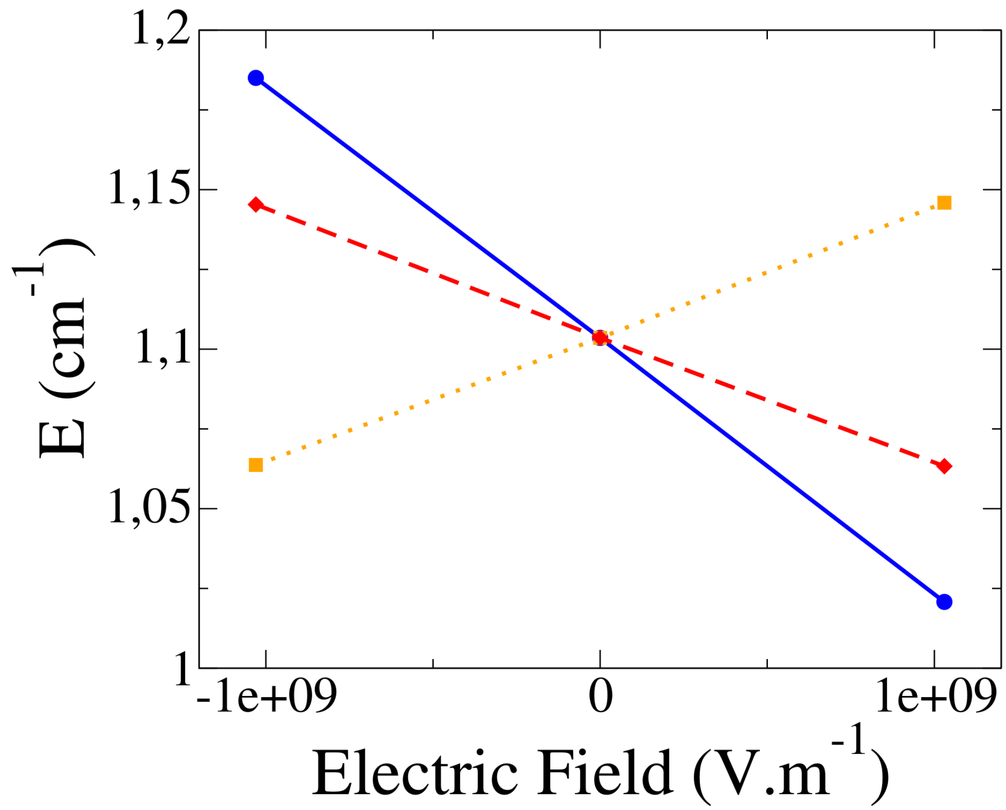


FIGURE 2.9 – Evolution of the rhombic E parameter (in cm^{-1}) as a function of the field $\vec{\mathbf{F}} \parallel Y$ in cases (a) (in blue and plain line), (b) (in orange and dotted line) and (c) (in red and dashed line)

Figure 2.8 shows the evolution of the D term with respect to the applied field. It once again follows a linear behavior in all three cases. The sign of dD/dF for case (a) (but now positive) is still opposite to that of for cases (b) and (c). The slopes for each case are shown in Table 2.7, once again the variation of the D term is dictated by the geometrical changes under the applied field, case (b). Note that the slopes dD/dF remain additive and are now much higher in the $\mathbf{F} \parallel Y$ case than in the $\mathbf{F} \parallel Z$ case, almost seven times larger.

The most significant contributions are reported in Table 2.8. Once again, the contribution that is the most affected by the electric field is that of the first excited state T_1 with good agreement between the ORCA and the variational/perturbative estimations. In case (a), note the increase in the energy difference $\Delta(\Delta E)$. This can be attributed

$Hz/(V.m^{-1})$	(a) Electronic structure	(b) Geometric structure	(c) Both	Sum of (a) and (b)
dD/dF	+25.32	-110.75	-85.45	-85.43
dE/dF	-2.39	+1.20	-1.19	-1.19

TABLE 2.7 – Slopes of the straight lines in cases (a), (b) and (c) for the D and E parameter for $\mathbf{F} \parallel Y$

	Excited state	**C(D) for -F**	**C(D) for +F**	$\Delta C(D)$ ORCA	$\Delta C(D)$ VAR.	$\Delta C(D)$ PERT.	$\Delta\vert\text{SOC}\vert^2$	$\Delta(\Delta E)$
(a)	T_1	-179.99	-178.26	1.73	1.74	2.21	36.05	24.40
	T_2	23.66	24.05	0.39				
	T_3	19.60	19.61	0.01				
	S_3	20.53	20.50	-0.03				
(b)	T_1	-174.84	-183.67	-8.83	-8.76	-11.10	-162.22	-122.40
	T_2	23.91	23.82	-0.09				
	T_3	19.56	19.65	0.09				
	S_3	20.44	20.60	0.16				
(c)	T_1	-175.65	-182.86	-7.11	-7.02	-8.91	-124.89	-98.20
	T_2	23.71	24.02	0.31				
	T_3	19.56	19.65	0.09				
	S_3	20.45	20.59	0.14				

TABLE 2.8 – Contributions to the axial parameter D (in cm^{-1}) of the most contributing excited states and their variations ΔC(D) between the -F and +F in the Y direction either provided by ORCA, calculated variationally and perturbatively. $\Delta|\text{SOC}|^2$ is the variation of the square of the module of the SOC between the ground and the first excited triplet states while Δ(ΔE) is the variation between the -F and +F of their energy difference (in cm^{-1}) obtained at the NEVPT2 level.

to the deformation of the electronic cloud around the Ni(II) ion in the direction of N3 for $+F$. This destabilises the $d_{x^2-y^2}$ orbital, which points directly towards N3 and stabilizes the d_{xy} orbital, resulting in an increase in the energy difference between the two states. The rationalization of the variations for case (b) can be explained by the evolution of the ligand field shown in Table 2.9. The application of the electric field along the Y direction tends to elongate the Ni-N3 bond and concomitantly reduces the other Ni-N bond lengths in the XY plane. The ligand field along the Y axis is therefore reduced stabilising the $d_{x^2-y^2}$ orbital while the d_{xy} orbital, which points directly towards N2 and N4, is destabilised. This results in a decrease in the orbital energy difference $\Delta(\Delta\epsilon(d_{x^2-y^2} - d_{xy}))$, thus decreasing the energy difference between the ground state T_0 and the excited state T_1. According to eq 2.2.1, this leads to an increase in the magnitude $|D|$, which can be observed in Figure 2.8 for case (b). Note that the impact of the electric field on the geometric structure results in the greatest change of magnitude $|D|$ with a slope as large as $-110.75Hz/(V.m^{-1})$. This is due to the large variation in the energy difference $\Delta(\Delta E)$ for case (b) which is one order of magnitude greater than when the field is along the Z axis. Similarly to the $\mathbf{F} \parallel Z$ case, the overall trend in the evolution of the axial anisotropy D is driven by the field's impact on the geometric structure.

Ligands	$-F$	$+F$	$\Delta(distance)$
Cl	2.32024	2.32043	$2.4.10^{-4}$
N1 (apical)	2.14947	2.14727	$1.7.10^{-4}$
N2	2.22518	2.22245	$-27.3.10^{-4}$
N3	2.1482	2.14596	$11.4.10^{-4}$
N4	2.2270	2.22229	$-4.1.10^{-4}$

TABLE 2.9 – Variation of the distances in Angstrom between the Ni(II) ion and the closest atoms of the ligand between $-F$ and $+F$ with $\vec{\mathbf{F}} \parallel Y$.

As for the case where the field is applied in the Z directions, changes in the electronic structure caused by the application of an electric field are once again responsible for the overall variation of E as indicated by Table 2.7. These variations are linear with good additivity but are smaller than observed when $\mathbf{F} \parallel Z$. Figure 2.9 shows that the rhombic parameter E decreases as the field increases, indicating that the difference between the X and Y axes diminishes. This can be explained by the evolution of the weights $w(d_{xz})$ and $w(d_{yz})$ of determinants with a single occupation in the d_{xz} and d_{yz} orbitals, which are found in Table 2.10. In case (a) and (c), the diffence $\Delta(X-Y)$ of the weights between X and Y decreases with the field, which reduces the rhombicity term E. Note that $\Delta(X-Y)$ is positive in case (b) which is consistent with the positive slope observed for geometric structure changes in Table 2.7 and Figure 2.9.

		$-F$	$-F$	$\Delta(\Delta w(X-Y))$
(a)	$w(d_{xz})$	96.603	96.738	
	$w(d_{yz})$	97.191	97.152	-0.174
	$\Delta w(X-Y)$	0.588	0.414	
(b)	$w(d_{xz})$	96.661	96.695	
	$w(d_{yz})$	97.159	97.195	+0.002
	$\Delta w(X-Y)$	0.498	0.500	
(c)	$w(d_{xz})$	96.597	96.762	
	$w(d_{yz})$	97.175	97.170	-0.170
	$\Delta w(X-Y)$	0.578	0.480	

TABLE 2.10 – Weight w(d_{xz}) and w(d_{yz}) of the determinants with single occupation in the respective orbitals, their difference $\Delta w(X-Y)$ and their overall variation $\Delta(\Delta w(X-Y))$ in all three case of calculation for $\vec{\mathbf{F}} \parallel Y$.

2.2.4 Conclusion

The aim of this study was to determine the influence of an external electric field on the anisotropy parameter of the ZFS Hamiltonian in a Ni(II) complex close to first-order spin-orbit coupling regime. A combination of DFT and wave-function theory was employed to evaluate the geometric changes induced by the field and extract the parameter using wave-function based method and effective Hamiltonian theory. Treating the electronic and geometric changes separately allowed us to identify which mechanisms are responsible for the variation in the parameters. Two orientations for the field were explored. Depending on which parameter it affected the most, both orientations proved to be of interest. The findings can be summarised in a few points :

(i) The rhombicity of the system, described by the rhombic term E, can be modulated by the electric field. The variation dE/dF can be significant, especially for $\mathbf{F} \parallel Z$ which was not expected at first. As rhombicity describes the anisotropy between the X and Y directions, it seems reasonable to assume that it would be more affected by the field oriented in the XY plane rather than in the Z direction. However, the most important contributions to the E term were found to come from excited states (T_2 and T_3) coupled to the ground state via excitations from the d_{xz} or the d_{yz} orbitals to the d_{z^2} or the $d_{x^2-y^2}$ orbitals. As these orbitals are mainly oriented along the Z axis, it is therefore natural that these excitations would be more impacted by the field along the Z directions than in the XY plane.

(ii) The electronic and geometric changes have an opposite effect on the variation dD/dF for both field orientations. In case (a) for $\mathbf{F} \parallel Z$, the large increase in magnitude $|D|$ can be related to a large variation of the SOC while the energy difference ΔE between T_0 and T_1 decreases. When $\mathbf{F} \parallel Y$, the variations are governed only the difference in energy ΔE between the two lowest triplets. In case(a), the electronic cloud displacement induced by the electric field tends to stabilise the d_{xy} orbital and to destabilise the $d_{x^2-y^2}$ orbital leading to an increase of ΔE.

(iii) Similarly to rhombicity, it was expected that the greatest change to the axial parameter D would come from the application of the electric field in the Z direction. However, as the changes in D are dictated by the behaviour of the two orbitals that belong to the XY plane, $d_{x^2-y^2}$ and d_{xy}, the most impressive impact comes from the field along the Y direction.

The main message of this work is that the magneto-electric effect can be of considerable magnitude on the ZFS parameters, and that modulation of magnetic properties by the electric field is not necessarily more effective when the field is oriented in the direction of the dipole moment. For both D and E parameters, the nature of the excitations that couple the ground state to the various excited states responsible for the strongest contributions is the determining factor. Finally, we have shown that most of the rationalization of these effects require analysis of the impact of the field on the orbitals involved in these excitations and can be related with the ligand field. These findings, even though they correspond to a particular case, are general which is promising for the design of new complexes whose magnetic properties can be controlled and tuned by the electric field. Finally, the large change on the ZFS parameters we predict can be useful for technological applications in the sense that electric fields may induce detectable magneto-electric effects.

Chapitre 3

Electric field on Exchange anisotropy

Numerous studies have been conducted to characterise the properties of magnetic anisotropies and many procedures have been developed with great success as discussed above. However, only a few studies have been applied to determine the properties of the exchange anisotropy using Quantum Chemistry theories. This study is part of a series of works that aims to rationalise the origin of exchange anisotropy in polynuclear molecules.

The spin Hamiltonian used to describe anisotropic behaviour in binuclear systems is as follows :

$$\hat{H}_{MS} = J_{AB}\hat{\mathbf{S}}_A \cdot \hat{\mathbf{S}}_B + \hat{\mathbf{S}}_A \cdot \overline{\overline{D}}_{AB} \cdot \hat{\mathbf{S}}_B + \mathbf{d}_{AB} \cdot (\hat{\mathbf{S}}_A \times \hat{\mathbf{S}}_B) \tag{3.0.1}$$

To achieve this, a procedure was developped for extracting values of the Dzyaloshinskii-Moriya pseudovector interaction $\mathbf{d_{AB}}$ and the symmetric exchange anisotropy $\overline{\overline{D}}_{AB}$ tensor.[18, 19] It relies on effective Hamiltonian theory in combination with wave-function based method. This procedure was tested on a simple model molecule, shown in Figure 3.1, consisting of two Cu^{2+} cations bond to Cl^- anions.

The reason for choosing this system is its simplicity, with only one magnetic electron per centre and a reasonable size, it allows for simple analytical derivation and high-level treatment of the correlation. Such a toy molecule can also be distorted to assess the impact of geometry, *i.e.* distances and angles, on these interactions. Initially, it was

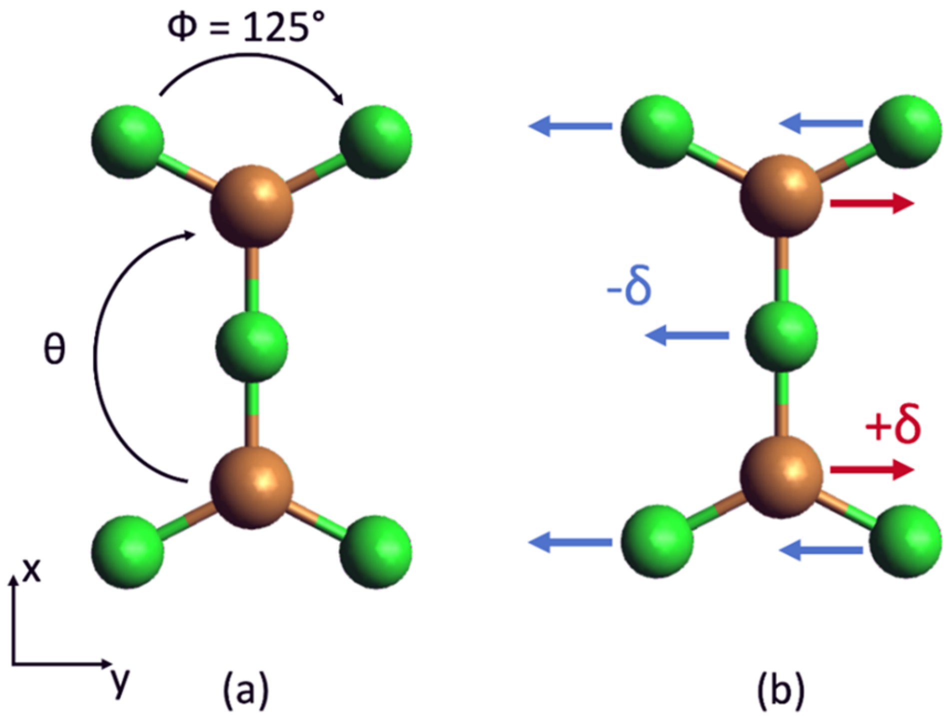


FIGURE 3.1 – (a) Cu_2Cl_5 model molecule of D_{2h} symmetry, (b) displacement δ of the atoms generated by the application of an electric field along the y direction.

determined that the most favourable setting for obtaining large values of the anisotropic interactions is to approach first-order SOC regime without reaching it. It was discussed in the previous chapter that such regime occurs when degeneracy of some of the d-orbitals allows for very strong SO couplings between the ground and excited states. In this system, this situation arises when the outside angle ϕ reaches 120°, at which point the local environment of each Cu^{2+} is close to a C_3 symmetry point group. This allows the condition of quasi-degeneracy of the $d_{x^2-y^2}$ and d_{xy} orbitals to be achieved. However, due to the D_{2h} symmetry of the system, it possesses an inversion centre, leading to a zero value of DMI. To induce a non-zero value of the DMI, the first two studies relied on the deformation of the molecule with a variation of the θ angle, which resulted in the hybridization of the two cartesian d-orbitals mentionned above. This mixing of the d-orbitals is key to creating DMI, as increasing the mixing was shown to allow the obtention of very large values of DMI. It was then thought that applying an external electric field could have a similar effect by acting on the electronic could itself rather than the nuclei's position.[20] As mentionned in the previous chapter, an electric field is an interesting option as it can be focused precisely at the molecular level. Using an external stimulus is also much easier than relying on the "intrinsic" geometry of the molecule and allows a larger number of candidates to be studied. This study focused on the impact of the electric field on the isotropic and antisymmetric exchange parameters of the spin Hamiltonian, it establishes that the Dzyaloshinskii-Moriya pseudo-vector is highly sensitive to the external field. Here, we follow the already established procedure to focus our attention on the symmetric exchange anisotropy tensor. The work proceeds along two axes :

(i) Derive an appropriate model to estimate the symmetric tensor under the condition of 1^{st} order SOC in the absence of an electric field.

(ii) Study how the symmetric tensor varies under the influence of an electric field, and its relation to the apparition of DMI.

3.1 Computational Informations

The electronic and relativistic state computation were performed using the MOLCAS code[48]. CASSCF calculations were carried out for various values of the angle Φ, ranging from $\Phi = 130°$ (close to 1^{st} order SOC) to $170°$ (far from it). The active space consists of six electrons occupying the $d_{x^2-y^2}$ and d_{xy} orbitals of each copper centre, CAS(6,4), with two sets of orbitals optimised in an average way. One for the four lowest triplet states and the other for the four lowest singlet states. An enlarged active space containing all d-orbitals and their electrons, CAS(18,10) was also considered, but a previous study showed that it had little impact on the extracted parameters. The conclusion was that the reduced active space CAS(6,4) was sufficient to capture the effect of first-order SOC. Nevertheless, the CAS(18,10) was used for the treatment of second-order spin orbit coupling. Dynamic correlation was introduced using the DDCI3 method, performed using the CASDI code[49] which is interfaced with the MOLCAS chain. Three types of SO-SI calculations were performed :

(i) CAS(6,4)SCF : the energies and wave-functions are obtained from CASSCF calculation.

(ii) CAS(6,4)DDCI/SCF-SO : the DDCI energies are used while the wave-functions remain that of the CAS(6,4)SCF.

(iii) CAS(6,4)DDCI/DDCI-SO : the energies and wave-functions are obtained from DDCI calculation.

This allows to study the impact of each component of dynamic correlation (energy or wave-function) on the exchange interactions. The TZVP basis set [50] were used throughout all calculations : 5s4p3d1p for Cu and 4s3p1d for Cl. The homogeneous electric field was added in the SEWARD stage of the calculation in MOLCAS, it was applied along the X,Y and Z directions, but most of the results reported here relates to the Y direction. The field values range from 0 $kV.cm^{-1}$ to 5.1724×10^{4} $kV.cm^{-1}$ for the maximum value.

3.2 Theoretical model

In order to derive the parameters of the Multi-Spin Hamiltonian, we must first define the model space. Let $a(a')$ and $b(b')$ be the magnetic orbitals located on each copper ion optimised for the triplet(singlet) state. These orbitals are obtained from state-specific calculations and may differ between triplet and singlet states by the tails on the Cl^- ligands (hence the difference of notation). Crystal field theory indicates that these orbitals should mainly be composed of the $d_{x^2-y^2}$ or d_{xy} orbitals respectively. With the application of an electric field, one can predict that these two orbitals will hybridise to form the orbitals $a(a')$ and $b(b')$. We refer to the orthogonal cartesians orbitals as $a_1(b_1)$ and $a_2(b_2)$ respectively depending wether the site is A or B so that :

$$\begin{aligned} a &= \alpha a_1 + \beta a_2 \qquad & b &= -\alpha b_1 + \beta b_2 \\ a' &= \alpha' a_1 + \beta' a_2 \qquad & b' &= -\alpha' b_1 + \beta b_2 \end{aligned} \tag{3.2.1}$$

where $\alpha(\alpha')$ and $\beta(\beta')$ are the mixing coefficients between the d-orbital in the triplet(singlet) state and a_1, a_2, b_1 and b_2 the $d_{x^2-y^2}$ and d_{xy} orbitals localised on sites A and B respectively. The coupling of the unpaired electron of each copper ion gives rise to a triplet and singlet state that will form the basis of the model space :

$$\begin{aligned}
T^{+} &= |ab| \\
T^{0} &= \frac{1}{\sqrt{2}}(|a\bar{b}| - |b\bar{a}|) \\
T^{-} &= |\bar{a}\bar{b}| \\
S &= \frac{1}{\sqrt{2}}(|a'\overline{b'}| + |b'\overline{a'}|)
\end{aligned} \tag{3.2.2}$$

The matrix representation of the Multi-Spin Hamiltonian in this basis is :

H_{MS}	T^{+}	T^{0}	T^{-}	S
T^{+}	$\frac{J_{AB}}{4} + \frac{D_{zz}}{4}$	$\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$\frac{(D_{xx}-D_{zz}-2iD_{xy})}{4}$	$\frac{d_y+id_x}{2\sqrt{2}}$
T^{0}	$\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J_{AB}}{4} - \frac{D_{zz}}{4} + \frac{(D_{xx}+D_{yy})}{4}$	$-\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$-\frac{id_z}{2}$
T^{-}	$\frac{(D_{xx}-D_{zz}+2iD_{xy})}{4}$	$-\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J_{AB}}{4} + \frac{D_{zz}}{4}$	$\frac{d_y-id_x}{2\sqrt{2}}$
S	$\frac{d_y-id_x}{2\sqrt{2}}$	$\frac{id_z}{2}$	$\frac{d_y+id_x}{2\sqrt{2}}$	$-\frac{3J_{AB}}{4} - \frac{D_{zz}}{4} - \frac{(D_{xx}+D_{yy})}{4}$

The evolution of J_{AB} and of the components d_x, d_y and d_z under the application of an external electric field has been studied previously. The components D_{xx}, D_{xy}, etc. are the components of the symmetric exchange anisotropy $\overline{\overline{D}}$-tensor which is the focus of this study. Its evolution will be compared to previous results, especially those relating to the DMI.

When approaching the condition of quasi-degeneracy of the $d_{x^2-y^2}$ and d_{xy} orbitals, *i.e.* first-order spin orbit coupling, second-order perturbation theory cannot be used to account for spin-orbit coupling due to the small differences in energy between the lowest states. Therefore, a variational treatment is mandatory. To this end, a model was derived to assess the contribution of 1^{st} order SOC based on the spin-free states generated by the occupation of these two orbitals. The following states form the uncoupled basis :

$$
\begin{aligned}
T_1^+ &= |a_1b_1|; \qquad T_1^0 = \frac{|a_1\bar{b}_1| - |b_1\bar{a}_1|}{\sqrt{2}} \\
S_1 &= \frac{|a_1\bar{b}_1| - |b_1\bar{a}_1|}{\sqrt{2}} \\
T_2^+ &= \frac{|a_1b_2| + |b_1a_2|}{\sqrt{2}}; \qquad T_2^0 = \frac{|a_1\bar{b}_2| + |b_1\bar{a}_2| - |a_2\bar{b}_1| - b_2\bar{a}_1|}{2} \\
S_2 &= \frac{|a_1\bar{b}_2| + |b_1\bar{a}_2| + |a_2\bar{b}_1| + b_2\bar{a}_1|}{2} \\
T_3^+ &= \frac{-|a_1b_2| + |b_1a_2|}{\sqrt{2}}; \qquad T_3^0 = \frac{-|a_1\bar{b}_2| + |b_1\bar{a}_2| - |a_2\bar{b}_1| + b_2\bar{a}_1|}{2} \\
S_3 &= \frac{-|a_1\bar{b}_2| + |b_1\bar{a}_2| + |a_2\bar{b}_1| - b_2\bar{a}_1|}{2} \\
T_4^+ &= |a_2b_2|; \qquad T_4^0 = \frac{|a_2\bar{b}_2| - |b_2\bar{a}_2|}{\sqrt{2}} \\
S_4 &= \frac{|a_2\bar{b}_2| - |b_2\bar{a}_2|}{\sqrt{2}}
\end{aligned}
\tag{3.2.3}
$$

For simplicity, the ionic part of these states was left out of the model, but will be taken into account in the *ab initio* calculations as they play an essential role in the antiferromagnetic coupling. Note that, in absence of electric field, crystal field theory indicates that the magnetic orbitals are the $d_{x^2-y^2}$ on each copper center. Thus the energy ordering should follow the order in which they are expressed in Eq. 3.2.3. The spin-orbit matrix can be derived from the SO-Hamiltonian :

$$
\hat{H}^{SO} = \zeta(\hat{l}_1\hat{s}_1 + \hat{l}_2\hat{s}_2) \tag{3.2.4}
$$

with $\hat{l}_i$ the angular momenta and $\hat{s}_i$ the spin momenta operators working on electron i and ζ the spin-orbit constant. The spin-orbit couplings between the states in Eq.3.2.3 originate from the $\hat{l}_z\hat{s}_z$ part of the $\hat{H}^{SO}$ that couples the determinants formed from orbitals of same spatial (m_l) and spin (m_s) component. The representation matrix of the SO Hamiltonian with the addition of electronic couplings is given in the basis developped in Eq. 3.2.3.

The electronic effects included are the isotropic couplings J_i, between triplet and

$\hat{H}^{SO}$	$\vert T_1^+\rangle$	$\vert T_2^+\rangle$	$\vert T_3^+\rangle$	$\vert T_4^+\rangle$	$\vert T_1^0\rangle$	$\vert T_2^0\rangle$	$\vert T_4^0\rangle$	$\vert S_1\rangle$	$\vert S_3\rangle$	$\vert S_4\rangle$
$\langle T_1^+\vert$	$E_1 + J_1$	h_1	$i\zeta\sqrt{2}$	δ_4	0	0	0	0	0	0
$\langle T_2^+\vert$	h_1	$E_2 + J_2$	0	h_2	0	0	0	0	0	0
$\langle T_3^+\vert$	$-i\zeta\sqrt{2}$	0	$E_3 + J_3$	$i\zeta\sqrt{2}$	0	0	0	0	0	0
$\langle T_4^+\vert$	δ_4	h_2	$-i\zeta\sqrt{2}$	$E_4 + J_4$	0	0	0	0	0	0
$\langle T_1^0\vert$	0	0	0	0	$E_1 + J_1$	h_1	δ_4	0	$-i\zeta\sqrt{2}$	0
$\langle T_2^0\vert$	0	0	0	0	h_1	$E_2 + J_2$	h_2	$-i\zeta\sqrt{2}$	0	$-i\zeta\sqrt{2}$
$\langle T_4^0\vert$	0	0	0	0	δ_4	h_2	$E_4 + J_4$	0	$-i\zeta\sqrt{2}$	0
$\langle S_1\vert$	0	0	0	0	0	$i\zeta\sqrt{2}$	0	E_1	h'_1	δ'_4
$\langle S_3\vert$	0	0	0	0	$i\zeta\sqrt{2}$	0	$i\zeta\sqrt{2}$	h'_1	E_3	h'_2
$\langle S_4\vert$	0	0	0	0	0	$i\zeta\sqrt{2}$	0	δ'_4	h'_2	E_4

TABLE 3.1 – Representation matrix of $\hat{H}^{SO}$ and electronic couplings in the uncoupled basis. T_3^0 and S_2 are not coupled to the others, as such they were left out for clarity. $h_1(h'_1)$ and $h_2(h'_2)$ are couplings induced by the electric field and differ between triplet and singlet states by their exchange integrals. δ_4 and δ_4 are small bielectronic integrals between $T_1(S_1)$ and $T_4(S_4)$. The M_S=-1 components of the triplets are left out but possess the same couplings as the M_S=1 components.

singlet states of the same spatial nature. The $h_i(h_i$') couplings are induced by the mixing of the $d_{x^2-y^2}$ and d_{xy} orbitals when the electric field is applied along the Y direction. This coupling gives rise to the DMI in this system, as it opens coupling paths between S_1 and T_1^0. The couplings can be estimated perturbatively at second-order, but as the system enters 1^{st} order SOC the best approach is to diagonalise the matrix, *i.e.* following a variational treatment. Anisotropic exchange is described by a symmetric tensor of rank two with six different components in a random set of axis. When expressed in its proper axis, it is reduced to simply two components, an axial parameter D_{AB} and rhombic term E_{AB}. In order to understand the origins of the anisotropic parameter, D_{AB} and the DMI, the couplings must be decomposed according to their contributions. Put simply, the axial anisotropy describes the difference in energy between the M_S=± 1 and the M_S=0 components. Within this model space, there is no difference in the couplings between the two M_S=±1 components of this matrix. As such there is no rhombicity, when considering only first-order SOC. Hence to estimate the D_{AB}, one has to look for states coupled to either T_1^0 or T_1^+ (T_1^- will not be mentionned but behaves the same). The T_1^+ component is coupled via SOC to T_3^+ which stabilises it, bringing a negative contribution to D_{AB} by lowering the energy of the M_S=±1 components without affecting the T_1^0 component. On the other hand, the S_3 which is coupled to T_1^0 will contribute positively to D_{AB}. One way to assess these contributions is by diagonalising the matrix with only the relevant couplings, T_3^+ or S_3, and see how the energies of T_1^+ or T_1^0 are affected. The small bielectronic integrals δ_4 and δ_4' also allow the contribution of T_4^+ and S_4 but with a much smaller impact.

In the presence of an electric field applied in the Y direction, the situation of Table 3.1, the h_i and h_i' electronic coupling are no longer zero and new pathways open up for the DMI. As shown in matrix 3.2, the d_z component is created by the coupling between the T_1^0 and the S_1 states which can be expressed using second-order perturbation theory :

$$\frac{id_z}{2} = \langle S| H^{SO} |T^0\rangle^{(0)} \tag{3.2.5}$$

$$= \frac{\langle S_1| H^{SO} |T_2^0\rangle \langle T_2^0| H^{el} |T_1^0\rangle}{E_2 + J_2 - E_1} + \frac{\langle S_1| H^{el} |S_3\rangle \langle S_3| H^{SO} |T_1^0\rangle}{E_3 - E_1} \tag{3.2.6}$$

This equation is only valid far from first-order SOC regime but helps visualising how the DMI appears due to a combination of electronic and spin-orbit couplings.

For a full extraction, one must also consider the effect of higher in energy states formed from determinants with the d_{xz} or d_{yz} orbitals, noted as $a_3(b_3)$ and $a_4(b_4)$, singly occupied.

$$\begin{aligned}
T_5^+ &= \frac{-|a_1b_3| + |b_1a_3|}{\sqrt{2}} \qquad T_5^0 = \frac{-|a_1\bar{b}_3| + |b_1\bar{a}_3| - |a_3\bar{b}_1| + |b_3\bar{a}_1|}{2} \\
S_5 &= \frac{-|a_1\bar{b}_4| + |b_1\bar{a}_3| + |a_3\bar{b}_1| - |b_3\bar{a}_1|}{2} \\
T_6^+ &= \frac{-|a_1b_4| + |b_1a_4|}{\sqrt{2}} \qquad T_6^0 = \frac{-|a_1\bar{b}_4| + |b_1\bar{a}_4| - |a_4\bar{b}_1| + |b_4\bar{a}_1|}{2} \\
S_6 &= \frac{-|a_1\bar{b}_4| + |b_1\bar{a}_4| + |a_4\bar{b}_1| - |b_4\bar{a}_1|}{2}
\end{aligned} \tag{3.2.7}$$

These states are coupled with T_1 and S_1 through the $\hat{l}_+\hat{s}_- + \hat{l}_-\hat{s}_+$ operator in $\hat{H}^{SO}$. Note that the determinants formed from the d_{z^2} orbital can be omitted because it cannot interact with either T_1 or S_1 via $\hat{H}^{SO}$. Taking these states into account enable to differentiate the two M_S=$\pm$1 of the triplet state and introduce rhombicity.

3.3 Impact of First-Order Spin-Orbit Coupling in the absence of electric field

Expressions for the anisotropic parameters have already been derived using second-order perturbation theory and involve the $T_5(S_5)$ and $T_6(S_6)$ states :

$$\begin{aligned} D_{AB} &= 4\frac{\zeta^2 J_3}{\Delta E_3^2} - \frac{1}{2}(\frac{\zeta^2 J_5}{\Delta E_5^2} + \frac{\zeta^2 J_6}{\Delta E_6^2}) \\ E_{AB} &= \frac{1}{2}(\frac{\zeta^2 J_5}{\Delta E_5^2} - \frac{\zeta^2 J_6}{\Delta E_6^2}) \end{aligned} \tag{3.3.1}$$

where $\Delta E_i^2 = \sqrt{\Delta T_i \Delta S_i}$ is the geometric mean of the excitation energies between the states T_1 and T_i or S_i and $J_i = E_{T_i} - E_{S_i}$ is the isotropic coupling between excited states of same spatial nature. As the first term of D_{AB} in eq 3.3.1 involves a state that contributes to 1^{st} order SOC, its estimation may deteriorate as ϕ decreases, *i.e.* entering first-order regime. This renders the equations valid only far from first-order regime and in the absence of external field. Note that the rhombic term requires the inclusion of the higher states for it to be different from zero.

Performing a rigorous extraction of the anisotropic parameters would require including all the excited states generated from single excitations within the d-orbital space, *i.e.* CAS(18,10). As this work focus solely on the contribution of 1^{st} order SOC, a CAS(6,4) is sufficient. To assess the impact of dynamic correlation on the D_{AB} parameter, calculations were performed with gradual inclusion of the DDCI3 corrections (energy and wave-function). These results are reported on Figure 3.2 with the angle ϕ ranging from 130° to 170°.

This shows that the magnitude $|D_{AB}|$ increases when entering 1^{st} order SOC, *i.e.* a decrease of the angle ϕ. A similar observation has been made for the DMI in previous study and in mononuclear complexes for local ZFS parameter D. It is important to

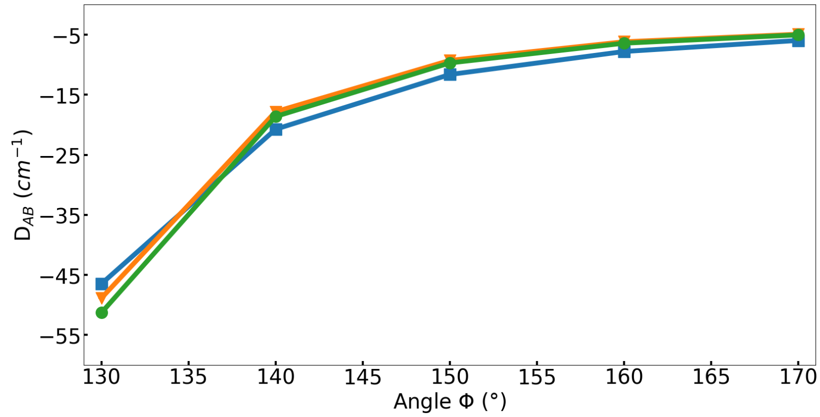


FIGURE 3.2 – D_{AB} as a function of the outside angle ϕ for (i) CAS(6,4)SCF/SCF-SO level (blue line with squares), (ii) CAS(6,4)DDCI/SCF-SO level (orange line with triangles), and (iii) CAS(6,4)DDCI/DDCI-SO level (green line with circles)

note that far from first-order SOC, the impact of electron correlation is minimal, with only slight variations between the three curves. This difference increases as the angle ϕ decreases. It should be noted that the curves (ii) and (iii), which differ in terms of the wave-function used (CASSCF-SO or DDCI-SO) are very similar and that the inclusion of the DDCI wave-function does not improve the results much unlike what was observed with the DMI. Therefore, the use of the correlated DDCI wave-functions in addition to the DDCI energies is not required and CASSCF wave-function should suffice for the SO-SI calculation in the absence of DMI.

Far from first-order SOC limit, the Eq. 3.3.1 can be used with the large active space CAS(18,10) taking into account the states from Eq. 3.2.7. At ϕ = 170°, the perturbative espressions give D_{AB}^{pert}=-4.86cm^{-1} and E_{AB}^{pert}=0.18cm^{-1}, which is a good estimate when compared to the value extracted from the CAS(18,10)DDCI/DDCI-SO and effective Hamiltonian theory D_{AB}=-4.09cm^{-1} and E_{AB}=0.10cm^{-1}. The slight ove-restimation may be due to the fact that a unique $\zeta = 830cm^{-1}$ value of the free ion Cu^{2+} was used,[42] in a complex this value is bound to decrease. As expected, these expressions lose their quality when entering first-order SOC, $\phi = 130°$, D_{AB}^{pert}=-83.31cm^{-1} and E_{AB}^{pert}=0.21cm^{-1} compared with the values extracted from CAS(18,10)DDCI/DDCI-SO and effective Hamiltonian theory D_{AB}=-45.20cm^{-1} and E_{AB}=0.02cm^{-1}. The rhombicity is negligible justifying the use of the smaller active space CAS(6,4) compared to the CAS(18,10).

$\hat{H}^{SO}$	$\|T_1^+\rangle$	$\|T_2^+\rangle$	$\|T_3^+\rangle$	$\|T_4^+\rangle$	$\|T_1^0\rangle$	$\|T_2^0\rangle$	$\|T_4^0\rangle$	$\|S_1\rangle$	$\|S_3\rangle$	$\|S_4\rangle$
$\langle T_1^+\|$	1585	0	1079i	12	0	0	0	0	0	0
$\langle T_2^+\|$	0	2362	0	0	0	0	0	0	0	0
$\langle T_3^+\|$	-1079i	0	3275	1079i	0	0	0	0	0	0
$\langle T_4^+\|$	12	0	-1079i	5036	0	0	0	0	0	0
$\langle T_1^0\|$	0	0	0	0	1585	0	12	0	-1078i	0
$\langle T_2^0\|$	0	0	0	0	0	2362	0	-1075i	0	-1076i
$\langle T_4^0\|$	0	0	0	0	12	0	5036	0	-1078i	0
$\langle S_1\|$	0	0	0	0	0	1075i	0	550	0	2
$\langle S_3\|$	0	0	0	0	1078i	0	1078i	0	3369	0
$\langle S_4\|$	0	0	0	0	0	1076i	0	2	0	4486

TABLE 3.2 – Numerical matrix of $\hat{H}^{SO}$ and electronic couplings in the uncoupled basis from a CAS(6,4)DDCI/DDCI-SO calculation with no electric field and $\phi = 130°$.

The matrix 3.1 was extracted from *ab initio* CAS(6,4)DDCI/DCCI-SO calculation at $\phi = 130°$ and in the absence of an electrical field. Its numerical matrix is shown in Table 3.2, which shows a good correspondence. Note that the coupling terms h_i are equal to zero making it impossible for DMI to occur. Diagonalising this matrix with or without the inclusion of T_3^+ and S_3 allows to determine their individual contributions to the D_{AB}, which are reported in Table 3.3. The sum of these contributions matches perfectly with the value obtained via H^{eff}.

Note that from the final value of D_{AB} and the perturbative expression 3.3.1, it is clear that the main contribution to the D_{AB} comes from T_3 and S_3 even with the large active space CAS(18,10). This justify once more the use of the small CAS(6,4) calculations.

	D_{AB}
Contribution of T_3^+ and T_3^-	-1176.74
Contribution of S_3	1124.28
Sum	-52.46
D_{AB} from H^{eff}	-52.46

TABLE 3.3 – Variational contribution in cm^{-1} of the excited states to D_{AB} and its value in cm^{-1} extracted from effective Hamiltonian at CAS(6,4)DDCI/DDCI-SO level

3.4 Impact of the electric field on the symmetric anisotropic exchange parameter

This section focuses on the behaviour of the anisotropic parameter in first-order spin orbit coupling, at $\phi = 130°$, with the application of an external electric field. Only changes in the electronic structure are studied, not the geometric ones. Indeed a previous study showed that the geometric changes only affect the values of the parameters and not their physical origin.

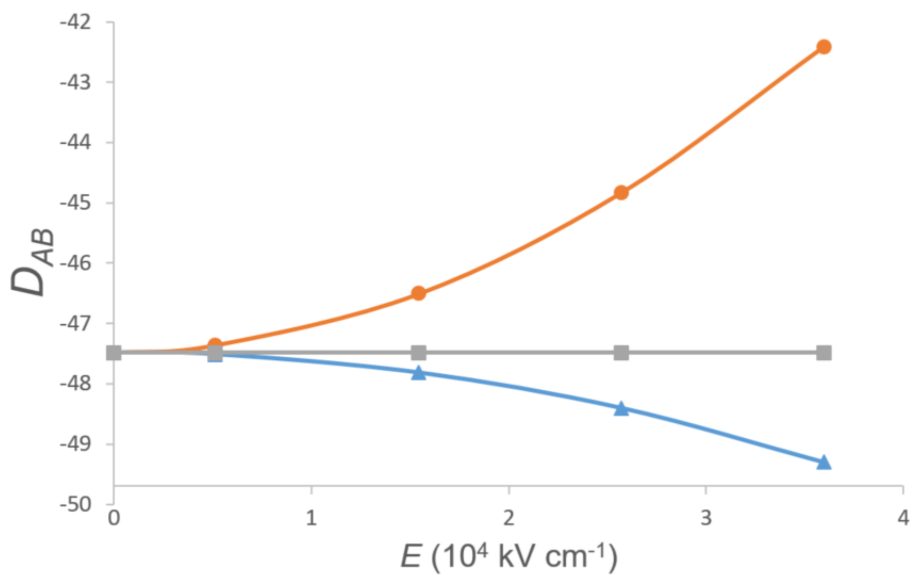


FIGURE 3.3 – D_{AB} (in cm^{-1}) as a function of the electric field applied in the X (blue line with triangles), Y (red line with circles) and Z (grey line with squares) directions, obtained at CAS(6,4)SCF/SCF-SO

To obtain an initial estimate of the impact of the electric field on the D_{AB} parameter, CAS(6,4)SCF/SCF-SO calculations were performed with the electric field applied in the three cartesian directions (the axes are represented on Figure 3.1). The results are shown in Figure 3.3, a slight variation is observed when the electric field is applied along the X axis, but no change when it is applied along the Z direction. However, when the electric field is applied in the Y direction, a significant variation is observed. This behaviour was also observed in the study of the DMI where the largest value was obtained with the electric field along the Y direction. Therefore, this direction of application was chosen for further investigation as it showed the greater change in the two parameters. The impact of electron correlation is shown on Figure 3.4 with the evolution of the D_{AB} parameter under an electric field. It shows a large impact of both correlated energies and wave-functions as the field increases. The importance of dynamic correlation is expected, because as the field increases so does the DMI value making the use of DDCI energies and wave-function, *i.e.* CAS(6,4)DDCI/DDCI-SO, mandatory as shown in previous studies.

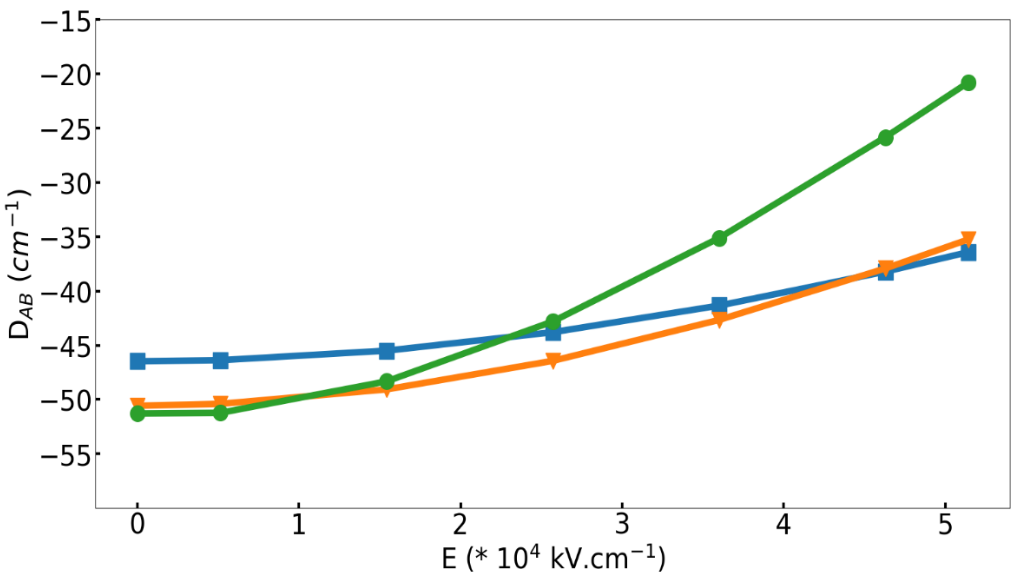


FIGURE 3.4 – D_{AB} (in cm^{-1}) as a function of the electric field applied in the Y direction, obtained at (i)CAS(6,4)SCF/SCF-SO (blue line with squares), (ii) CAS(6,4)DDCI/SCF-SO (orange line with triangles) and (iii) CAS(6,4)DDCI/DDCI-SO (green line with circles)

Applying the electric field in the Y direction results in a decrease of the axial parameter magnitude $|D_{AB}|$. To identify the cause of this decrease, the numerical matrix 3.4 was extracted.

Note that now, the h_i terms are no longer zero, allowing DMI via the coupling from eq 3.2.5. The same diagonalisation procedure as in section 3.3 is applied, selecting states that may contribute to either the D_{AB} or the DMI. This enables the impact of the electric field on the axial parameter and its behaviour in the presence of DMI to be determined. The decomposition of the different contributions is reported in Table 3.5.

$\hat{H}^{SO}$	$\|T_1^+\rangle$	$\|T_2^+\rangle$	$\|T_3^+\rangle$	$\|T_4^+\rangle$	$\|T_1^0\rangle$	$\|T_2^0\rangle$	$\|T_4^0\rangle$	$\|S_1\rangle$	$\|S_3\rangle$	$\|S_4\rangle$
$\langle T_1^+\|$	1580	-201	1078i	12	0	0	0	0	0	0
$\langle T_2^+\|$	-201	2394	0	203	0	0	0	0	0	0
$\langle T_3^+\|$	-1078i	0	3294	1077i	0	0	0	0	0	0
$\langle T_4^+\|$	12	203	-1077i	5081	0	0	0	0	0	0
$\langle T_1^0\|$	0	0	0	0	1580	-201	12	0	-1078i	0
$\langle T_2^0\|$	0	0	0	0	-201	2394	203	-1075i	0	-1075i
$\langle T_4^0\|$	0	0	0	0	12	203	5081	0	-1077i	0
$\langle S_1\|$	0	0	0	0	0	1075i	0	556	202	1
$\langle S_3\|$	0	0	0	0	1078i	0	1077i	202	3369	-207
$\langle S_4\|$	0	0	0	0	0	1075i	0	1	-207	4541

TABLE 3.4 – Numerical matrix of $\hat{H}^{SO}$ and electronic couplings in the uncoupled basis from a CAS(6,4)DDCI/DDCI-SO calculation in the presence of an electric field of 0.01 a.u (5.1724×10^4 kV.cm^{-1}) and $\phi = 130°$.

The contributions from the $T_3^{\pm 1}$ and S_3 as well as the sum of the two, are comparable to the previous case in the absence of electric field. The effect of the electric field is mainly evident in the h_1 and h_1' terms, which couple the components of the T_1 and T_2 states, thereby enabling DMI to emerge. These two couplings have large contributions to the $|d_{AB}|$ term, accounting for 99% of the value of the DMI parameter extracted from H^{eff}. On the other hand, these couplings have positive contributions to the D_{AB}, leading to a decrease in its magnitude $|D_{AB}|$. This demonstrates that there is an interferential behaviour between the two anisotropic parameters when the field is applied in the Y direction. This behaviour is not observed for other orientations. When the field is applied along the X direction, the D_{AB} term is affected (see Figure 3.3) while the d_{AB} remains zero as previously shown. Along the Z direction, it is the D_{AB} parameter that stays constant while a d_{AB} term can be observed, albeit much smaller.

	D_{AB}	$\|d_{AB}\|$
Contribution of T_3^+ and T_3^-	-1162.0	0
Contribution of S_3	1111.4	0
Sum	-50.6	0
Contribution of h_1'	-7	-86.6
Contribution of h_1	29.3	329.7
Contribution of h_2'	5.9	29.1
Contribution of h_2	-0.4	-24.0
Total sum	-22.8	248.2
Parameters from H^{eff}	-21.03	245.0

TABLE 3.5 – Variational contribution in cm^{-1} of the excited states to D_{AB} and $|d_{ab}|$ given in value in cm^{-1} extracted from effective Hamiltonian at CAS(6,4)DDCI/DDCI-SO level

3.5 Conclusion

This work studied the behaviour of the symmetric anisotropy parameters in a binuclear Cu(II) model complex, identifying different mechanisms as the source of their variations. The first part of the study focused on the impact of first-order coupling on the axial D_{AB} and rhombic E_{AB} parameters under certain geometry conditions. It showed that as the local environment of each metallic centre approaches a C_3 symmetry, the condition for first-order SOC are met, allowing for large values of axial anisotropy. Initially, parameters were extracted perturbatively using previously derived expressions. However, this method of extraction proved to be no longer valid in first-order SOC regime and a variational approach was devised instead. This approach relies on the diagonalisation of the electronic and spin-orbit matrix in the basis of the three M_S components of the four triplet and four singlet states generated by a reduced active space consisting of the two highest d-orbitals. In this first part, the importance of the use of dynamically correlated wave-function during the extraction was determined. It showed that in the absence of DMI terms, reasonnable values of the D_{AB} parameter can be obtained using the non-dynamically correlated CASSCF wave-functions using corrected energies during the SO-SI step of the calculation. Once a DMI term appears, this is no longer true and the use of both the correlated energies and wave-functions are necessary for a rigorous extraction. The second part introduces the application of an external electric field, focusing on its impact on the axial parameter. As with the DMI, it was demonstrated that the D_{AB} can also be influenced by an electric field and could contribute to the magneto-electric coupling. Three orientations of the electric field were explored and it was shown that, in one instance an interferential behaviour could be observed between the D_{AB} and d_{AB} terms. Analysis of the numerical matrices showed that the coupling at the DMI source for this field orientation also impacted the magnitude of the axial parameter : when the former increased, the latter decreased.

This work complements a series of studies attempting to define a recipe for obtaining large values for the parameters of both types of anisotropy (symmetric and antisymmetric). Two key elements have been identified as necessary in order to obtain such results :

(i) geometries around the metal must allow the near-degeneracy of d-orbitals to approach first-order SOC. This can be achieved with geometries containing symmetries such as a C_3 axis.

(ii) hybridisation of the degenerate orbitals must be possible via the application an electric field. The choice of orientation of the field leads different type of hybridisation due to the deformation of the electronic cloud.

These considerations could pave the way for the design of compounds with strong magneto-electric coupling.

Chapitre 4

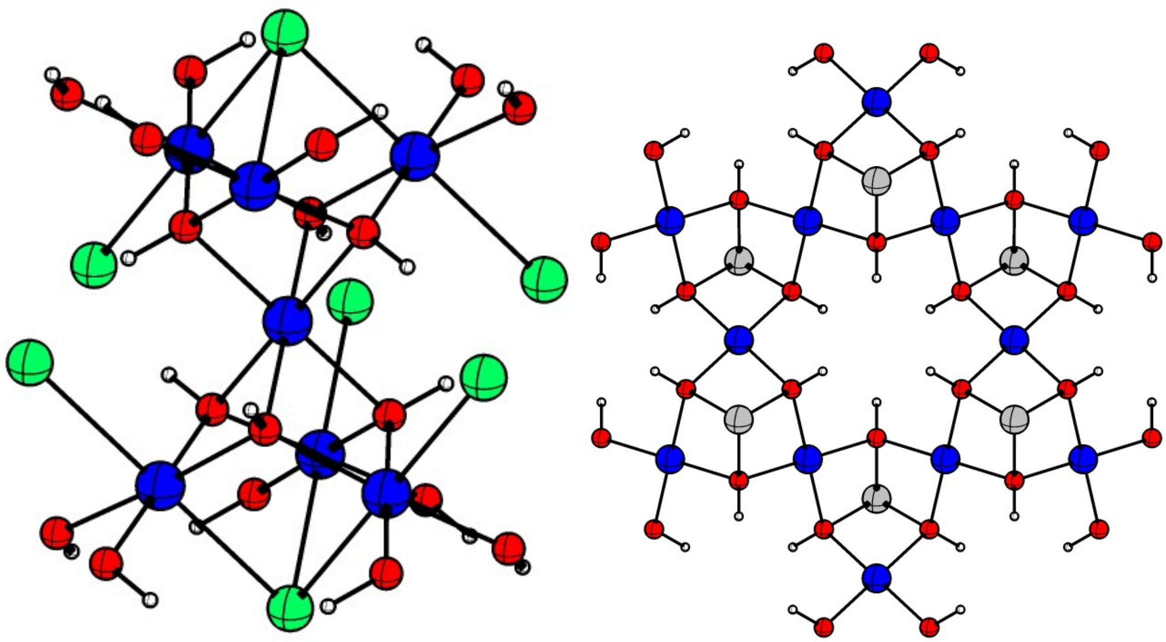
Herbertsmithite

The following work is part of an effort to open up quantum chemistry methods to more theoretically oriented physical systems, in particular to the frustrated magnetism field. This work applies the previously presented methods to Quantum Spin Liquids (QSLs), a subject of long-standing interest to theorists. This concept was first introduced by P.W Anderson[51] in 1973, in an attempt to indentify the ground state in a triangular lattice of spin-1/2 with an antiferromagnetic Heisenberg type interaction. The ground state of such a system is far from trivial, the infinite number of spin orientation configurations all lead to spin frustration. Consequently, unlike more conventional antiferromagnetic systems, the spins do not order magnetically at low temperatures. This absence of magnetic order would not come from thermal fluctuations but from purely quantum ones. Contrary to paramagnets, the disorder would be a direct consequence of highly correlated quantum states. The picture of magnetic disorder is a good starting point to define a QSL, but it remains imperfect. Several other properties have been predicted as occurring in such systems, namely fractionalised excitations and long-range quantum entanglement. These properties can be used to refine the definition. Such concerns are beyond the scope of this work, but should serve as an explanation to the challenges posed by the study of QSLs. Another obstacle to characterising such objects is the difficulty of obtaining reliable experimental data on real systems.

One of the lattice in which QSL behaviour is suspected to occur is called Kagomé lattice. It consists of corner-sharing triangles, with metallic ions occupying the vertices.

This work focuses solely on Herbertsmithite which has long been considered to be a perfect realisation of such a lattice with an antiferromagnetic first-neighbour interaction. In a first approximation, it can be described by one of the simplest Hamiltonians : the Heisenberg model, which consists in magnetic exchange couplings of S = 1/2 spins on neighboring sites of the Kagomé lattice. Even at the classical level, this model remains unordered down to zero temperature (classical spin liquid), with an extensive ground state degeneracy. At the quantum level, the exact nature of the S = 1/2 Kagomé antiferromagnet is today still an open question, which has stimulated many theoretical and numerical studies (high temperature series expansions [52, 53], exact diagonalizations[54], tensor network methods[7, 8], mean-field approaches [55], variational studies [56-58]...). These methods describe the collective effects and aim to identify the characteristics of the ground state. Back to Herbertsmithite, the excitement surrounding the discovery of a compound that realises such an interesting theoretical model has led to increasingly advanced synthesis methods (high-quality crystals) and measurements (specific heat under high magnetic fields [59], NMR magnetic susceptibility [60, 61], thermal conductivity, magnetic structure factors via neutron scattering [62]). While magnetic susceptibility measurements show no evidence of magnetic ordering at low temperatures with a nearest-neighbor interaction $J \approx 180K$[63], some deviations from the behaviour of an ideal S=1/2 Heisenberg antiferromagnet have been observed. This has motivated numerous new studies, from both theory and experiment, to identify the sources of these deviations from the model. Two possibilities were investigated, the first was that more than one isotropic coupling should be taken into account in the HDvV Hamiltonian. Long-range in-plane couplings (beyond nearest-neighbours) have been mentioned as a possible source, but more importantly the presence of defects in the structure with magnetic impurities has been considered. From an experimental point of view, it is difficult to distinguish the contribution of the latter from the inherent physics of the Kagomé lattice. The second possibility is the presence of exchange anisotropy in the system, which was confirmed by an ESR study[64] that took into account the Dzyaloshinskii-Moriya couplings. Both of these avenues present a real challenge to theoretical physicists, but we believe that theoretical chemists may be able to provide answers. Although unable to reproduce the collective effects of the crystal, methodologies have been developed using quantum chemistry theories to extract the local interactions and validate models. This work aims to do exactly that. The combinated use of embedded cluster method and DFT or wave-function methods allows the estimation of multiple isotropic couplings. The deformation of the structure following the consideration of defects will be studied using DFT for geometry optimisation. And finally, anisotropic exchange interactions will be extracted using the procedure detailed in the previous chapter.

Crystal Structure



(a) "4+2" structure　　　(b) "2+4" structure

FIGURE 4.1 – Crystal structure of Herbertsmithite from side (left) and top view (right). Left : Local coordination environment of intralayer $Cu_3(OH)_6$ triangles and interlayer Zn^{2+} ion. Right : $Cu_3(OH)_6$ Kagomé lattice from top view (Cl^- ions are hidden). Copper in blue, Zinc in grey, Oxygen in red, Chlorine in green and Hydrogen in white

Herbertsmithite has the formula $ZnCu_3(OH)Cl_2$, its crystal stucture is trigonal rhombohedral (space group $R\overline{3}m$) is represented in Figure 4.1. The Kagomé lattice sites are occupied by Cu^{2+} ions which forms CuO_4 plaquettes with the bridging oxygens. Each Kagomé planes are separated by Cl^- and Zn^{2+} ions forming distorted ZnO_6 octahedra. Note that the hydroxyde groups, and Zn^{2+} ions, alternate above and below the Kagomé plane. Each Kagomé plane is rotated 120° relative to the others, forming a three-layer stacking pattern.

4.1 Computational Informations

The structure studied was taken from a X-ray study[21], the Hydrogen position were optimised using periodic DFT with PBE functional [65] implemented in the Quantum Espresso code [66].

Density functional theory calculations were performed using the ORCA package [43] with def2-TZVP basis set[44] on all atoms and the ωB97X-D3 functional[67] which has shown to provide very good value of magnetic couplings[68]. Wave-function based method calculations were carried out with the MOLCAS[48] and CASDI[14] codes with the extended ANO-RCC basis set[69, 70] (6s5p3d2f for Cu and Zn, 4s3p2d for O, 4s3p1d for Cl and 2s1p for H)

The embedding's pseudopotential were set to the SDD Effective Core Potentiel (ECP) for ORCA[43] calculations and *ab initio* model potentials (AIMP) for MOLCAS calculations. The quality of the embedding was checked by comparing the value of J_1 using the B3LYP functional[47] for embedded cluster and periodic calculations (using the Crystal code[71]). The J_1 values are in perfect agreement : 240K for periodic and 239K for embedded cluster. This demonstrates the adequacy of the embedding adopted in this study even if, as we will see below, B3LYP slightly overestimates the coupling intensity.

4.2 Density-Functional Theory study

In a first approximation, the isotropic behavior of Herbertsmithite is described by a unique nearest-neighbor coupling. The highly-correlated nature of such systems suggests the possibility of longer range interaction between copper ions separated by one or more neighbors. To explore this possibility, the need to compute large fragments arises making the application of wave-function based method complicated. First the number of Cu^{2+} into the fragment has a direct impact on the size of the active space, even restricting to one orbital/one electron per copper centres leads to convergence difficulties in the calculations. Also, the number of determinants and spin states generated prohibits the use of post-CASSCF methods which are crucial for a good estimation of magnetic couplings based on wave-function methods. Finally, the number of spin states render an analytical derivation complicated with large matrices to handle. As such, Density Functional Theory provides a good middle ground to compute large systems. It also presents some disadvantages, the DFT wave-function is monodeterminantal and usually not an eigenfunction of the $\hat{S}^2$ operator which prevents the mapping on the HDvV Hamiltonian. Nevertheless, the combinated use of Broken-Symmetry DFT (BS-DFT) and the Ising Hamiltonian provides a good way to obtain an estimation of the couplings in the system. The Broken-Symmetry approach relies on the computation of DFT solutions where the M_S value of spin moment of each copper ion of the fragment have been imposed beforehand to $\pm\frac{1}{2}$. It is then possible to obtain multiple solutions per fragments to make a connection between the energy differences of each solution and the Ising Hamiltonian energies. Futhermore, the time necessary for a DFT calculation and for the derivation of its analytical solution allows for the exploration of different properties of the embedding such as the size of the fragments and its overall quality.

4.2.1 In-plane couplings

In a first description, a fragment presenting only the nearest-neighbor coupling was introduced with three copper ions as shown on Figure 4.2(a). Four solutions can be computed with $M_S \geq 0$, first a $M_S = 3/2$ solution where all magnetic moment of the Cu^{2+} ions are aligned creating a fully ferromagnetic solution. Then, three other solutions are possible by switching the M_S value of one of the Cu^{2+} ion to $-1/2$. If the choice of fragment and construction of the embedding are done correctly, this should lead to three equal energy differences as the system presents a C_3 symmetry axis.

Let us call the highest occupied orbital of each copper ion a, b and c. The Slater

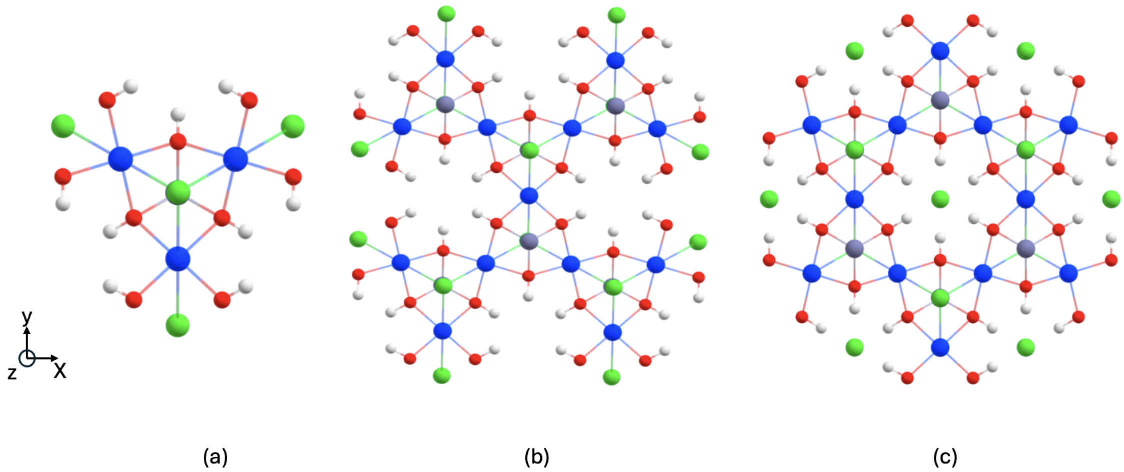


FIGURE 4.2 – The three fragments (based on 3, 13 and 12 Cu(II) centers, from left to right) used in DFT calculations of the in-plane couplings. Cu are represented in blue, Zn in grey, Cl in green, O in red and H in white.

determinant describing the high spin solution (M_S=+1/2 on each Cu^{2+}) can then be written $|abc|$ while the low spin solutions (one of the M_S value is switched to $-1/2$) are written $|ab\bar{c}|$,$|a\bar{b}c|$ and $|\bar{a}bc|$. Applying $\hat{H}_{Ising} = \sum_{i,j} J_1 \hat{S}_{i,z}\hat{S}_{j,z}$ to the high spin (HS) solution $|abc|$ gives the eigenvalue $\frac{3}{2}J_1$ while the application on the low spin (LS) solutions gives the eigenvalue $\frac{1}{2}J_1$. Note that in this work, the convention most commonly used by physicists is employed with a plus sign in front of the Hamiltonian. A positive value of J indicates antiferromagnetism while negative indicates ferromagnetism. The Ising Hamiltonian's matrix representation is diagonal :

H_{Ising}	$\|abc\|$	$\|ab\bar{c}\|$	$\|a\bar{b}c\|$	$\|\bar{a}bc\|$
$\|abc\|$	$\frac{3}{2}J_1$	0	0	0
$\|ab\bar{c}\|$	0	$\frac{1}{2}J_1$	0	0
$\|a\bar{b}c\|$	0	0	$\frac{1}{2}J_1$	0
$\|\bar{a}bc\|$	0	0	0	$\frac{1}{2}J_1$

TABLE 4.1 – Matrix representation of the Ising Hamiltonian in the basis of determinants of M_S=1/2 and M_S=3/2

The value of J_1 can then be extracted from the energy difference between the high spin and low-spin solutions :

$$J_1 = E_{HS} - E_{LS} \tag{4.2.1}$$

The same procedure of extraction can now be applied to larger fragments allowing the evaluation of couplings between more distant magnetic centres. The second fragment created included nine copper ions which allows for the consideration of two new couplings, J_2 and J_3 which are represented on Figure 4.3 and whose value can be found in Table 4.2. It indicates that these couplings are small ($|J_2| \approx |J_3| \approx 1cm^{-1}$ each) with a large standard deviation. While these results provide good insight on the nature of these two couplings, the extraction is unsatisfactory. The source of this deviation is believed to come from the environment of Cu(II) ions taking part in the interaction. The extraction of these couplings requires to flip the spin of at least one of the Cu(II) ion at the edge of the fragment whose neighboring Cu(II) ions are not always explicitely computed (treated with ECP's). This has a significant impact on the convergence which leads to large variation relative to the small values obtained. It was decided to increase the size of the fragment to twelve and thirteen copper ions (fragment (c) and (b) from Figure 4.2) so that there would be ways to account for these interactions between magnetic centers far from the edges and thus weakly affected by the substitution of atoms by ECPs.

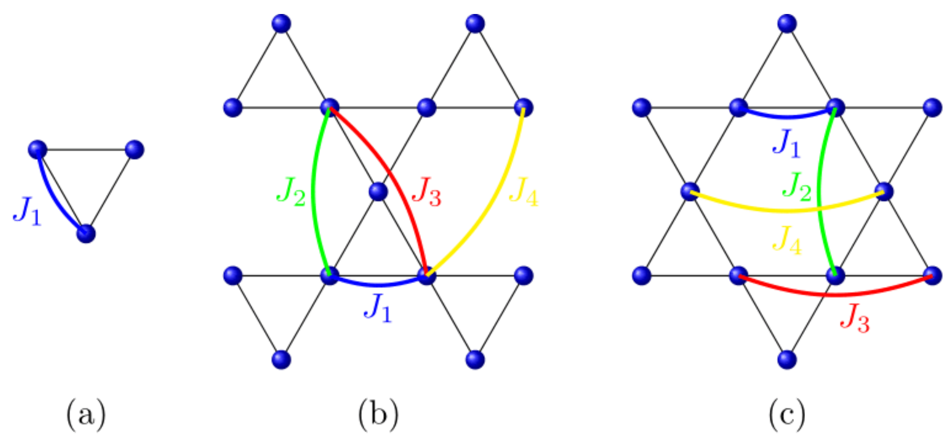


FIGURE 4.3 – Schematic representation of the fragments and the couplings introduced

The results obtained for the couplings in the three fragments are presented in Table 4.2. The first thing to comment is the good transferability with a small relative deviation from one cluster to the other. Moreover, the nearest-neighbor interaction J_1 obained is in good agreement with values found in the litterature while J_2,J_3 and J_4

$J_i(K)$	(a)	(b)	(c)	d_{Cu-Cu} (Å)
J_1	178.0	191.2	181.0	3.42
J_2	-	0.5	0.4	5.91
J_3	-	-1.1	-1.0	6.83
J_4	-	-0.1	-0.2	6.83

TABLE 4.2 – Values of the J_i couplings in Kelvin in the three fragments and the distance associated to each couplings.

are very weak.

These results highlight the mainly antiferromagnetic nature of these interactions with a leading interaction J_1, at least two order of magnitude higher than the others. This confirms the use of models composed of only one J interaction between nearest neighbors and the neglect of other couplings should cause no more than a small error.

4.2.2 Out-of-plane coupling

Another preoccupation of the model is the occupation disorder. Cristallographic studies have identified two types of defects present at low temperatures in the crystal samples [72]. The first is the substitution of the Zn^{2+} inter-plane site with a Cu^{2+} ion, estimated to occur at a rate of 0.15, and the second is Zn^{2+} ions entering the Kagomé plane with a much smaller occurance rate (<0.05). The latter is suspected to appear even less (<0.01) because of the difficulty of the Zn^{2+} ion to enter the Cu^{2+} site in the Kagomé plane.[73] As such, our study focuses on the first type of disorder. Following the same approach, an embedding with the defect at its centre was constructed to include the interplan ion, the fragment is shown in Figure 4.4. This cluster contains two layers of Kagomé planes, each with three Cu^{2+} ions and hydroxo ligands of their coordination sphere while the interstitial Zn^{2+} ion is replaced by a Cu^{2+} ion. In a structuraly perfect Kagomé lattice, the inter-site is occupied by a Zn^{2+} ion which is not susceptible to the Jahn-Teller effect because of its doubly occupied d-orbitals, allowing for a C_3 symmetry axis. However, replacing this ion with a Cu^{2+} (with one singly occupied d-orbital) wich is susceptible to the Jahn-Teller effect, induces structural deformation in the vicinity of the defect.

To account for the deformations, the structure of the fragment was optimised using

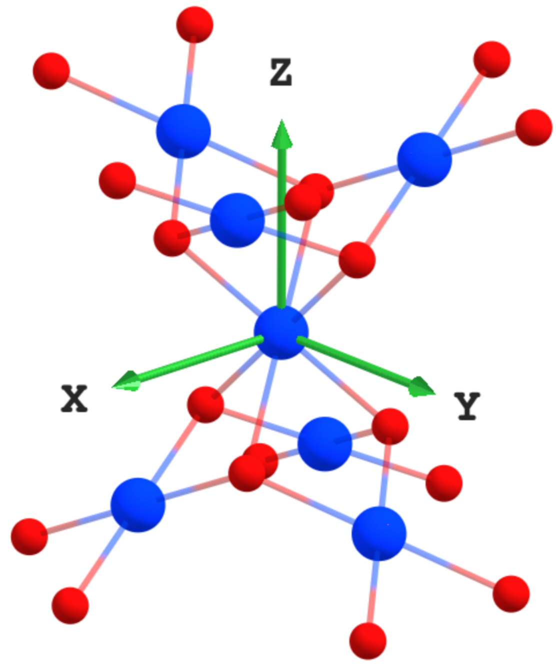
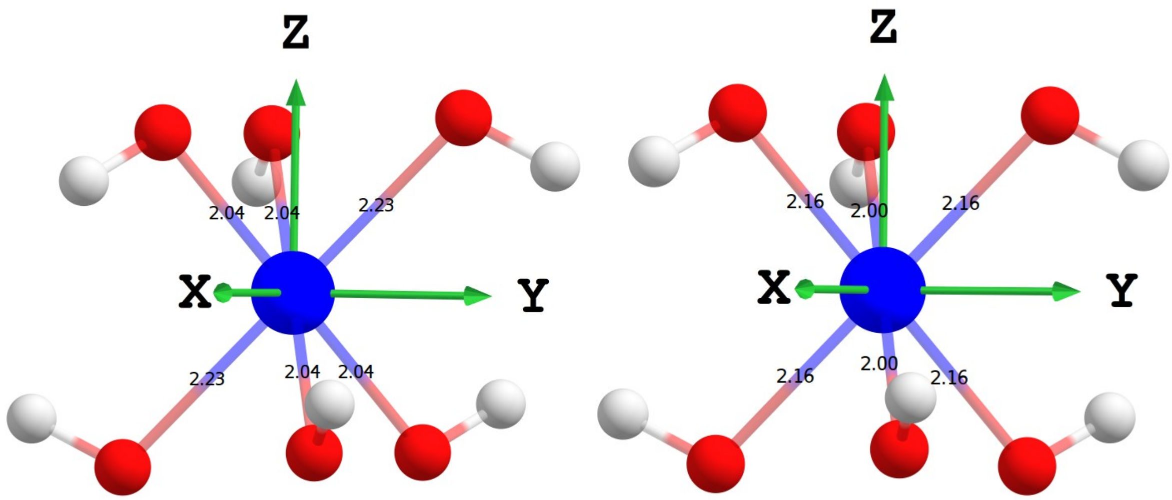


FIGURE 4.4 – Embedded cluster used to compute the out of plane coupling before geometry optimisation. The three copper ions (blue) at the top and bottom of the fragment each belong to their own Kagomé plane. Hydrogens atoms are hidden.

the ORCA package via DFT calculations with the PBE or B3LYP functionals. It is expected that local impurities act only locally and should not have any long range effect, thus no changes were made to the embedding. In order to maintain the rigidity of the crystal structure only the coordination sphere of the defect, *i.e.* the hydroxide groups, was left free during optimisation, while the positions of the remaining ions were constrained. Starting from the symmetric structure, two different types of structure were obtained depending of the functional used. The PBE functional calculation converges towards a structure with four short Cu-O bonds and two long ones, this structure is referred to as the “4+2” structure. While the B3LYP functional calculation converges towards the opposite, with two short Cu-O bonds and four long ones. This structure is labelled “2+4”. In order to determine the most stable structure, a second set of optimisations was performed starting from the previously obtained PBE (resp. B3LYP) structure but now using the B3LYP (resp. PBE) functional. Both calculations converged towards the “4+2” structure which was found to be slightly lower in energy than the “2+4” structure with an energy difference of $\Delta E = 100K$ using the SCF energies obtained with the B3LYP functional. A similar study[74] was carried out using a different computational protocol, the optimisation was performed via periodic DFT in a structure where all Zn^{2+} ions were replaced by Cu^{2+}. In this study, the environment around the Cu^{2+} defect is taken as octahedral, so the highest occupied orbital is either the $d_{x^2-y^2}$ or d_{z^2} which lead to two structures that are similar to ours. The naming of the structures in this study's results has been reused. As the two structures are quite close in energy, a difference lower than the estimated first neighbour coupling, both structures were explored for the extraction of J couplings.



(a) "4+2" structure (b) "2+4" structure

FIGURE 4.5 – Close environment of intersite Cu^{2+} in both structure obtained via DFT with the B3LYP functional. Bond lengths are in Angstrom

Table 4.3 shows the distances between the oxygen atoms and the different Cu^{2+} ions in the symmetric structure and the relative displacements induced by the defect. In both the optimised structures, one type of oxygen moves away from the in-plane copper while the other moves closer. The lift of degeneracy of the ground state results from that of the d-orbitals. In the "4+2" structure, the essentially d_{yz} orbital is stabilised by the distancing of the ligands in the Y direction, in accordance with the ligand field theory. The magnetic orbital is hence of d_{xz} nature with a lower contribution of the d_{xy} orbital. The same reasing can be applied to the "2+4" structure with the d_{yz} orbital destabilised by bringing the ligand closer in the Y direction, which result in a mainly d_{yz} magnetic orbital. Note that for the analysis the main elongation/shortening, found in the respective structure, is set to be along the Y axis for comparison. It does not match the representation of Figure. 4.5b.

	C_3	"4+2"	"2+4"
$Cu_{defect} - O$ (1)	2.119	-7.50×10^{-2}	4.24×10^{-2}
$Cu_{defect} - O$ (2)	"	11.55×10^{-2}	-11.72×10^{-2}
$Cu_{plane} - O(1)$	1.984	2.86×10^{-2}	0.40×10^{-2}
$Cu_{plane} - O(2)$	"	-1.17×10^{-2}	4.65×10^{-2}

TABLE 4.3 – Distance between the oxygens and the copper ions, defect or in plane, in the C_3 structure and the relative displacement in both optimisation scheme given in Å.

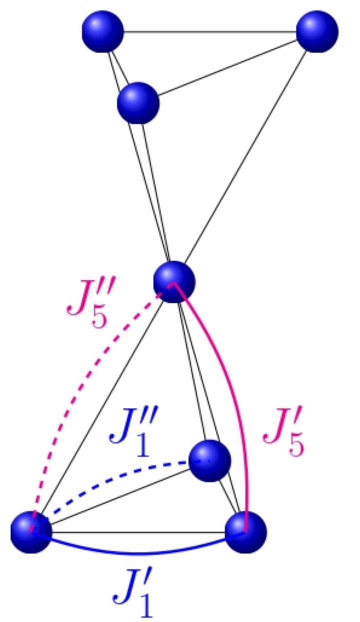


FIGURE 4.6 – Schematic representation of the J_i couplings in the cluster with defect.

Focusing on a tetra-nuclear fragment with a triad of Cu^{2+} ions in the same Kagomé plane and the defect ion, two of the bridging oxygens undergo an equivalent displacement while the third one differs. Whereas the symmetric structure has only one in-plane coupling, J_1, the displacements of the oxygen require the definition of new couplings :

— J_1' is the in-plane coupling defined between two Cu^{2+} ions bridged by one of the two equivalent oxygens. It appears twice in the the triad of copper ions.

— J_1'' is the in-plane coupling which involves the remaining oxygen, appearing once.

— J_5' is the out-of-plane coupling between two Cu^{2+} bridged by two equivalent oxygens. It appears once in the triad involving the defect.

— J_5'' is the out-of-plane coupling between two Cu^{2+} through two non-equivalent bridges. It appears twice in the triad involving the defect.

These couplings are represented in Figure 4.6, the dashed-line indicates couplings that involves the non-equivalent oxygen. Note that the couplings are represented only once on the schematic.

Structure	J_1'	J_1''	J_5'	J_5''
"4+2"	92.9	262.9	-47.5	-73.0
"2+4"	185.3	66.14	-81.61	-63.4

TABLE 4.4 – Values of the J_i interactions given in Kelvin in presence of a Cu^{2+} defect in the intersite for both optimised structure.

Table 4.4 reports the values extracted using BS-DFT for the two optimised structure. Calculations were also performed on the C_3 structure in order to provide comparisons. The C_3 structure indicates a strong $J_1 = 184.2K$ antiferromagnetic coupling which is consistent with previously extracted values, as well as a non negligible out-of-plane ferromagnetic coupling $J_5 = -80.6K$ (the symmetric structure has only one type of out-of-plane coupling). Extractions from the optimised structure show a significant impact of the deformations on the J_1 coupling. Firstly, coupling within the plane decreases as oxygen atoms move away from it, whereas it increases as they move closer. This can be rationalised by the decrease of the ligand-metal hopping integral when the distance increases, resulting in the decrease of the antiferromagnetic superexchange contribution to J coupling (see Eq. 1.1.2 and 1.1.3).

The most important result of this study is the identification of ferromagnetic couplings between the in-plane copper ion and the defect. These couplings are not negligible and can go up to $0.4|J_1|$ which should have an impact on the physics of the system. Note that with a distance of 3.07Å separating the defect from the in-plane Cu^{2+} ions compared to the distance of 3.42Å between two in-plane Cu^{2+}, the out-of-plane couplings become first-neighbor coupling.

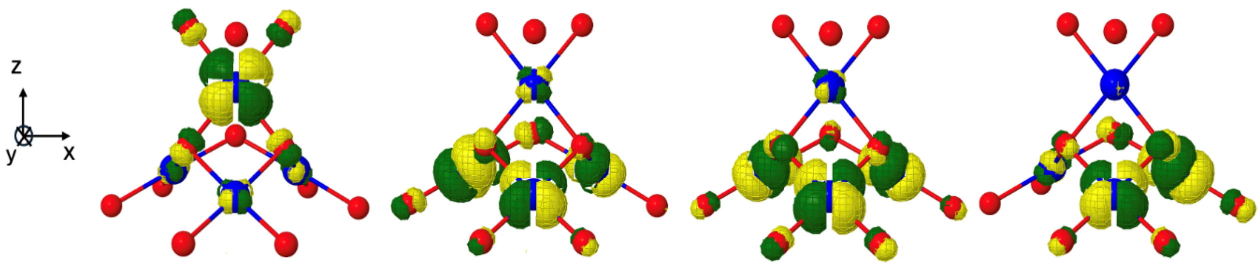


FIGURE 4.7 – CAS(4,4)SCF active orbitals calculated for a tetra-nuclear fragment constituted of three in-plane (below) and an inter-plane (above) copper ions.

Around the defects, these two types of interactions are of the same order of magnitude which should impact the collective properties of the whole material. To obtain insight into the ferromagnetic character of the inter-plane couplings, we have plotted in Figure 4.7 the magnetic orbital optimized using a CAS(4,4)SCF calculation on a tetra-nuclear embedded fragment (obtained by removing the three uppermost copper ions of the "4+2" structure cluster to get a clearer view).

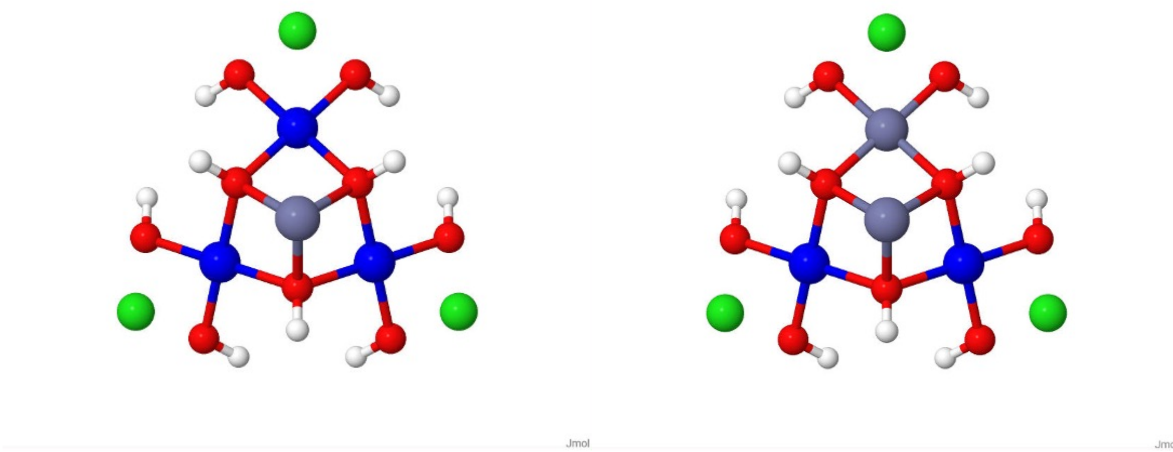
The nature of the coupling is roughly determined by three ingredients :

(i) the orbital overlap increasing the kinetic exchange and hence the antiferromagnetic character of the interaction.

(ii) the direct exchange, a purely ferromagnetic interaction.

(iii) the spin polarization, that may favor ferro- or antiferromagnetism depending on the system.

The three active orbitals on the right of the Figure 4.7 are strongly delocalised on the three in-plane copper ions. On the contrary, the orbital (left) localized on the impurity shows a very small overlap with the orbitals of the in-plane copper ions, excluding a large kinetic exchange contribution. The direct exchange integral is generally quite small between distant atoms, and therefore, large values of J_5' and J_5'' must be attributed to spin polarization. The appearance of such large interactions calls into question the validity of considering this material as two-dimensional.



(a) Trimer (b) Dimer

FIGURE 4.8 – Trinuclear and dinuclear embedded clusters used for wave-function computation

4.3 Wave-Function Theory study

The second purpose of this study was the determination of anisotropic interaction in the crystal. The extraction of such interactions follows the same procedure as described in Chap. 3. The computation of SO-states being very computationally demanding, this extraction will be restricted to two fragments of small size.

The first fragment, Figure 4.8 (left), is similar to the one used for DFT calculation, with three copper ions arranged in an equilateral triangle. A second fragment Figure 4.8 (right) was designed by substituting one of the Cu^{2+} with a diamagnetic Zn^{2+}, both share rather close structural properties (charge, mass, ionic radius). As the d shell of the zinc ion is completely filled it will not take part in the magnetic interactions, leaving us with essentially a dimer to capture the interaction of only two Cu^{2+} ions. This approach has been studied in several other materials and we will see from the results that this substitution has little impact on the extracted values but can significally reduce the overall study time.

4.3.1 Isotropic Coupling

To check the applicability of WFT calculation to this system, the first step was to try and reproduce the J_1 value. In both cluster, two types of active space were considered :

— one electron and orbital per magnetic center.

— all electrons occupying the d shell of each magnetic center.

The first type of active space results in a CAS(2,2)SCF for the dimer calculation which generates one triplet (S=1) and one singlet (S=0) state.

$$\begin{aligned}
|1,1\rangle &= |ab| \\
|1,0\rangle &= \frac{1}{\sqrt{2}}(|a\bar{b}| - |b\bar{a}|) \\
|1,-1\rangle &= |\bar{a}\bar{b}| \\
|0,0\rangle &= \frac{1}{\sqrt{2}}(|a\bar{b}| + |b\bar{a}|)
\end{aligned} \tag{4.3.1}$$

These functions are eigenfunctions of the $\hat{\mathbf{S}}^2$ operator, as such their energies can be mapped on the HDvV Hamiltonian's eigenvalues. The value of J_1 is then given by the difference in energy between the triplet and singlet state $J_1 = E(S=1) - E(S=0)$. The enlarged active space CAS(18,10)SCF generates 25 Triplet and 25 Singlet states.

In the case of the trimer cluster, CAS(3,3)SCF represents the minimum active space which generates one quartet (S=3/2) state and two doublet (S=1/2) states. The two doublets are degenerate if only one value of J is taken into account as is our case with an equilateral repartition of the copper ions.

$$\begin{aligned}
|3/2,3/2\rangle &= |abc| \\
|3/2,1/2\rangle &= \frac{1}{\sqrt{3}}(|ab\bar{c}| + |a\bar{b}c| + |\bar{a}bc|) \\
|1/2,1/2\rangle_1 &= \frac{1}{\sqrt{6}}(|ab\bar{c}| + |a\bar{b}c| - 2|\bar{a}bc|) \\
|1/2,1/2\rangle_2 &= \frac{1}{\sqrt{2}}(|ab\bar{c}| - |a\bar{b}c|)
\end{aligned} \tag{4.3.2}$$

The expressions of the negative m_s value component were left out as they only differ by the number of up and down spins. The value of J_1 is then given by $J_1 = \frac{2}{3}(E(S = 3/2) - E(S = 1/2))$. The enlarged active space CAS(27,15)SCF generates 125 quartet and 250 doublets states.

Table 4.5 shows the results obtained at different levels of calculation for different active spaces. CASSCF calculation tends to overlocalise the magnetic orbitals on the metal ions, reducing the ligand participation, thus underestimating mechanism at the origin of the antiferromagnetic contribution to the J integral. As such it does not provide a good estimate but can nonetheless give information about the nature of the coupling.

Fragment	CAS(2,2)SCF CAS(3,3)SCF	CAS(18,10)SCF CAS(27,15)SCF	CASPT2	DDCI1	DDCI3
Dinuclear	15.4	11.9	100.4	71.9	115.3
Trinuclear	11.8	11.1	80.6	72.9	86.3

TABLE 4.5 – Values of J_1 in K extracted on both fragment. Dynamic correlation calculation are performed on the minimal active space.

These results indicate an antiferromagnetic J coupling with a large impact of dynamic correlation, yet these values do not match with the DFT calculations and the litterature. An active space only composed of d-orbitals is not enough to achieve a correct estimation of the isotropic coupling in this system. To improve the description of the antiferromagnetic mechanisms it was then thought to include the bridging ligand orbitals into the active space. The inclusion of the p-orbitals of the bridging oxygen presents a challenge, as they stay doubly occupied at CASSCF level, reaching a stable active space becomes troublesome because of convergence issue. Moreover, when achieved, the inclusion in the active space by a direct inclusion does not improve the estimated value of J_1. To avoid these troubles, a methodology based on the projection of model vectors onto the inactive orbital space[75] is applied. This allows to obtain orbitals that are concentrated on the bridging ligand and whose shape is optimal for interactions with the active orbitals. These orbitals are then introduced into the active space and are represented in Figure 4.9 in the dinuclear calculation and Figure 4.10 in the trinuclear fragment.

It has been previously established that performing a DDCI1 calculation on top of an

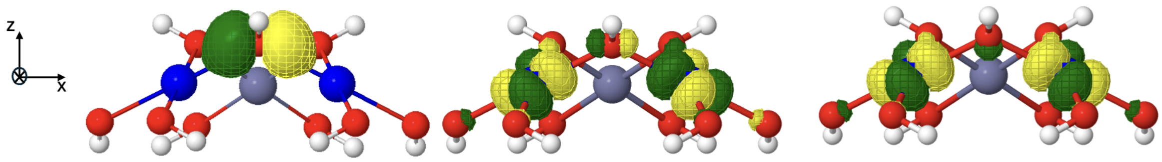
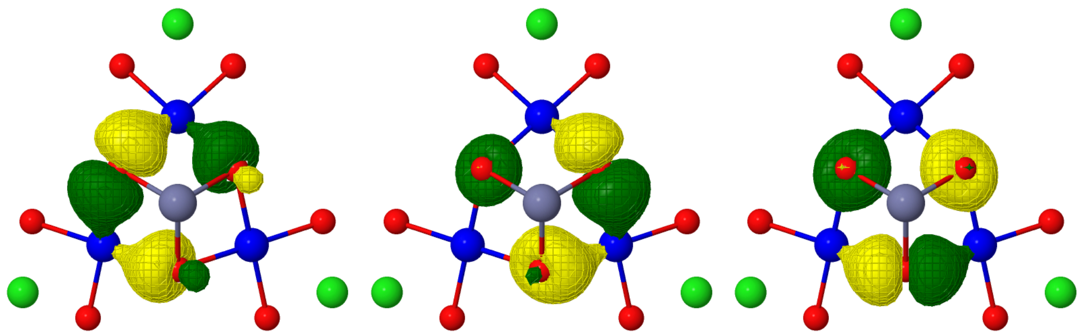
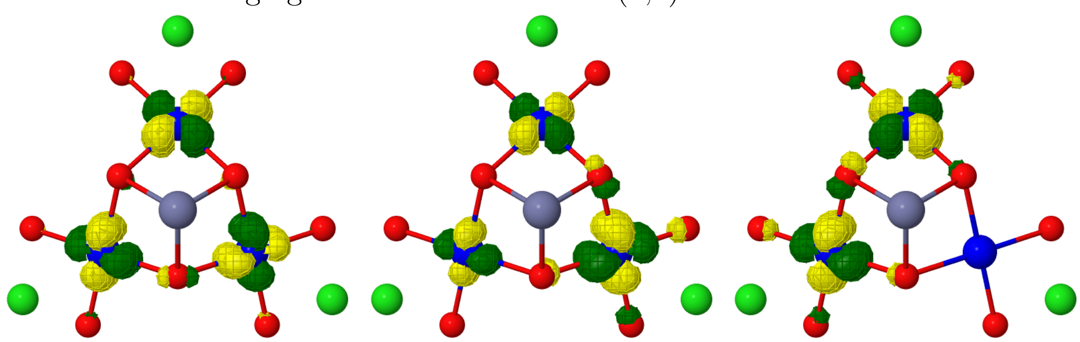


FIGURE 4.9 – CAS(4,3)SCF, bridging orbital and magnetic orbitals calculated on the dinuclear fragment



Bridging orbitals from the CAS(9,6)SCF calculation.



Magnetic orbitals from the CAS(9,6)SCF calculation.

FIGURE 4.10 – Active space orbitals from CAS(9,6)SCF calculation.

active space composed of the ligands orbitals in addition to the magnetic orbitals allows to obtain value of J coupling on par to a DDCI3 calculation. This allows to greatly reduce computational time, the active space is now composed of one electron/orbital per metallic centre and two electrons in one orbital per oxygen (CAS(4,3)SCF in the dinuclear and CAS(9,6)SCF in the trinuclear fragment). This procedure has allowed the extraction of new values for J_1 in both fragment, $J_1 = 151.5K$ in the dinuclear cluster and $J_1 = 181.7K$ in the trinuclear cluster. Even if the dinuclear value of J_1 is lower, the coupling extracted from the trinuclear fragment is well within range of the expected value by both DFT and experiment as such the extraction is considered satisfactory.

4.3.2 Anisotropic interactions

The computation of the isotropic couplings was a way to assertain the validity of our methods and calibration of the embedded cluster method. The electronic structure of the compound seems to be well replicated with a satisfactory value of the J_1 coupling. We can now address the main purpose of this study, which is to extract anisotropic interactions. Theoretical models propose the inclusion of a weak Dzyaloshinskii-Moriya pseudo-vector along the out of plane direction to account for the behavior at low temperature but the symmetric tensor of anisotropy $\overline{\overline{D}}_{ij}$ is usually left out. Following the procedure explained in Chap. 3, all three components of the DMI were extracted as well as the D_{ij}-tensor.

Bi-nuclear calculation

Starting from the CASSCF wave-function obtained with the enlarged active space of all 18 copper d-shell electrons and the 10 d-orbitals, computation of the SO-states was performed using the SO-RASSI method implemented in MOLCAS. The lowest part of the energy spectrum was corrected using the isotropic coupling value previously obtained. No assumption on the magnetic axes was made and so the extraction of all anisotropic parameters (9 in total) was performed using effective Hamiltonian theory with a model space composed of the three M_S components of the triplet and singlet states. The matrix representation of the model Hamiltonian in the model space basis is :

H_{MS}	$\vert 1,-1\rangle$	$\vert 1,0\rangle$	$\vert 1,1\rangle$	$\vert 0,0\rangle$
$\langle 1,-1\vert$	$\frac{J}{4}+\frac{D_{zz}}{4}$	$\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$\frac{(D_{xx}-D_{zz}-2iD_{xy})}{4}$	$\frac{d_y+id_x}{2\sqrt{2}}$
$\langle 1,0\vert$	$\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J}{4}-\frac{D_{zz}}{4}+\frac{(D_{xx}+D_{yy})}{4}$	$-\frac{D_{xz}-iD_{yz}}{2\sqrt{2}}$	$-\frac{id_z}{2}$
$\langle 1,1\vert$	$\frac{(D_{xx}-D_{zz}+2iD_{xy})}{4}$	$-\frac{D_{xz}+iD_{yz}}{2\sqrt{2}}$	$\frac{J}{4}+\frac{D_{zz}}{4}$	$\frac{d_y-id_x}{2\sqrt{2}}$
$\langle 0,0\vert$	$\frac{d_y-id_x}{2\sqrt{2}}$	$\frac{id_z}{2}$	$\frac{d_y+id_x}{2\sqrt{2}}$	$-\frac{3J}{4}-\frac{D_{zz}}{4}-\frac{(D_{xx}+D_{yy})}{4}$

Note that the $\mathbf{d_{ij}}\cdot\hat{\mathbf{S}}_i\times\hat{\mathbf{S}}_j$ operator is equivalent to the antisymmetric tensor $\overline{\overline{T}}_{ij}$ of the operator $\hat{\mathbf{S}}_i\cdot\overline{\overline{T}}_{ij}\cdot\hat{\mathbf{S}}_j$. There is a one to one correspondance between its components, with d_x=T_{yz}, d_y=$-T_{xz}$ and d_z=T_{xy}. This remark will be useful for identifying the non-zero components of the vector via a symmetry analysis.

The values obtained for both interaction expressed in Kelvin in the axis system of the cluster are :

$$\overline{\overline{D}}_{12} = \begin{pmatrix} 0.02 & 0 & 0 \\ 0 & 0.28 & 0 \\ 0 & 0 & -0.31 \end{pmatrix} \qquad \mathbf{d}_{12} = \begin{pmatrix} 0 \\ -4.73 \\ -1.70 \end{pmatrix}$$

The symmetric tensor can be diagonalized to extract the $D = -0.46K$ axial parameter and a small $E = 0.13K$ rhombic term following the usual definition of these terms. These values are small and should have only little impact on the magnetic properties of this system. However, the DMI components are non-negligible. The di-nuclear fragment studied presents a C_S symmetry point group, with (YZ) being the symmetry plane. The symmetry rules for this point group have been derived to identify the non-zero components of the antisymmetric tensor $\overline{\overline{T}}_{ij}$.[76, 77] When the symmetry plane is (XY), these components are T_{xz} and T_{yz}. Replacing the X and Z axes to match our axes frame, one gets that the non-zero components are $T_{xz} = d_y$ and $T_{xy} = d_z$. This rationalises the value obtained for the component along the X-axis, *i.e.* d_x, which is equal to zero. Note that the in-plane component (along the y-axis) is larger than the out of plane component (along the z-axis). This result can be surprising but can be rationalised by looking at the physical origins of the DMI component. It was demonstrated analytically that the DM-interaction comes from the hybridization of the metal's d-orbitals. It was shown that :

— a mixing between $d_{x^2-y^2}$ and d_{yz} or d_{xy} and d_{xz} generates a d_x component.

— a mixing between $d_{x^2-y^2}$ and d_{xz} or d_{xy} and d_{yz} generates a d_y component.

— a mixing between $d_{x^2-y^2}$ and d_{xy} or d_{xz} and d_{yz} generates a d_z component.

This mixing can be clearly observed on Figure 4.9 (with the axis system), and from the expression of one of these magnetic orbitals on the Cartesian d-orbitals of one copper ion : $0.4889d_{x^2-y^2} + 0.2120d_{xz} - 0.2841d_{xy} - 0.3666d_{yz}$.

Different configurations for the dinuclear cluster were explored to make sure that the values extracted were not affected by the choice of di-nuclear fragment inside the triad of Cu^{2+} ions. The extra zinc ion was swapped around all three in-plane sites. Note that the symmetry rules only apply if the symmetry plane coincides with the axis frame of the dinuclear fragment under consideration. This is not the case for the other two dimer clusters, therefore, it is expected that all three components of the DMI will be non-zero. Therefore, it will be more adequate to refer to the in-plane projection as $d_{\parallel}$ and the out of plane component as $d_{\perp}$. In all three calculations, the respective value of $|d_{\perp}|$ and $|d_{\parallel}|$ were found to be equals.

Trinuclear calculations

The extraction was then performed on the tri-nuclear fragment with an active space of 27 electrons in 15 orbitals. In the case of three magnetic centres, the model Hamiltonian is now written :

$$\begin{aligned}\hat{H}_{MS} =& J_{12}\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_2 + \hat{\mathbf{S}}_1 \cdot \overline{\overline{D}}_{12} \cdot \hat{\mathbf{S}}_2 + \mathbf{d}_{12} \cdot (\hat{\mathbf{S}}_1 \times \hat{\mathbf{S}}_2) \\ & J_{12}\hat{\mathbf{S}}_2 \cdot \hat{\mathbf{S}}_3 + \hat{\mathbf{S}}_2 \cdot \overline{\overline{D}}_{23} \cdot \hat{\mathbf{S}}_3 + \mathbf{d}_{23} \cdot (\hat{\mathbf{S}}_2 \times \hat{\mathbf{S}}_3) \\ & J_{13}\hat{\mathbf{S}}_1 \cdot \hat{\mathbf{S}}_3 + \hat{\mathbf{S}}_1 \cdot \overline{\overline{D}}_{13} \cdot \hat{\mathbf{S}}_3 - \mathbf{d}_{13} \cdot (\hat{\mathbf{S}}_1 \times \hat{\mathbf{S}}_3)\end{aligned} \tag{4.3.3}$$

Note the negative sign in front of the last term d_{13} which originates from the permutation relationship specific to the DMI (d_{13}=-d_{31}). On the other hand, we set $J_{12} = J_{23} = J_{13}$ and let the D_{ij}-tensor be expressed in the axis system of the cluster with 6 parameters each. The first step is to obtain the matrix representation of this Hamiltonian in the basis of the determinants formed from the three magnetic orbitals on each site (a, b and c) located on each copper center. To express this matrix in the coupled basis, *i.e.* in the basis formed from the m_s components of the quartet and two doublet states, one has to build the transformation matrix. It can be obtained using the Clebsch-Gordan coefficients or by diagonalizing the HDvV Hamiltonian representation matrix and using the eigenvectors as the transition matrix.

	$\left\|\frac{3}{2},\frac{3}{2}\right\rangle$	$\left\|\frac{3}{2},\frac{1}{2}\right\rangle$	$\left\|\frac{3}{2},-\frac{1}{2}\right\rangle$	$\left\|\frac{3}{2},-\frac{3}{2}\right\rangle$	$\left\|\frac{1}{2},\frac{1}{2}\right\rangle_1$	$\left\|\frac{1}{2},-\frac{1}{2}\right\rangle_1$	$\left\|\frac{1}{2},\frac{1}{2}\right\rangle_2$	$\left\|\frac{1}{2},-\frac{1}{2}\right\rangle_2$
$\left\langle\frac{3}{2},\frac{3}{2}\right\|$	$\frac{3J}{4}$							
$\left\langle\frac{3}{2},\frac{1}{2}\right\|$	0	$\frac{3J}{4}$						
$\left\langle\frac{3}{2},-\frac{1}{2}\right\|$	0	0	$\frac{3J}{4}$					
$\left\langle\frac{3}{2},-\frac{3}{2}\right\|$	0	0	0	$\frac{3J}{4}$				
$\left\langle\frac{1}{2},\frac{1}{2}{}_1\right\|$	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(2d_{12}-d_{13}-d_{23})e_z}{2\sqrt{6}}$	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{6}}$	0	$-\frac{3J}{4}$			
$\left\langle\frac{1}{2},-\frac{1}{2}\right\|_1$	0	$\frac{i(2d_{12}-d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{6}}$	$\frac{i(2d_{12}-d_{13}-d_{23})e_z}{2\sqrt{6}}$	$\frac{(2d_{12}-d_{13}-d_{23})}{4\sqrt{2}}$	0	$-\frac{3J}{4}$		
$\left\langle\frac{1}{2},\frac{1}{2}\right\|_2$	$\frac{\sqrt{3}(d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(d_{13}-d_{23})e_z}{2\sqrt{2}}$	$-\frac{i(d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{2}}$	0	$\frac{i(d_{12}+d_{13}+d_{23})}{2\sqrt{3}}$	$-\frac{i(d_{12}+d_{13}+d_{23})(e_x-ie_y)}{2\sqrt{3}}$	$-\frac{3J}{4}$	
$\left\langle\frac{3}{2},\frac{3}{2}\right\|_2$	0	$\frac{i(d_{13}-d_{23})(e_x+ie_y)}{4\sqrt{2}}$	$\frac{i(d_{13}-d_{23})e_z}{2\sqrt{2}}$	$-\frac{i\sqrt{3}(d_{13}-d_{23})(e_x-ie_y)}{4\sqrt{2}}$	$-\frac{i(d_{12}+d_{13}+d_{23})(e_x+ie_y)}{2\sqrt{3}}$	$-\frac{ie_z(d_{12}+d_{13}+d_{23})}{2\sqrt{3}}$	0	$-\frac{3J}{4}$

The symmetric tensor terms (6 each, 18 total) are left out of the matrix for clarity but are taken into account during the extraction. The extracted Dzyaloshinskii-Moriya pseudo vectors are :

$$\mathbf{d}_{12} = \begin{pmatrix} 0 \\ -4.73 \\ -1.78 \end{pmatrix} \qquad \mathbf{d}_{13} = \begin{pmatrix} -4.1 \\ 2.36 \\ -1.78 \end{pmatrix} \qquad \mathbf{d}_{23} = \begin{pmatrix} 4.1 \\ 2.36 \\ -1.78 \end{pmatrix}$$

The axis system was not changed between the dinuclear and trinuclear fragments. Hence, the two Cu^{2+} ions numbered 1 and 2 in the trinuclear fragment are the one that are included in the dinuclear calculation. Note that the extracted DMI-vector is almost the same $\mathbf{d}_{12}^{dinuclear} \approx \mathbf{d}_{12}^{trinuclear}$ which validate the expansion to a trinuclear fragment. The three DMI vectors share the same out of plane $d_z^{ij} = 1.78K$ and in plane component $|d_{xy}^{ij}| = 4.73K$. Their orientations is depicted in Figure 4.11. It is important to mention that the out-of-plane component $d_\perp$ oscillates between below and above the (XY) plane of copper ions from one triangle to the next, following the alternating positions above and below of the oxygens

Fragment	$\lvert d_{ij}^{\parallel} \rvert$	$\lvert d_{ij}^{\parallel} \rvert$	$\lvert d \rvert$	D	E
Bi-nuclear	4.73	1.70	5.03	-0.46	0.13
Tri-nuclear	4.73	1.78	5.05	-0.51	0.16

TABLE 4.6 – Values in Kelvin of the $|d|$ DM magnitude and its components, axial (D) and rhomic (E) anisotropic exchange parameters, obtained for both fragments.

Table 4.6 provides the results obtained from the extractions performed on the bi-nuclear and tri-nuclear fragments. It shows good agreement between the two fragments for all of the parameters considered in the model.

4.4 Conclusion

The purpose of this work was the application of *ab initio* calculations to extract magnetic interactions in Herbertsmithite which has not been done before. The first part of the study focused on the isotropic couplings using embedded cluster method,

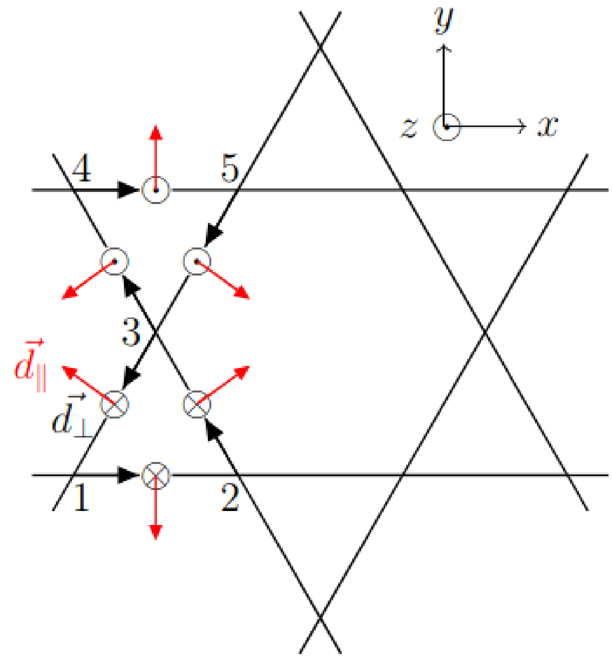


FIGURE 4.11 – Schematic representation of the DMI vector components in two adjacent Cu^{2+} triangles of the lattice. The black arrows between the magnetic centers indicates the order of the first and second spins in the vector products of the DM interactions. The out-of-plane components alternate from one triangle to its neighbors and point in the opposite direction to the position of the oxygens. The angle between the in-plane components of the vectors within the triangle is 120°.

the exploration of fragments with different sizes has been done using DFT calculations, it allowed the determination of the possible isotropic couplings. Extractions performed on multiple fragments showed good transferability of the couplings from one cluster to the other. This justifies the validity of the application of the embedded cluster method for this material. The study of the structure without defect indicates that the only relevant in-plane coupling is between first neighbour. A reported value of $J_1 \approx 180K$ matches very well with experimental data, other couplings were found to be two orders of magnitude smaller. The subject of defects was addressed as well with the study of structural deformation around the most recurring defect. The substitution of the intersite Zn^{2+} ion by a additional Cu^{2+} ion tends to modify the environment near this site due to a strong Jahn-Teller effect. Two types of structure were identified, which differ by the displacement of the hydroxo ligands around the interplane site. As the two structures are close in energy, $\approx 100K$, the extraction of isotropic couplings was performed for both geometries. It was found that the deformations have a significant impact on the in-plane coupling, replacing the unique J_1 by two different values. The most remarkable result of this study is the introduction of new ferromagnetic couplings between in-plane copper ions and the interplane defect. These coupling are non negligibles, approximately $0.4|J_1|$, which could have a strong impact on collective properties. The second part of this study focused on the extraction of anisotropic interactions on a dinuclear and a trinuclear fragments. First, the extraction of the first neighbour coupling using wave-function based method was performed. It was found that an active space composed only of the d-orbitals of the copper ions does not allow a good estimation of J_1. To solve this problem, some orbitals of the bridging ligand were included in the active space following a projection procedure. This allowed the extraction of a J_1 value that compares well with DFT calculations. Then, the Dzyaloshinskii-Moriya pseudovector and the symmetric tensor of anisotropic exchange were extracted on both fragments. A significant value of DMI was obtained with both an out-of plane and in-plane components, the latter usually overlooked in theoretical models. Results between fragment compares perfectly which validate the use of the Multi-Spin Hamiltonian to describe the anisotropy of this system.

Conclusion

Several remarkable results emerge from this work. Firstly, during the study of the series of iron complexes, we obtained, in collaboration with experimentalists and for the first time, certain complexes that fall under the first-order spin-orbit regime. These results are in perfect agreement with experimental observations and have enabled us to understand why the magnetisation and magnetic susceptibility curves could not be correctly reproduced by the D and E interaction fitting software. Indeed, the presence of a significant angular momentum means that the magnetic moment cannot be defined from spin alone. The study of the impact of the electric field $\mathbf{F}$ on parameters D and E in the Ni(II) complex has led to remarkable and unexpected results. Indeed, the response to the electric field is considerable. The slope dD/dF is nearly 100 times greater than that calculated and measured for a Mn(II) complex, the study of which was very recently published [22]. Furthermore, applying the field in the Z direction, parallel to the principal magnetic axis, induces a smaller response on D than in the Y direction. These results have been rationalised and show that the determining factor is the nature of the excitation in the states that contribute significantly to anisotropy. In the case considered, we have seen that this excitation, which in this instance is between the $d_{x^2-y^2}$ and d_{xy} orbitals, is located in the (XY) plane. It is therefore in this plane that the application of the electric field will induce the most significant response. Another important result is the impact of the field on the geometry of the complex. Indeed, very small distortions lead to considerable variations in the D parameter value. The work carried out on the Cu_2Cl_5 model complex enabled us to understand the physical origin of the symmetric exchange anisotropy tensor. As the parameters of this tensor are generally very weak, we considered a model molecule in a geometry inducing spin-orbit coupling close to the first-order regime. We also studied the impact of the electric field on the anisotropy parameters in this binuclear complex, which enabled us to understand the interference effect between the symmetric and antisymmetric (Dzyaloshinskii-Moriya) tensors. The important conclusion from this work is that all the parameters studied, *i.e.* the symmetric tensor of single-nuclear complexes, the symmetric exchange tensor and the Dzyaloshinskii-Moriya tensor that appear in poly-nuclear complexes, contribute to magnetoelectric coupling. This shows that it is possible to control all these magnetic properties using an electric field. The most substantial part of this thesis is the study of Herbertsmithite material. It should be noted that, unlike several strongly correlated materials studied previously (cuprates [23], nickelates [78], manganites [25], vanadates [26], etc.), this material has proven particularly difficult to describe. The DDCI method, which is currently the most accurate method for calculating magnetic couplings, significantly underestimates the antiferromagnetic coupling between nearest-neighbours. This forced us to use an alternative method, which consists of optimising the orbital of the bridging ligand, that dictates the coupling, and introduce it into the active space in the dynamic correlation calculation step. Here too, several interesting physico-chemical results were obtained :

- The coupling with interplanar Cu^{2+} impurities resulting from an exchange between interplanar Zn^{2+} and intraplanar Cu^{2+} is ferromagnetic and of the same order of magnitude as the nearest-neighbour J_1. This result, combined with an occurrence of approximately 15% of these impurities, calls into question the two-dimensional nature of the material and could explain why it is impossible to reproduce its collective properties using a 2D model.
- The presence of these impurities, which occupy a C_3 symmetry site and produce Cu^{2+} prone to Jahn-Teller effect, induces local distortion of the lattice. These distortions cause two very different couplings between the first-neighbour intraplane Cu^{2+} ions.
- The symmetric anisotropy tensor and longer-range isotropic interactions are negligible in amplitude. On the other hand, the Dzyaloshinskii-Moriya is relatively significant and its strongest component lies in the plane. A detailed analysis of

the wave functions and symmetries has made it possible to rationalise this result. It should be noted that this source of anisotropy is often neglected in studies of collective properties and, when taken into account, it is generally reduced to its single component along the z-axis, *i.e.* perpendicular to the Kagomé lattice plane. Taking it into account in models could improve the description of this material.

Many perspectives are now open. One of them has already been pursued and could lead to a forthcoming publication following a similar study to Chap 3. It involves a study of all the anisotropy parameters in a binuclear compound containing Cu^{2+} and Ni^{2+}. In this compound, the local anisotropy tensor is added to the other tensors studied in Cu_2Cl_5. The analytical derivations of a first-order model show similarities with those of the previous study, but also some qualitative differences. Furthermore, the study of the impact of the electric field in such a system could reveal possible interference effects between all the sources of anisotropy in the Hamiltonian of equation 0.0.7. Finally, an important perspective concerns strongly correlated materials. Other materials related to Herbertsmithite, such as paratacamite [27] or claringbullite [28], are worth studying, on the one hand to test commonly used models and possibly propose alternative models, and on the other hand for the challenge posed by studying these systems, which are so difficult to describe using our best methods. This last proposal would be a more methodological endeavour aimed at designing and developing a method that would be more precise than DDCI and applicable to a wider range of systems.

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